



H2O Sonar Model Interpretation Report

This report presents the results of the machine learning model interpretation. The report summarizes the results of the explainers that were run to interpret the model in order to describe various model properties such as model problems, approximate model behavior, the most important model features and their behavior, fairness model metrics and issues, model insights, and recommendations for model debugging.

Summary

Problems:	2
Model:	<u><daimojo.model.model object at 0x7f1787e90a50></u>
Target column:	default payment next month
Dataset:	<u>data/creditcard.csv</u>
Interpretation ID:	e11b8df0-8abf-4168-8e83-1ceb2bef706e
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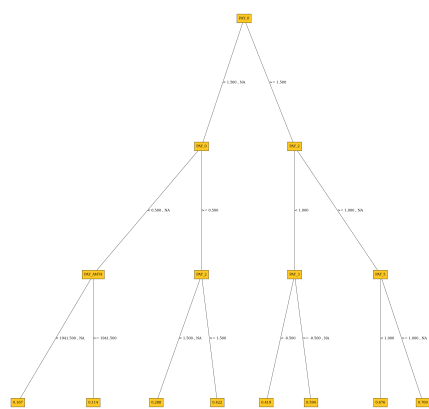
- Residual Surr...Decision Tree
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Explainers overview

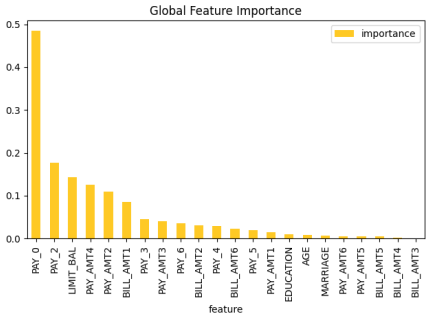
Representatives All explainers



Approximate model behavior

Surrogate Decision Tree

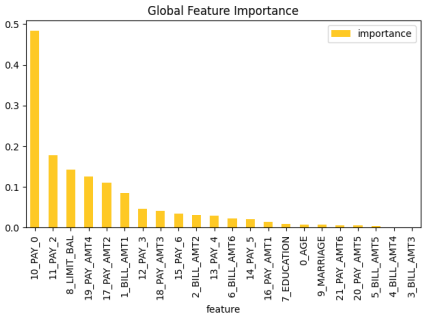
The surrogate decision tree is an approximate overall flow chart of the model, created by training a simple...



Original feature importance

Shapley Values for Original Features of MOJO Models (Naive Method)

Shapley values for original features of (Driverless AI) MOJO models are approximated from the accompanying ...



Transformed feature importance

Shapley Values for Transformed Features of MOJO Models

Shapley explanations are a technique with credible theoretical support that presents consistent global and ...

Interpretation

92%

Scheduled explainers (13):

- [Disparate Impact Analysis](#)
- [Shapley Values for Original Features of MOJO Models \(Naive Method\)](#)
- [Shapley Values for Transformed Features of MOJO Models](#)
- [Residual Surrogate Decision Tree](#)
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Finished explainers (13):

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Successful explainers (12):

- [Disparate Impact Analysis](#)
- [Shapley Values for Original Features of MOJO Models \(Naive Method\)](#)
- [Shapley Values for Transformed Features of MOJO Models](#)
- [Residual Surrogate Decision Tree](#)
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- [Shapley Values for Original Features \(Kernel SHAP Method\)](#)
- [Partial Dependence Plot for Two Features](#)
- [Friedman's H-statistic](#)
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Failed explainers (1):

- [Retrieval Augmented Generation Assessment \(DeepEval\)](#)

Configuration and parameters

Interpretation parameters:

- **model** = <class 'daimojo.model.model'>
 - *Interpreted model.*
 - Type: any
 - Default value:
- **dataset** = data/creditcard.csv
 - *Dataset used to interpret the model.*
 - Type: any
 - Default value:
- **validset** = None
 - *Optional validation dataset used to interpret the model.*
 - Type: str
 - Default value:
- **testset** = None
 - *Optional test dataset used to interpret the model.*
 - Type: str
 - Default value:
- **use_raw_features** = True
 - *Whether to use original features for the training of Surrogate model explainers (True, default) or force transformed features (False).*
 - Type: bool
 - Default value:
- **target_col** = default payment next month
 - *Target column of the interpreted model.*
 - Type: str
 - Default value:
- **weight_col** =
 - *Optional name of the dataset column with weights of examples to be used in the model interpretation.*
 - Type: str
 - Default value:
- **prediction_col** =
 - *Optional name of the dataset column with predictions of examples to be used in the (standalone ~ no model, just predictions) model interpretation.*
 - Type: str
 - Default value:
- **drop_cols** = []
 - *Optional list of the dataset column names to be drop and not used in the model interpretation.*
 - Type: list
 - Default value: []
- **sample_num_rows** = 0
 - *The sample size, number of rows, to be used for the surrogate models. This setting overrides global library sampling configuration.*
 - Type: int
 - Default value: 0
- **results_location** = /tmp/pytest-of-dvorka/pytest-0/test_talk2report_all_explainer0
 - *Filesystem path (database connectio or other storage type location) which specifies where to store interpretation results. Current directory is used by default*
 - Type: str
 - Default value:
- **used_features** = None
 - *Optional specification of the features (dataset columns) used by the interpreted model (in case that model doesn's support used features introspection).*
 - Type: list
 - Default value: None

H2O Sonar library configuration:

Config parameter	Value	Description	Type	Default value
h2o_host	localhost	The host of the H2O-3 server that should be used for the explanation that requires it.	str	localhost
h2o_port	45373	The port of the H2O-3 server that should be used for the explanation that requires it.	int	12349
h2o_auto_start	True	Automatically start H2O-3 server on the interpretation start (True), or do not start the server (False).	bool	True
h2o_min_mem_size	2G	Minimum memory specification for H2O-3 server started by H2O Sonar.	int	2G
h2o_max_mem_size	4G	Maximum memory specification for H2O-3 server started by H2O Sonar.	int	4G
custom_explainers	[]	List of custom "Bring Your Own Explainer" string locators to be registered on H2O Sonar run. The location has the following structure: "[PACKAGE and MODULE]::[EXPLAINER-CLASS-NAME]" where PACKAGE and MODULE is dot (.) separated path to the the module (installed on PYTHONPATH) and EXPLAINER-CLASS-NAME is the name of explainer class. Example: ["my_package.explainer_module::MyExplainerClass", "their_package.explainer_module::TheirExplainerClass"]	customlist	[]
look_and_feel	h2o_sonar	Charts theme (look and feel) - one of: 'h2o_sonar', 'blue', 'driverless_ai'.	str	h2o_sonar
per_explainer_logger	True	Create new logger for each explainer (which logs to explainer sandbox) or reuse one logger and use library logger for all log messages.	bool	True
create_html_representations	True	Indicate that explainers can create HTML representation (True), or request to skip it (False) from performance/resource consumption reasons.	bool	True
connections	[]			
licenses	[]			

Dataset

Dataset description:

- **Dataset** = <class 'datatable.Frame'>
- **Dataset file path** = data/creditcard.csv
 - Path to the dataset.
- **Dataset size** = 944719B
 - Size of the dataset.
 - Dataset data frame type.
- **Dataset frame shape** = (10000, 25)

- Dataset data frame shape.
- Row count = 10000
 - The number of the dataset rows.
- Column names = ['ID', 'LIMIT_BAL', 'SEX', 'EDUCATION', 'MARRIAGE', 'AGE', 'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1', 'BILL_AMT2', 'BILL_AMT3', 'BILL_AMT4', 'BILL_AMT5', 'BILL_AMT6', 'PAY_AMT1', 'PAY_AMT2', 'PAY_AMT3', 'PAY_AMT4', 'PAY_AMT5', 'PAY_AMT6', 'default payment next month']
 - Dataset column names.
- Unique values = [10000, 72, 2, 7, 4, 54, 11, 11, 11, 11, 10, 10, 8371, 8215, 8072, 7913, 7764, 7550, 3763, 3581, 3305, 3247, 3258, 3174, 2]
 - The number of unique values for dataset columns.
- Column types = ['int', 'int']
 - Dataset column types.
- Dataset columns:
 - ID
 - Data type = int
 - ID column = True
 - Numeric column = True
 - Count = 10000
 - Unique = 10000
 - LIMIT_BAL
 - Data type = int
 - Numeric column = True
 - Count = 72
 - Unique = 72
 - SEX
 - Data type = int
 - Numeric column = True
 - Count = 2
 - Unique = 2
 - EDUCATION
 - Data type = int
 - Numeric column = True
 - Count = 7
 - Unique = 7
 - MARRIAGE
 - Data type = int
 - Numeric column = True
 - Count = 4
 - Unique = 4
 - AGE
 - Data type = int
 - Numeric column = True
 - Count = 54
 - Unique = 54
 - PAY_0
 - Data type = int
 - Numeric column = True
 - Count = 11
 - Unique = 11
 - PAY_2
 - Data type = int
 - Numeric column = True
 - Count = 11
 - Unique = 11
 - PAY_3
 - Data type = int

- Numeric column = True
- Count = 11
- Unique = 11
- PAY_4
 - Data type = int
 - Numeric column = True
 - Count = 11
 - Unique = 11
- PAY_5
 - Data type = int
 - Numeric column = True
 - Count = 10
 - Unique = 10
- PAY_6
 - Data type = int
 - Numeric column = True
 - Count = 10
 - Unique = 10
- BILL_AMT1
 - Data type = int
 - Numeric column = True
 - Count = 8371
 - Unique = 8371
- BILL_AMT2
 - Data type = int
 - Numeric column = True
 - Count = 8215
 - Unique = 8215
- BILL_AMT3
 - Data type = int
 - Numeric column = True
 - Count = 8072
 - Unique = 8072
- BILL_AMT4
 - Data type = int
 - Numeric column = True
 - Count = 7913
 - Unique = 7913
- BILL_AMT5
 - Data type = int
 - Numeric column = True
 - Count = 7764
 - Unique = 7764
- BILL_AMT6
 - Data type = int
 - Numeric column = True
 - Count = 7550
 - Unique = 7550
- PAY_AMT1
 - Data type = int
 - Numeric column = True
 - Count = 3763
 - Unique = 3763
- PAY_AMT2
 - Data type = int
 - Numeric column = True
 - Count = 3581
 - Unique = 3581

- PAY_AMT3
 - Data type = int
 - Numeric column = True
 - Count = 3305
 - Unique = 3305
- PAY_AMT4
 - Data type = int
 - Numeric column = True
 - Count = 3247
 - Unique = 3247
- PAY_AMT5
 - Data type = int
 - Numeric column = True
 - Count = 3258
 - Unique = 3258
- PAY_AMT6
 - Data type = int
 - Numeric column = True
 - Count = 3174
 - Unique = 3174
- default payment next month
 - Data type = int
 - Numeric column = True
 - Count = 2
 - Unique = 2
- *Dataset columns metadata.*

Model

Model description:

- **Model type** = driverless_ai
 - *Model type.*
- **Experiment type** = binomial
 - *Experiment type.*
- **Target column** = default payment next month
 - *Target column.*
- **Labels count** = 2
 - *The number of interpreted model labels.*
- **Labels** = ['0', '1']
 - *Interpret model labels.*
- **Used features** = ['LIMIT_BAL', 'SEX', 'EDUCATION', 'MARRIAGE', 'AGE', 'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1', 'BILL_AMT2', 'BILL_AMT3', 'BILL_AMT4', 'BILL_AMT5', 'BILL_AMT6', 'PAY_AMT1', 'PAY_AMT2', 'PAY_AMT3', 'PAY_AMT4', 'PAY_AMT5', 'PAY_AMT6']
 - *Features (dataset columns) used by the interpreted model.*
- **Transformed features** = ['0_AGE', '10_PAY_0', '11_PAY_2', '12_PAY_3', '13_PAY_4', '14_PAY_5', '15_PAY_6', '16_PAY_AMT1', '17_PAY_AMT2', '18_PAY_AMT3', '19_PAY_AMT4', '1_BILL_AMT1', '20_PAY_AMT5', '21_PAY_AMT6', '22_SEX', '2_BILL_AMT2', '3_BILL_AMT3', '4_BILL_AMT4', '5_BILL_AMT5', '6_BILL_AMT6', '7_EDUCATION', '8_LIMIT_BAL', '9_MARRIAGE']
 - *Transformed features created by/used by the interpreted model.*

Directories, files and logs

Directories and files:

- [Interpretation summary in HTML format](#)
- [Interpretation summary in JSON format](#)
- [This model interpretation directory](#)
- [H2O Sonar library results directory](#)
- [H2O Sonar library log](#)

Explainer: Disparate Impact Analysis

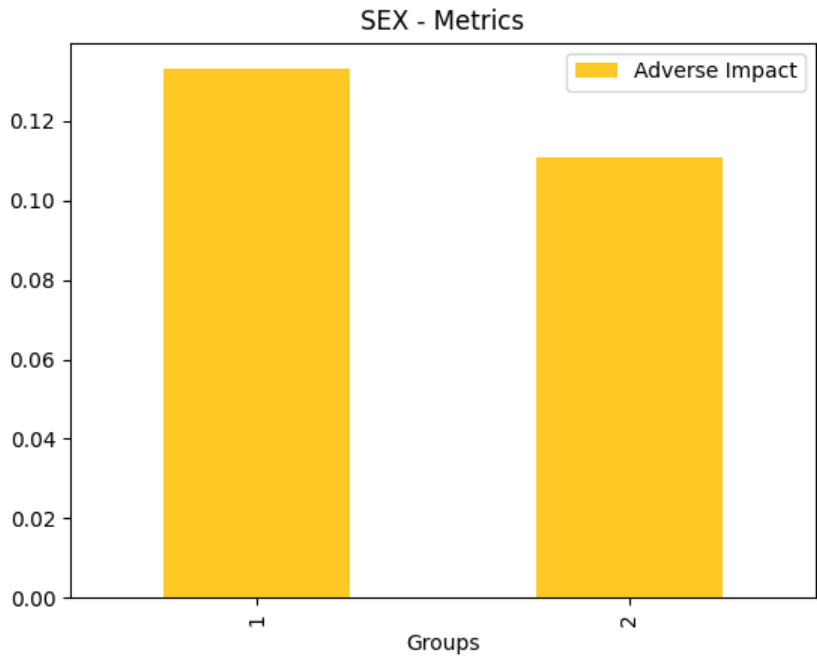
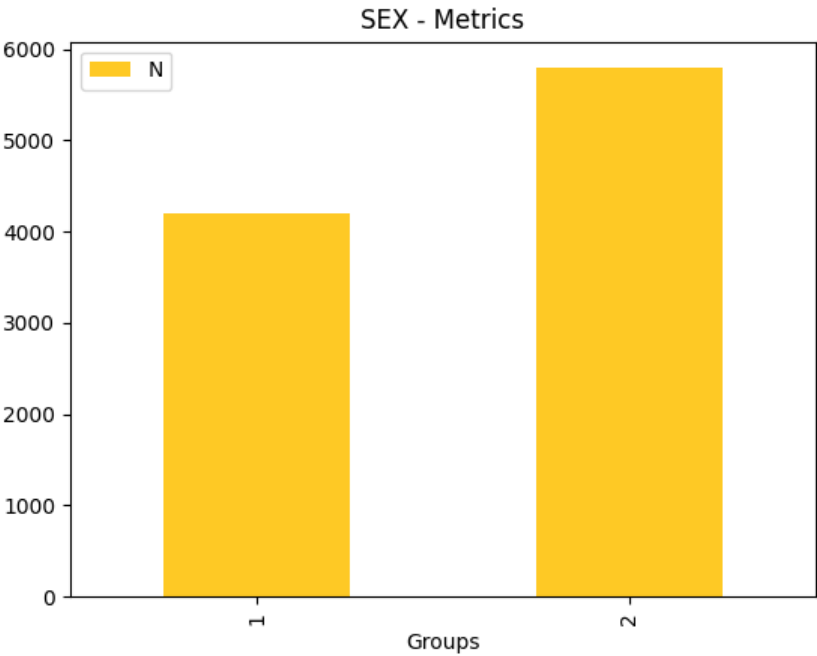
- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

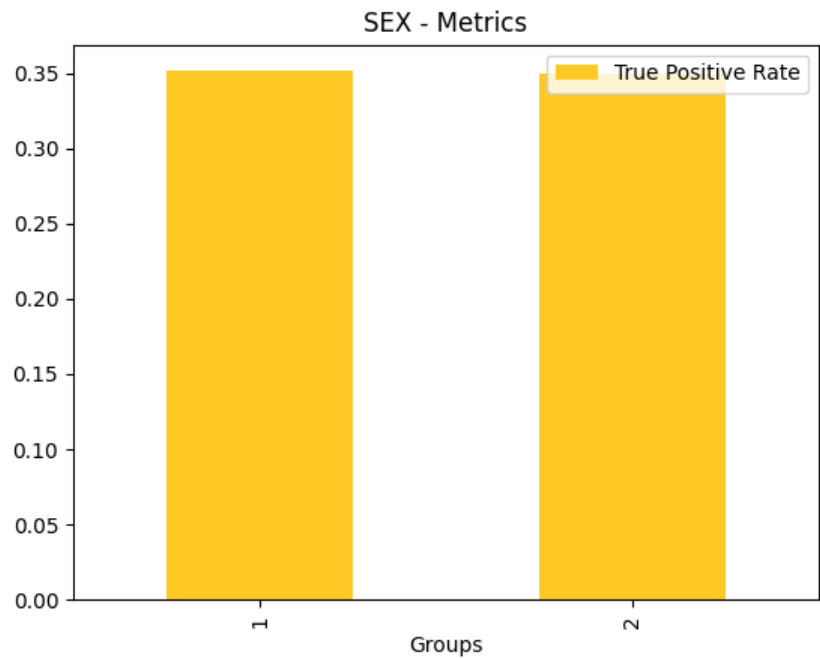
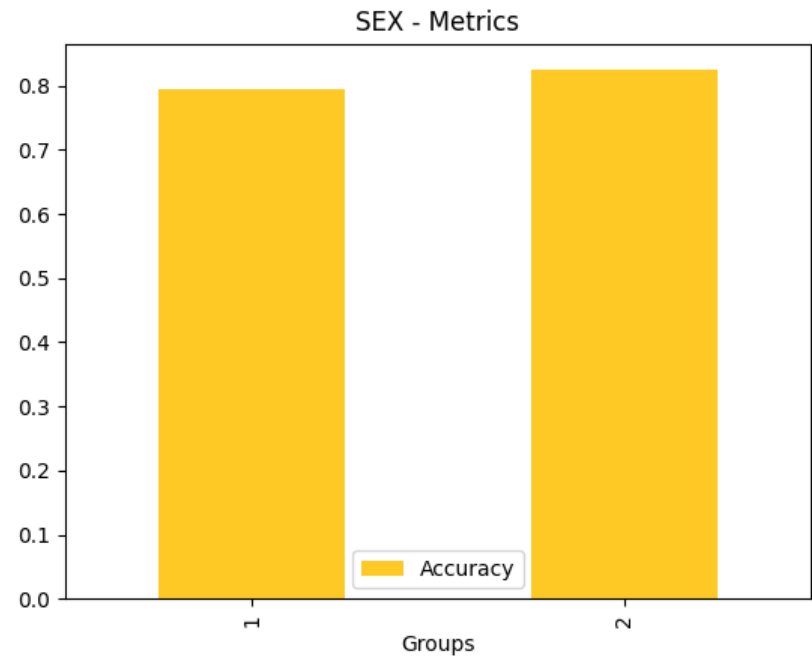
Explainer description

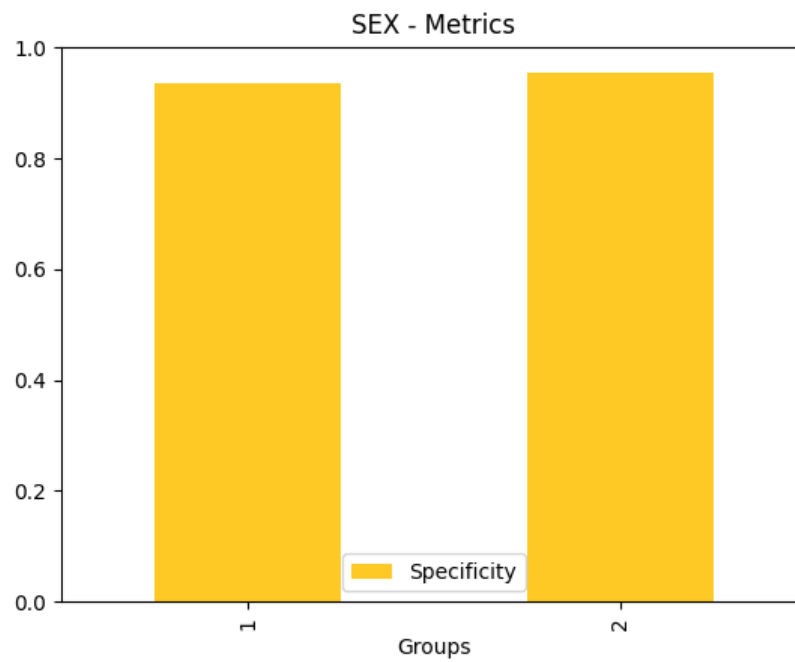
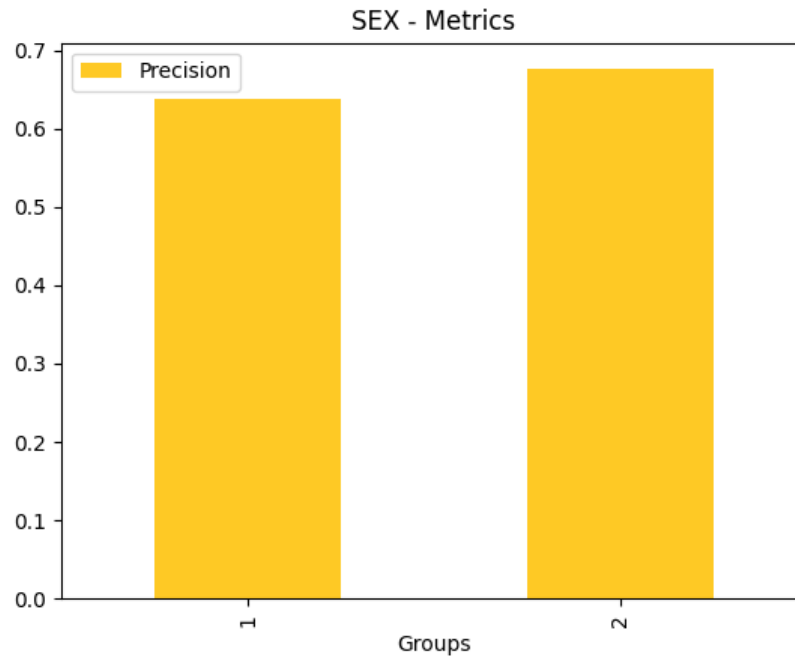
Disparate Impact Analysis (DIA) is a technique that is used to evaluate fairness. Bias can be introduced to models during the process of collecting, processing, and labeling data as a result, it is important to determine whether a model is harming certain users by making a significant number of biased decisions. DIA typically works by comparing aggregate measurements of unprivileged groups to a privileged group. For instance, the proportion of the unprivileged group that receives the potentially harmful outcome is divided by the proportion of the privileged group that receives the same outcome - the resulting proportion is then used to determine whether the model is biased.

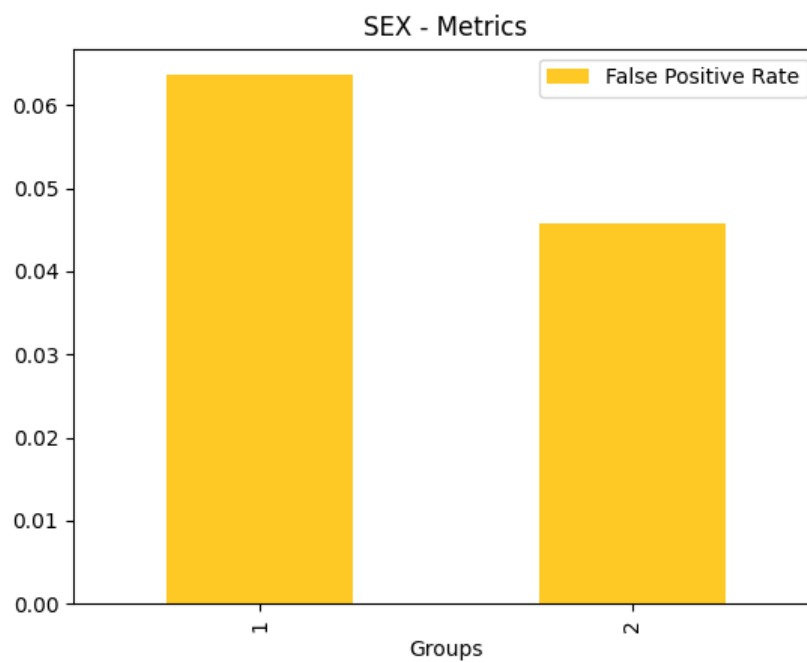
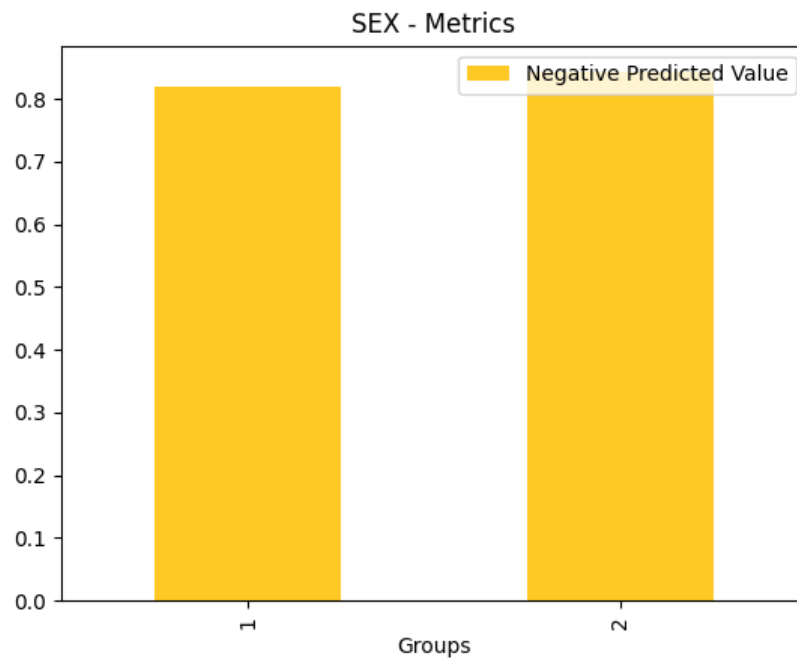
Explanations

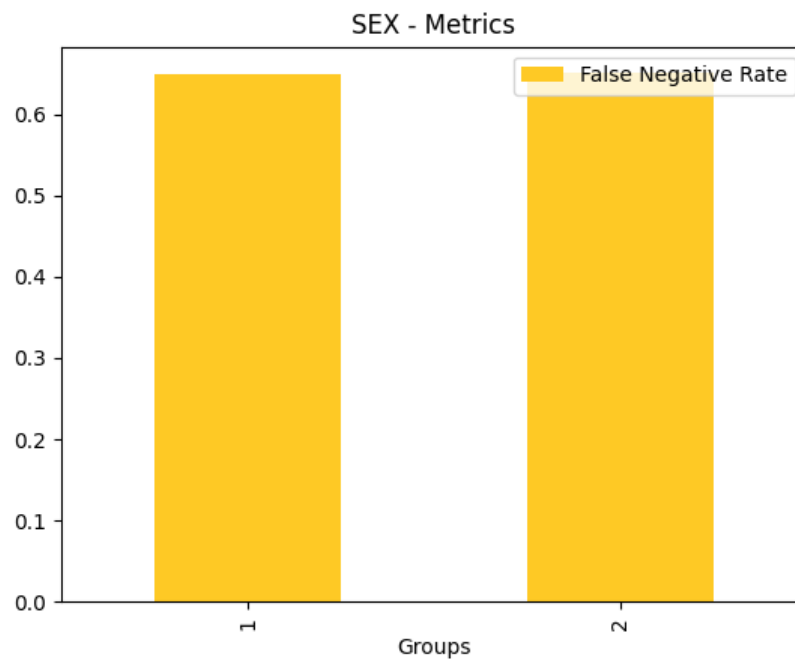
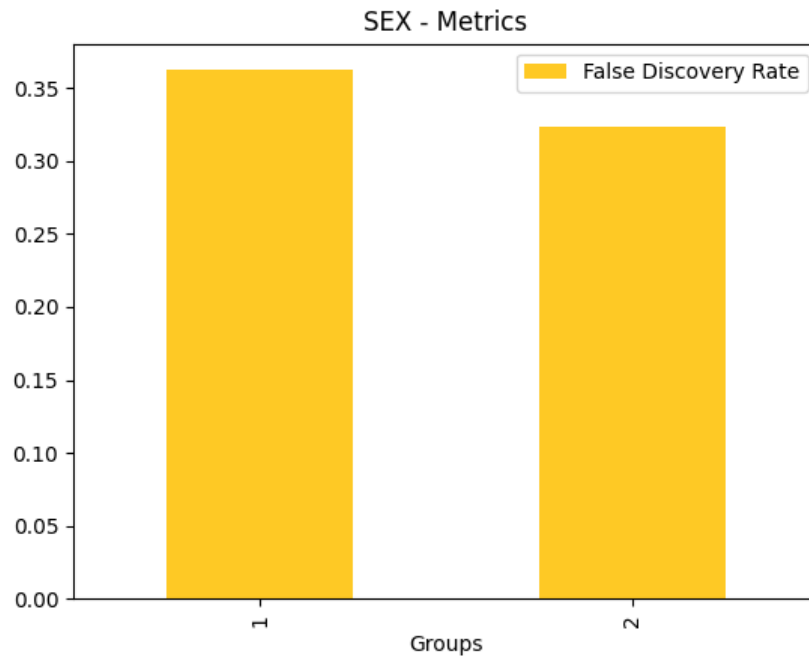
Fairness metrics for the feature: SEX

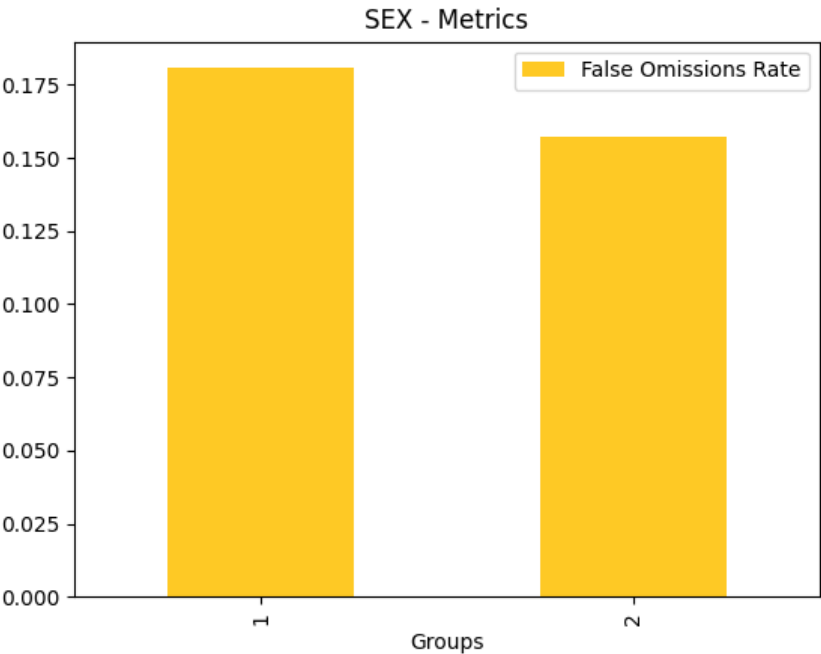




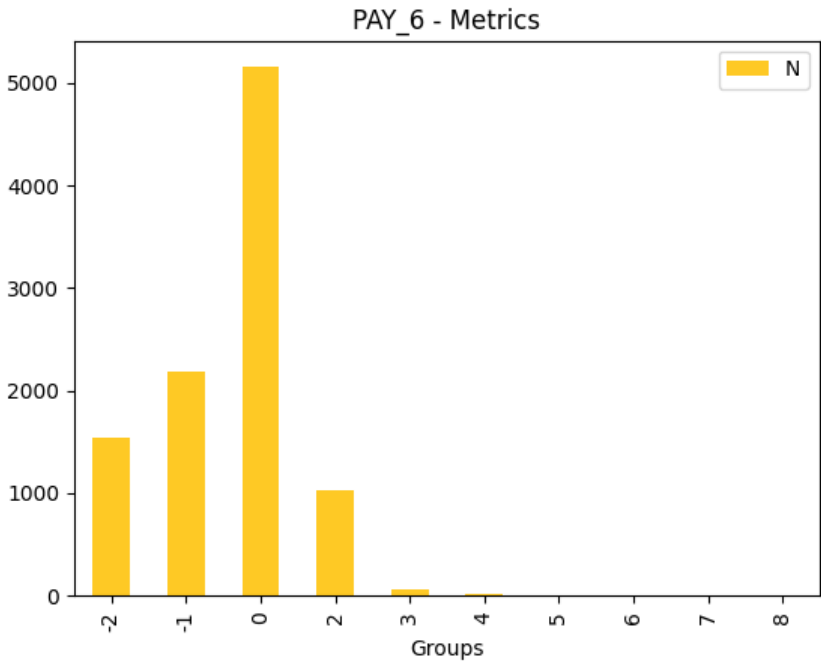


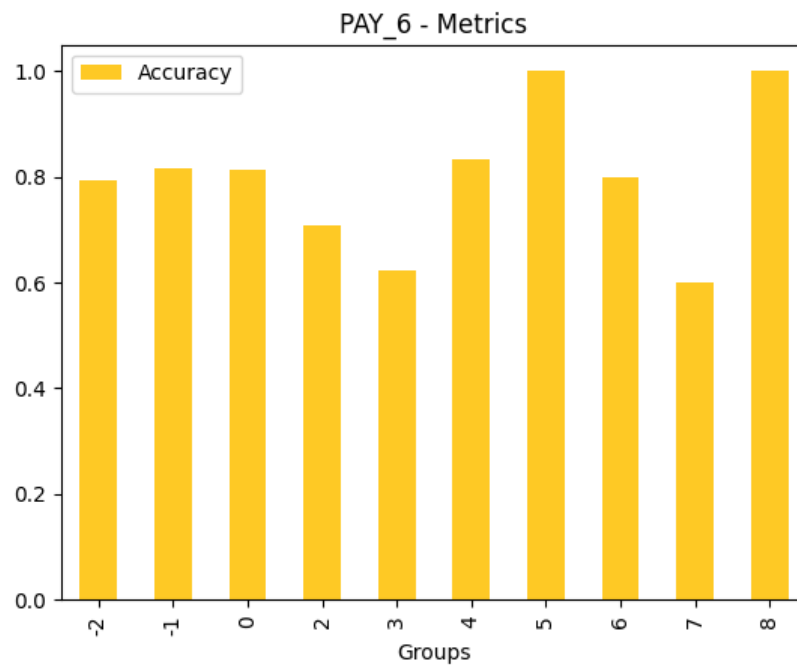
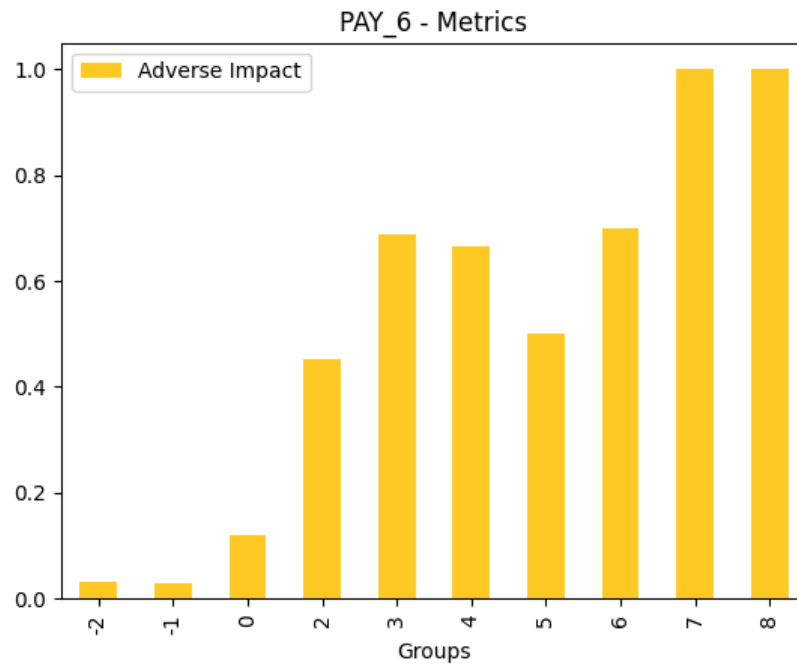


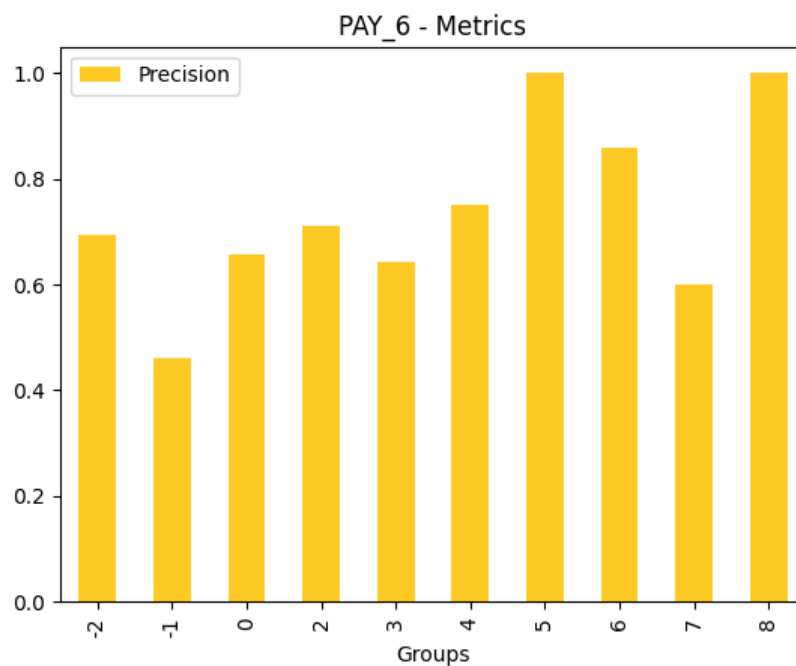
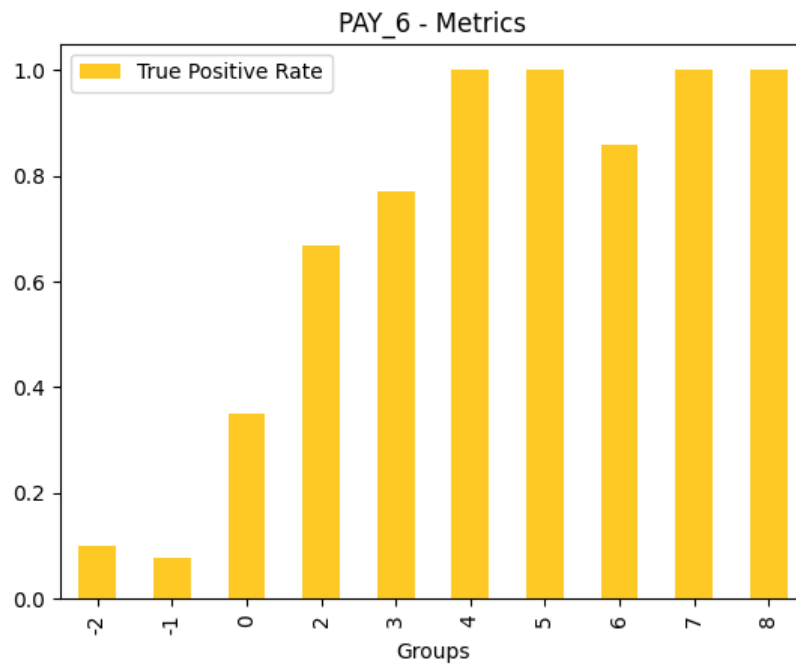


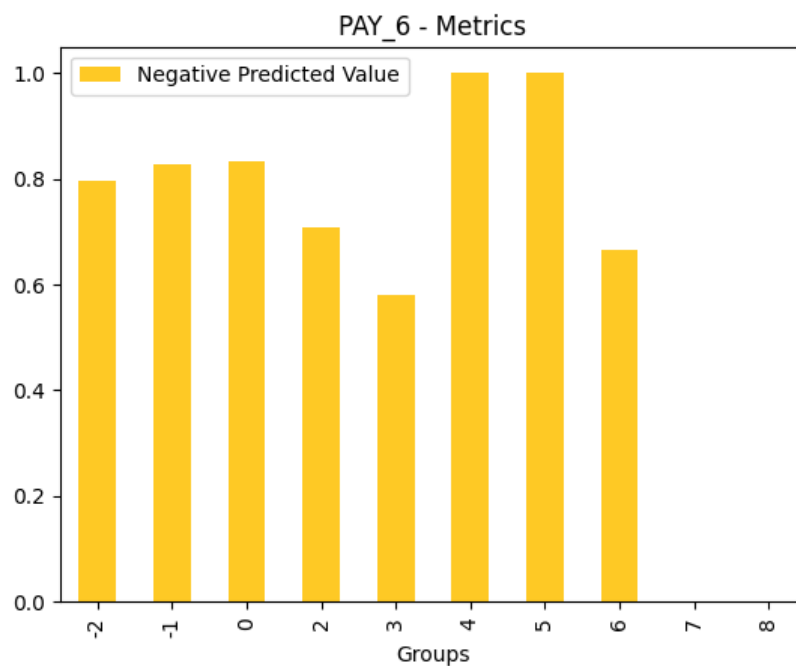
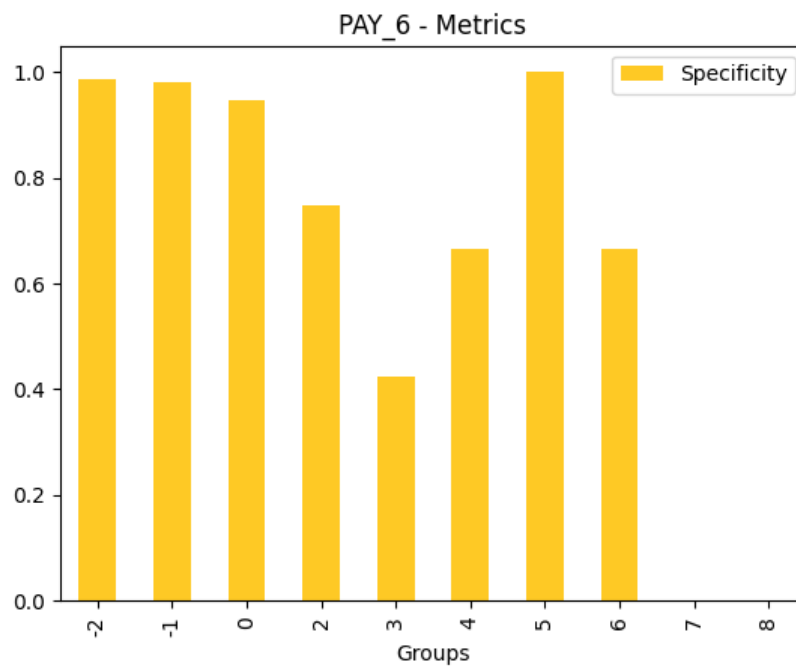


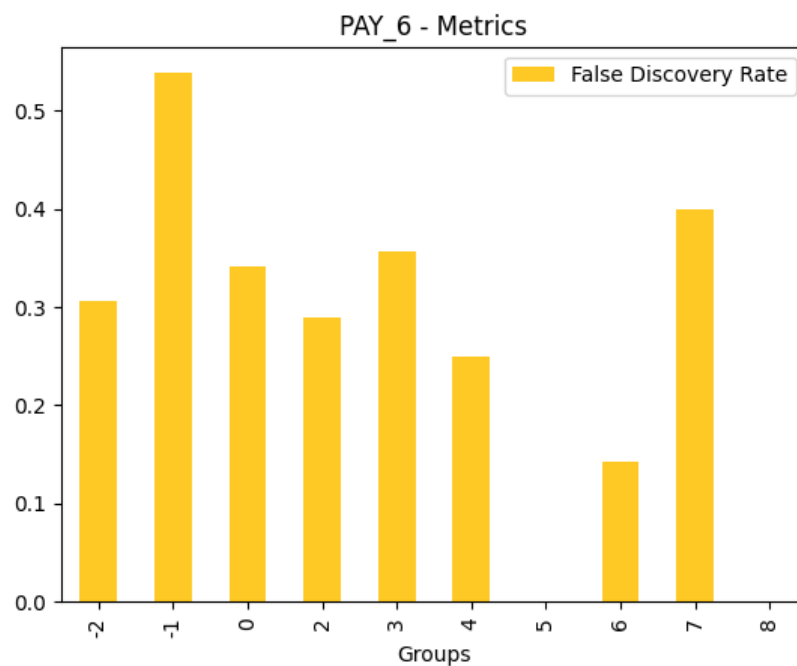
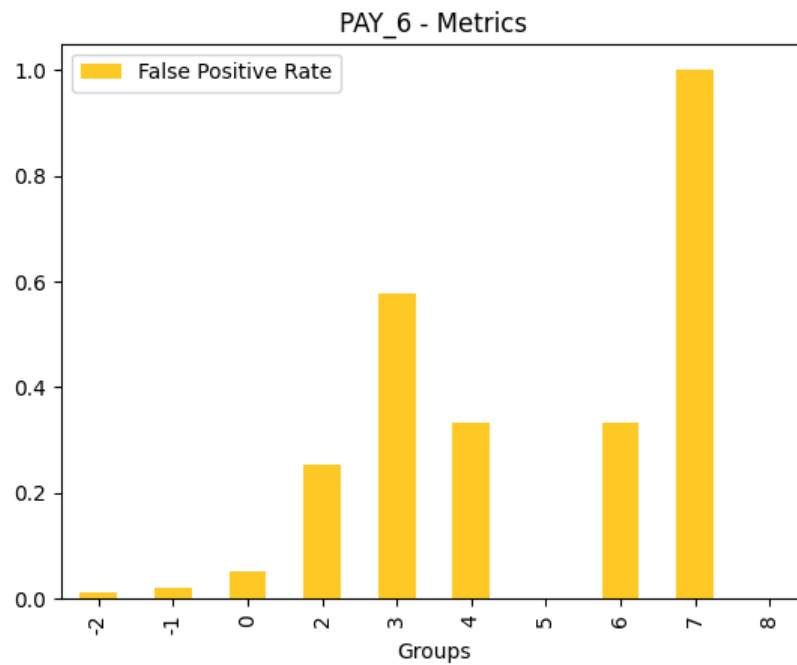
Fairness metrics for the feature: PAY_6

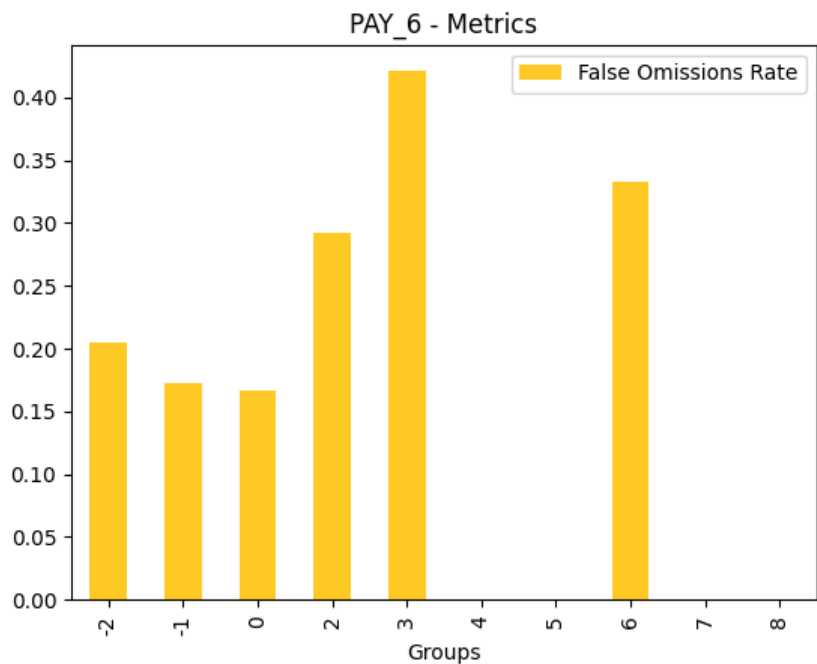
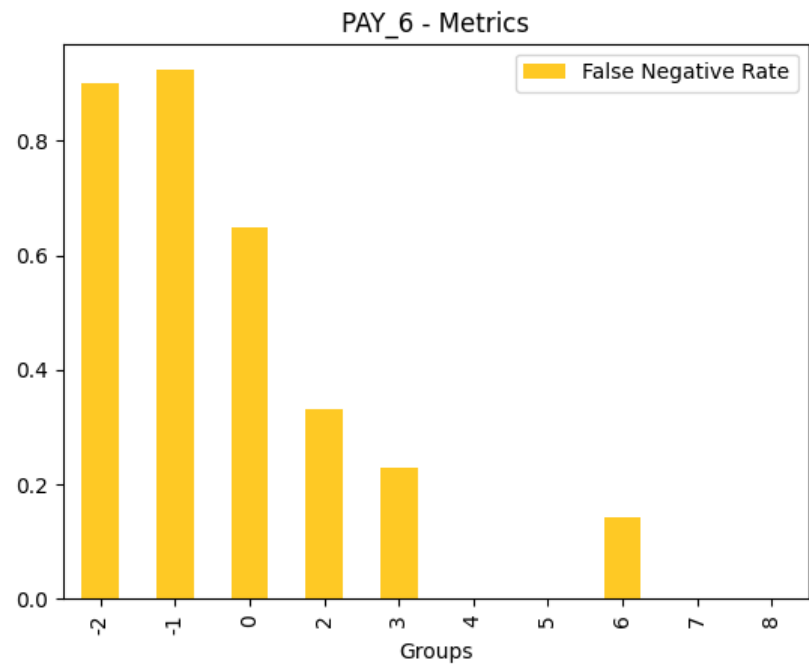




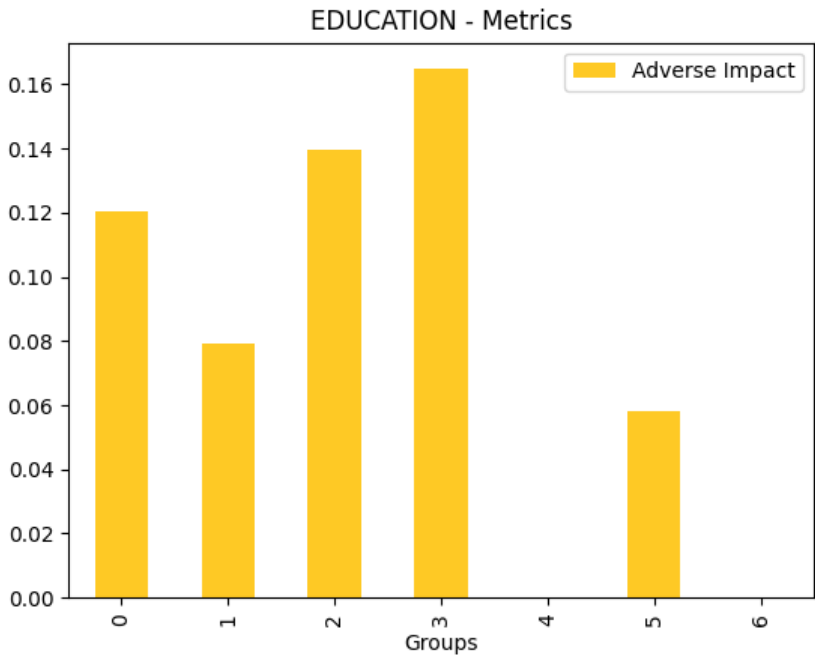
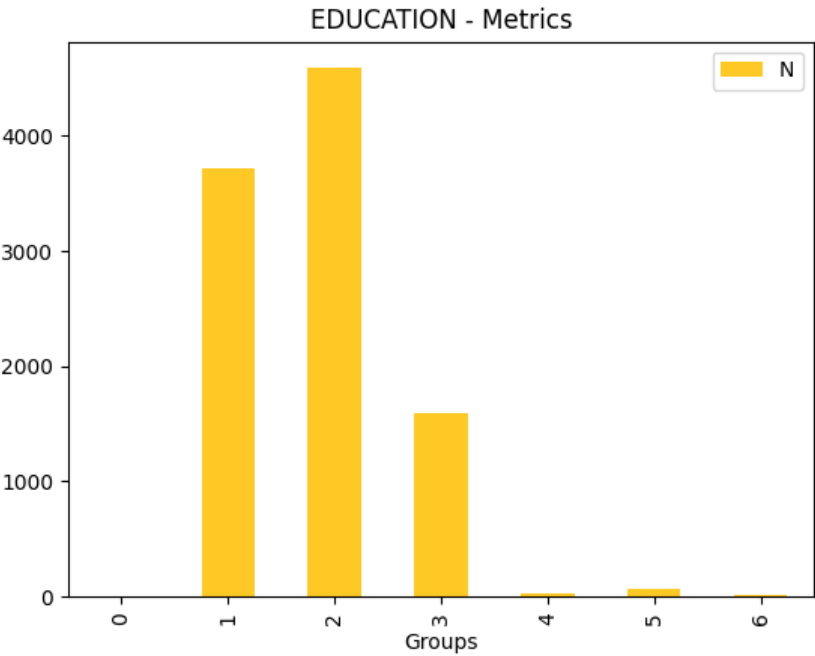


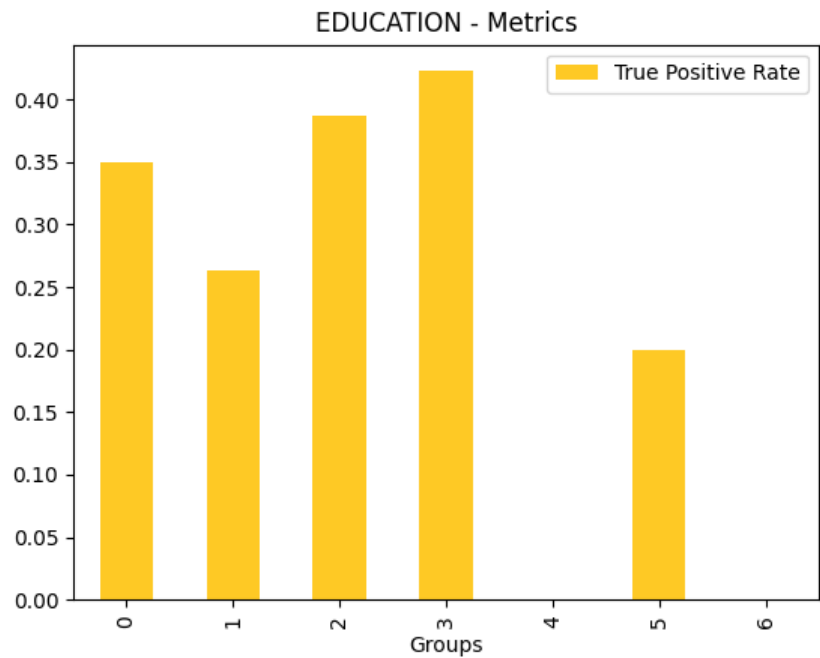
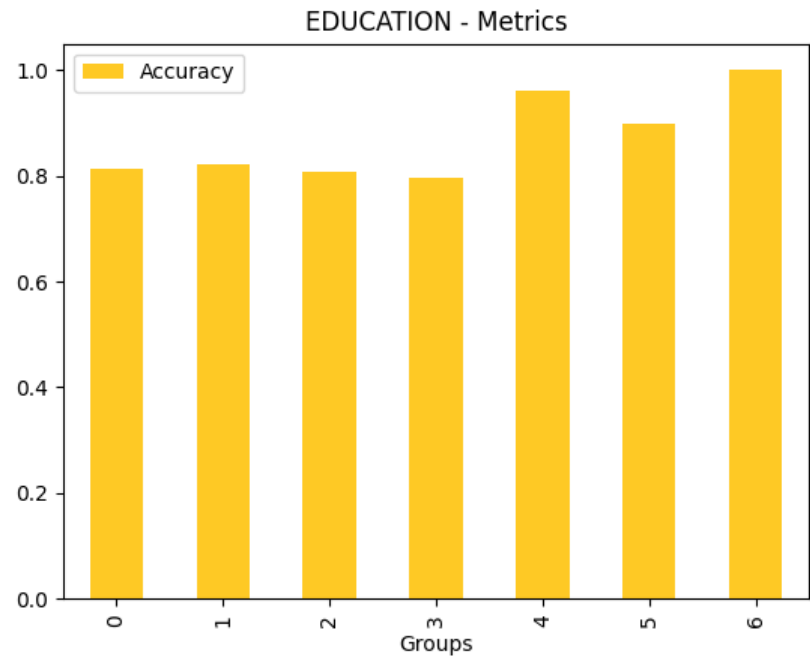


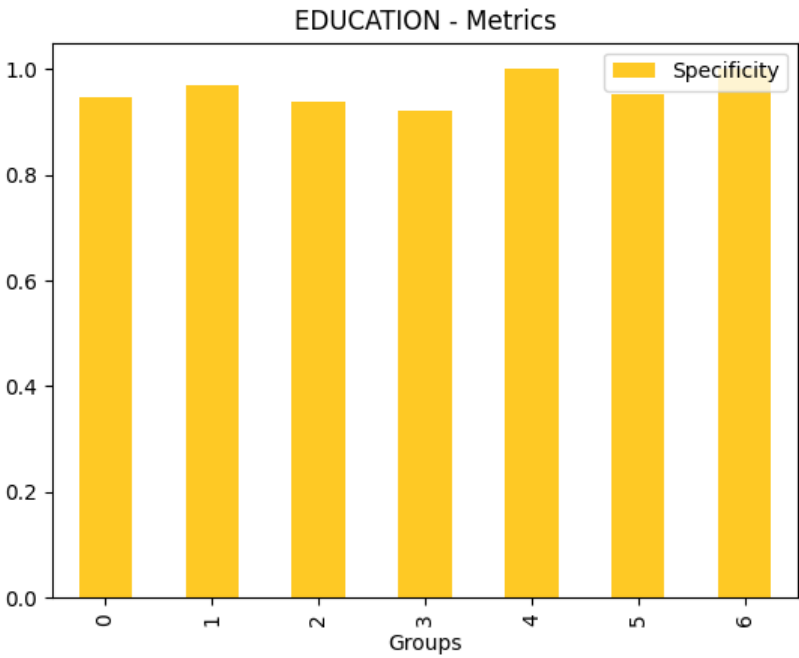
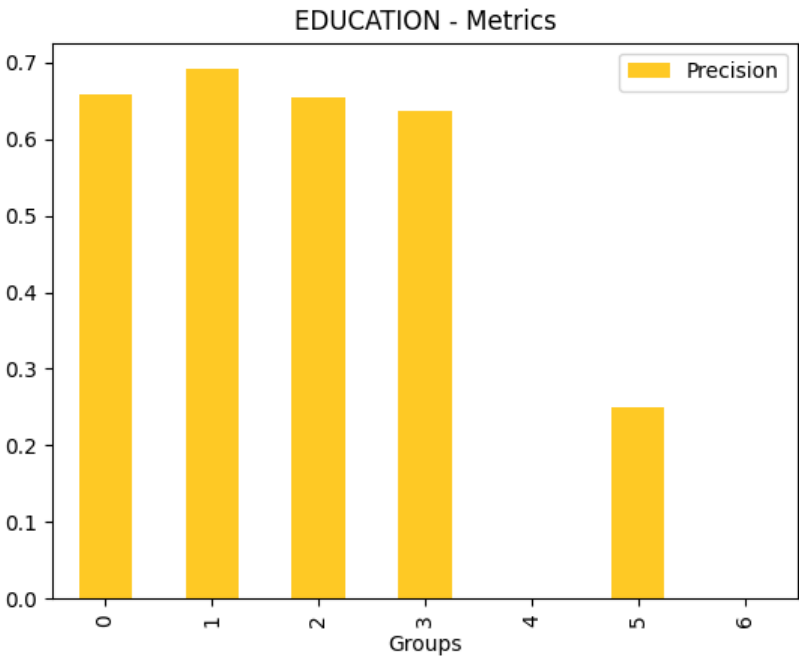


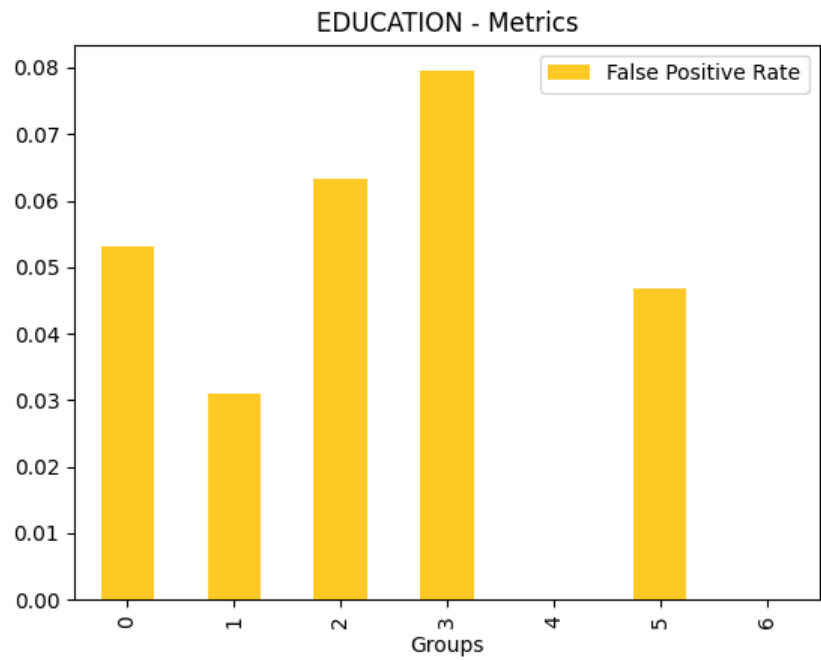
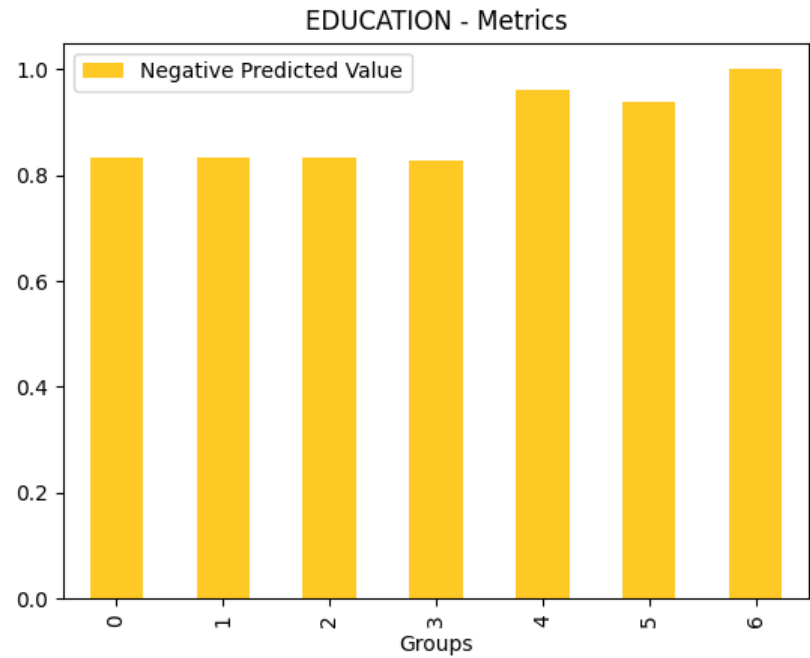


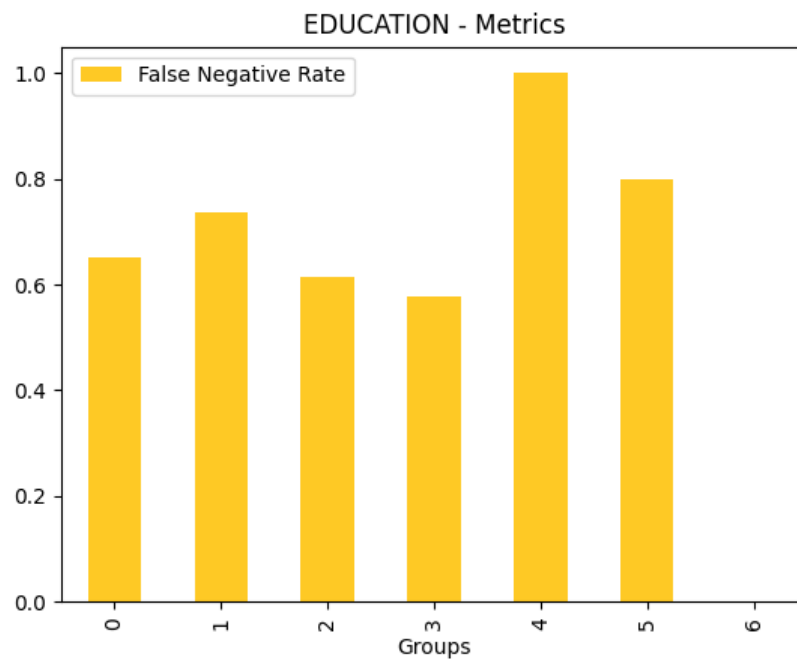
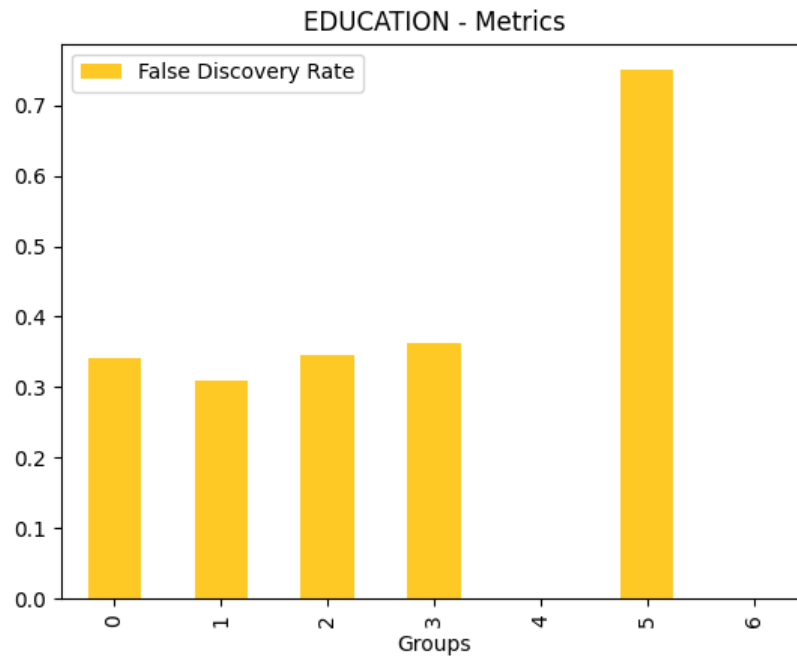
Fairness metrics for the feature: EDUCATION

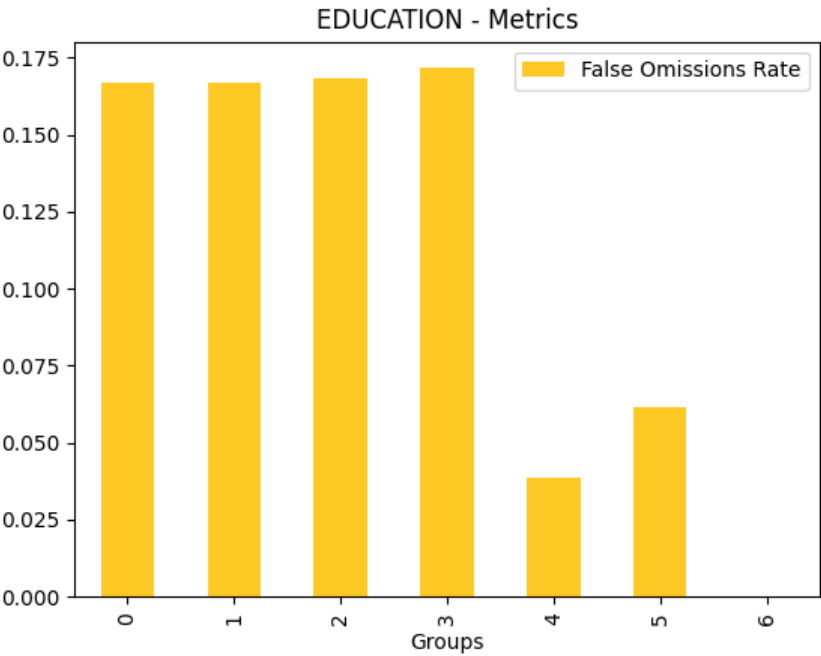




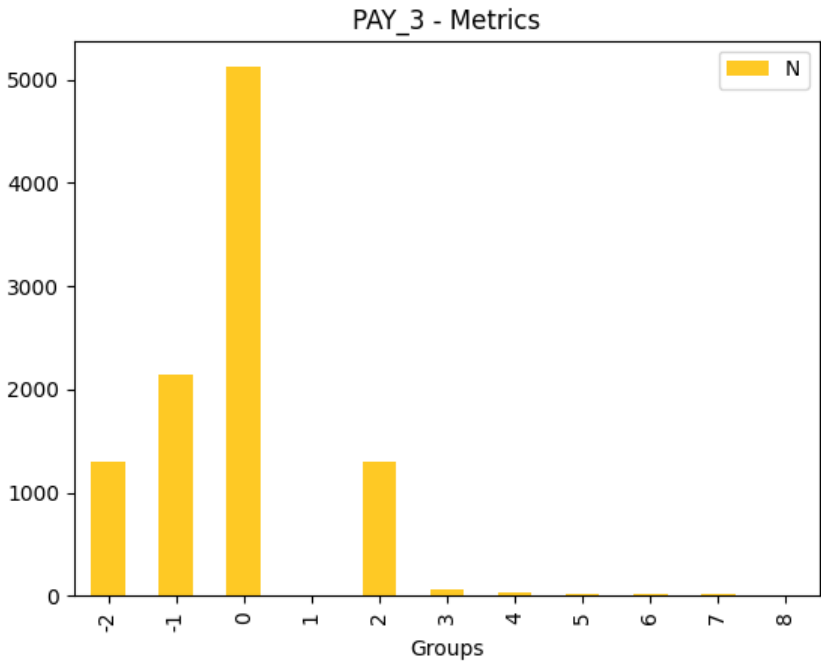


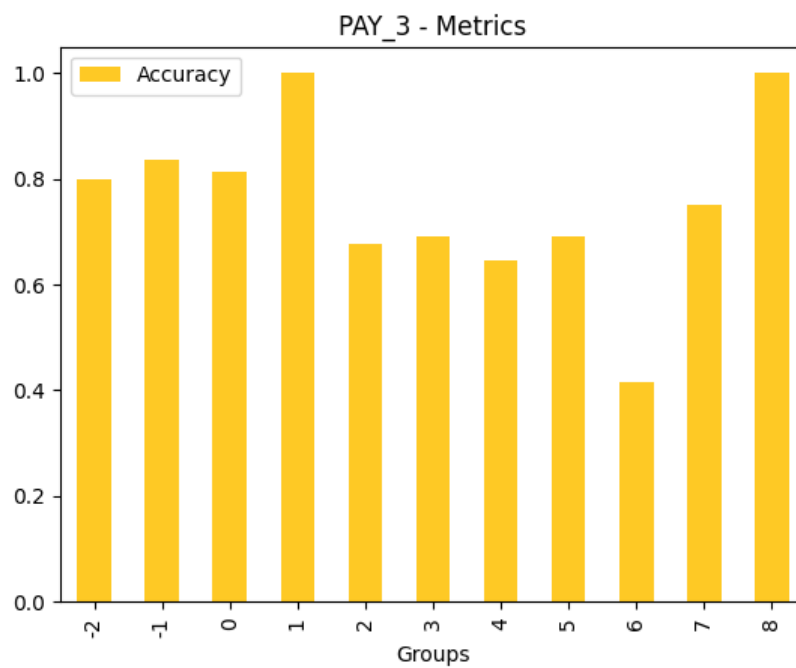
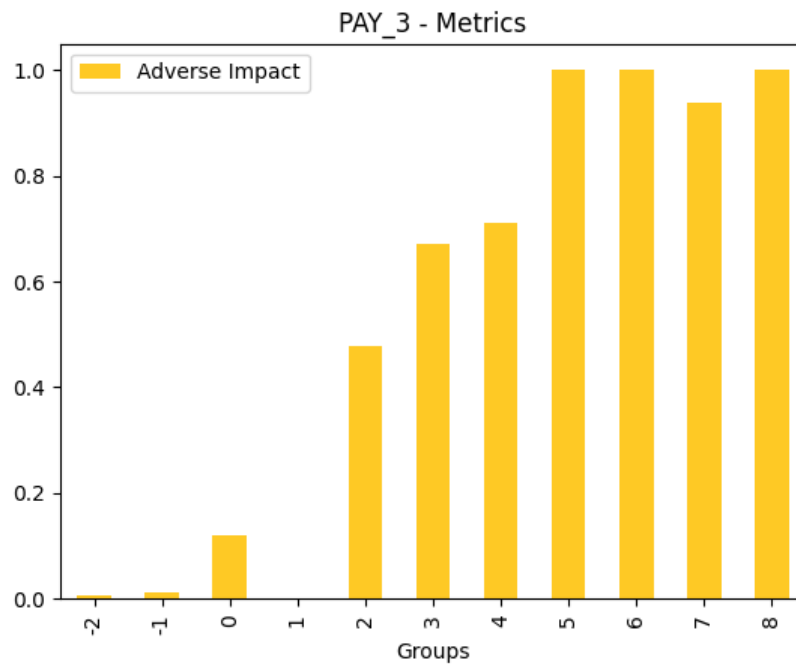


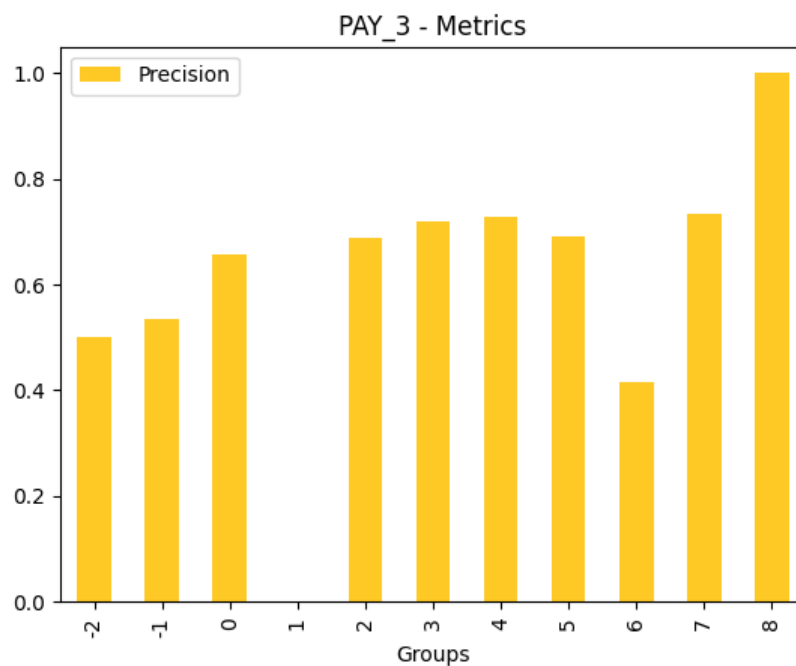
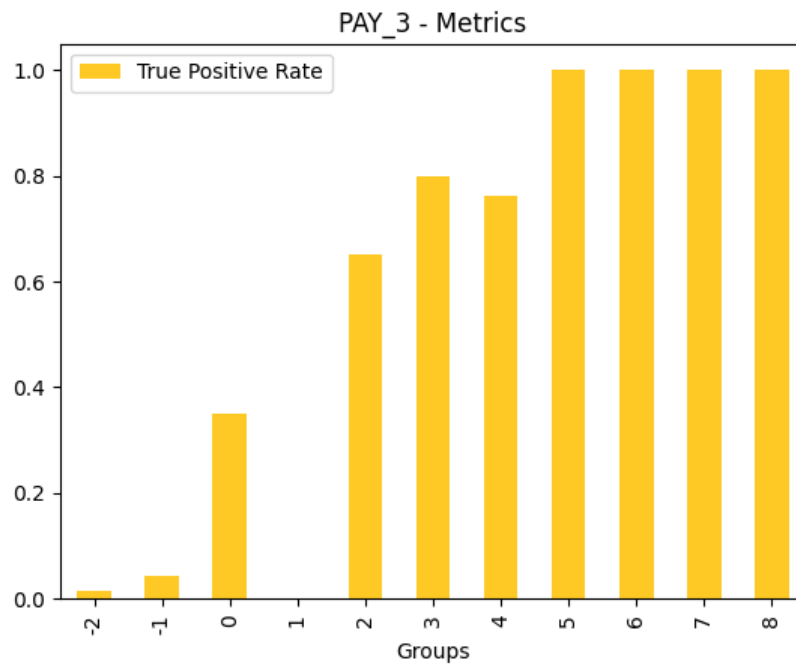


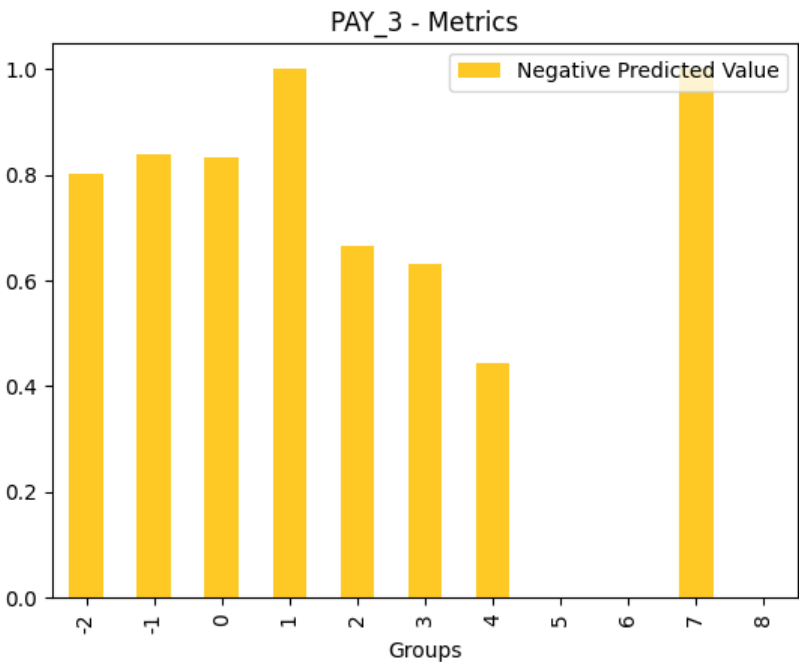
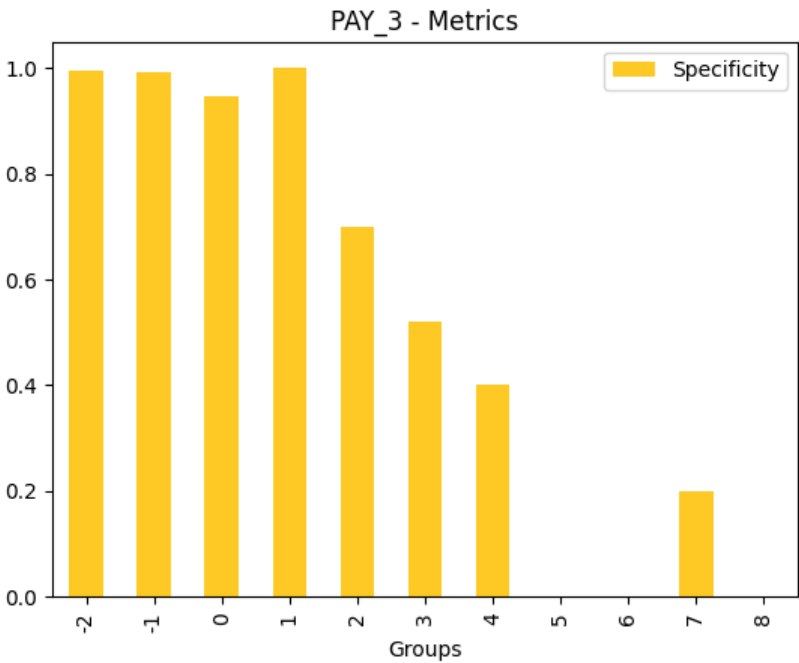


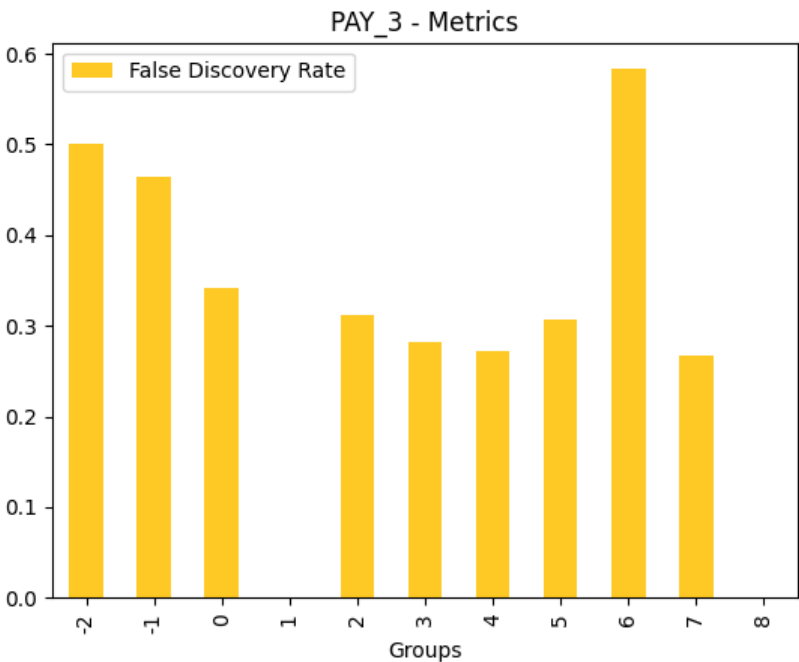
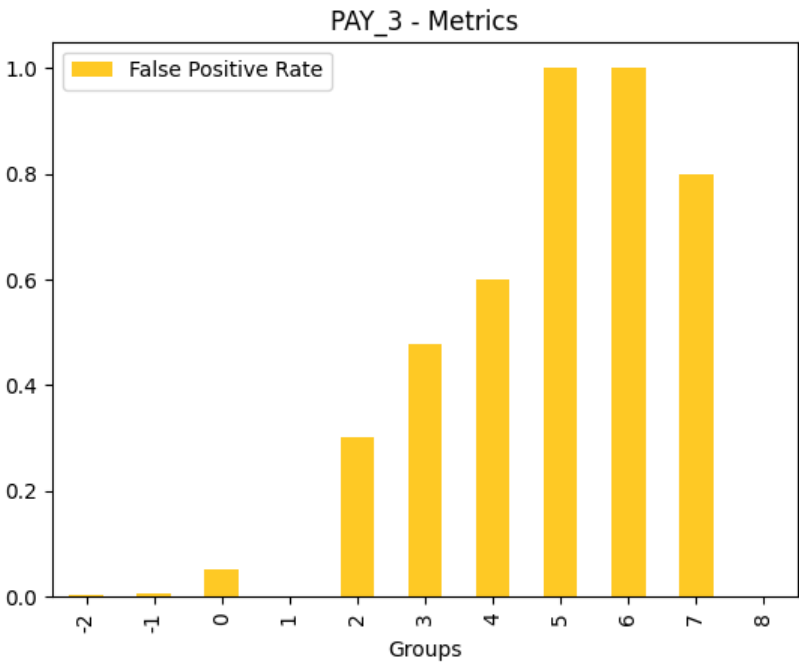
Fairness metrics for the feature: PAY_3

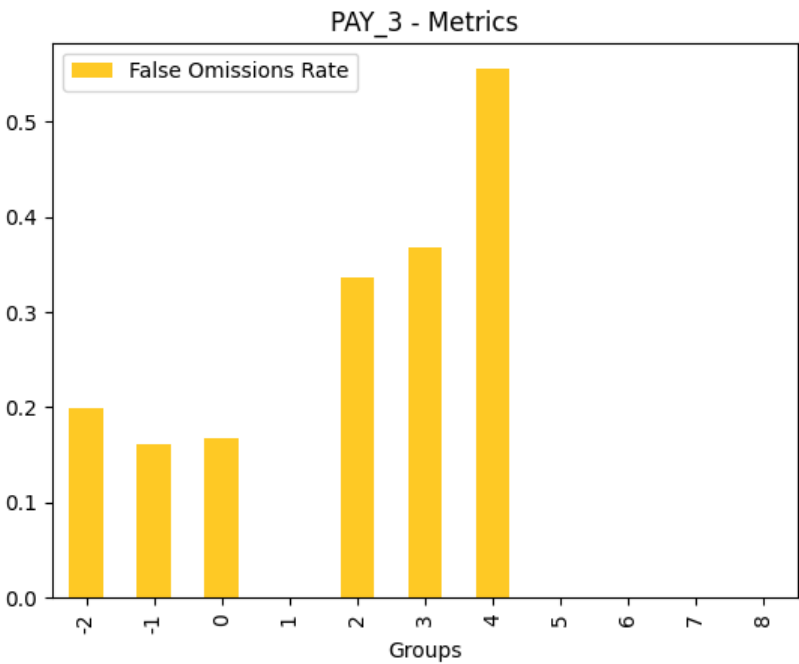
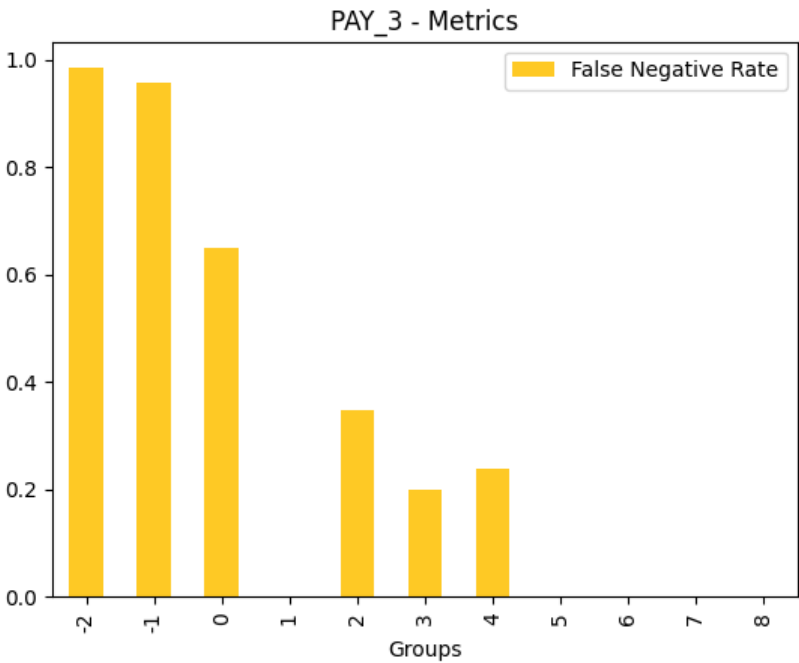




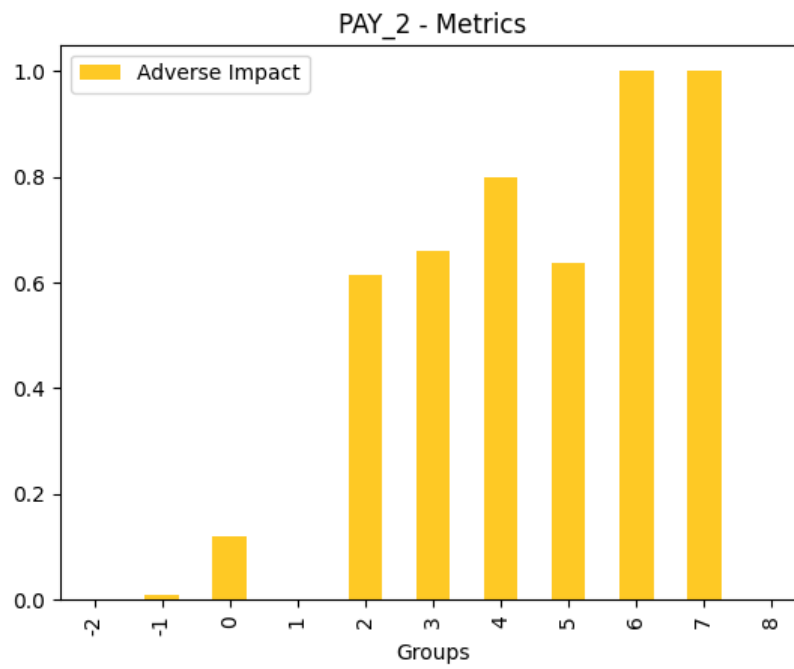
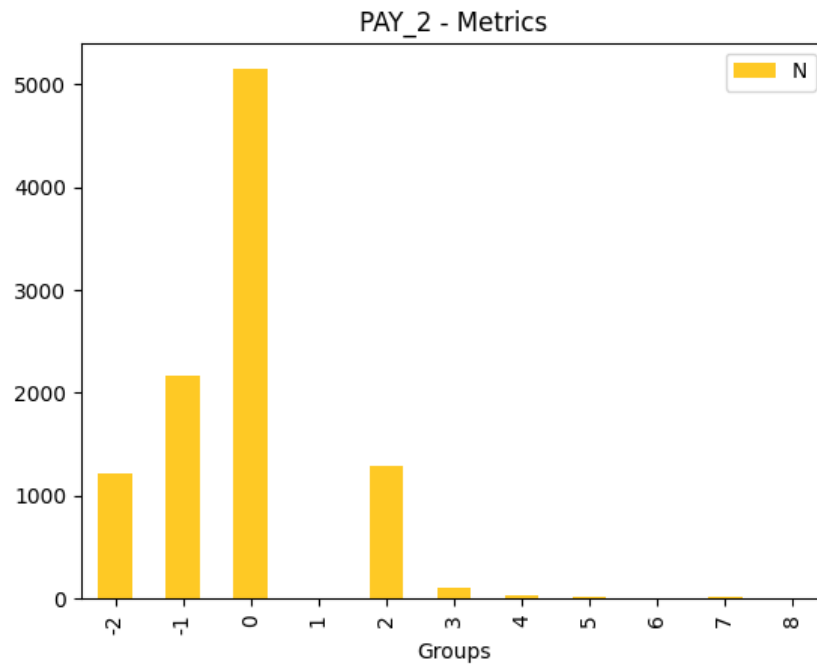


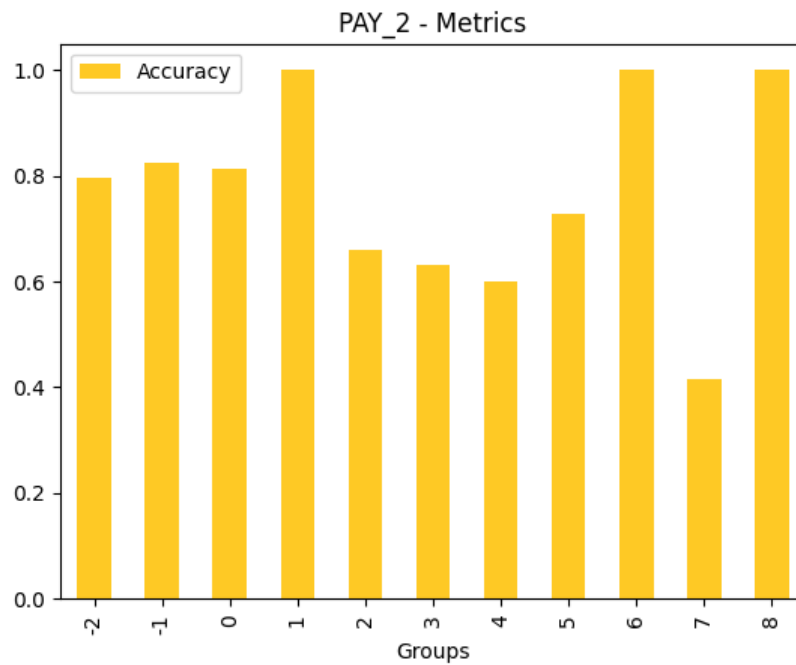


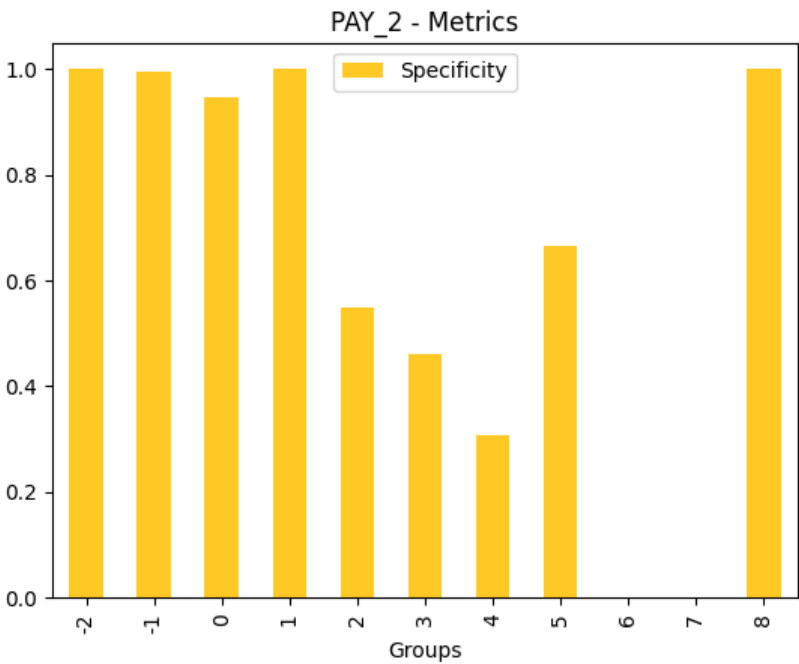
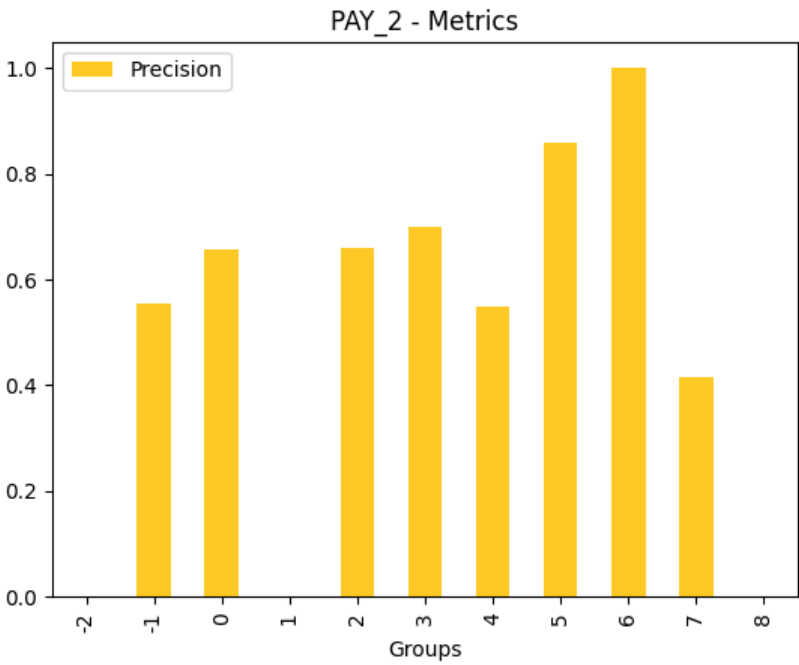


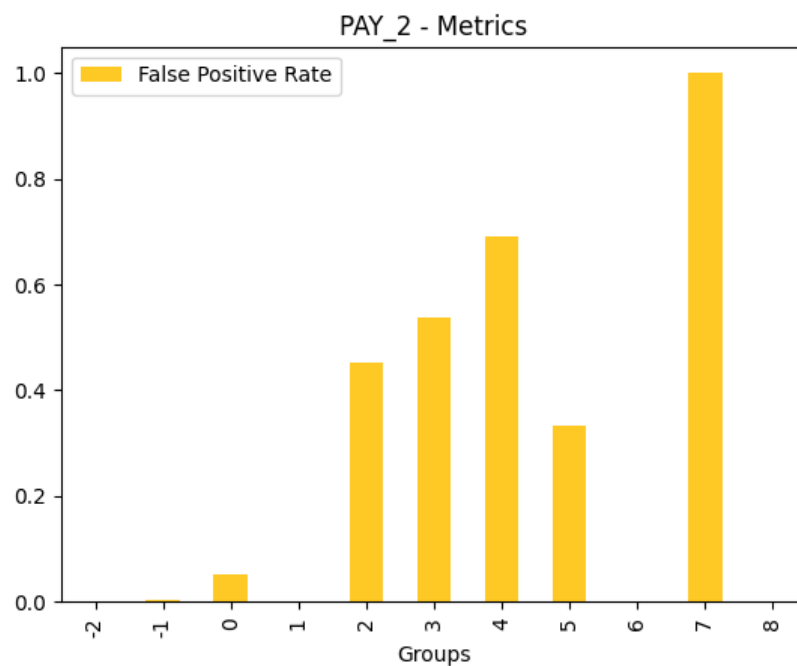
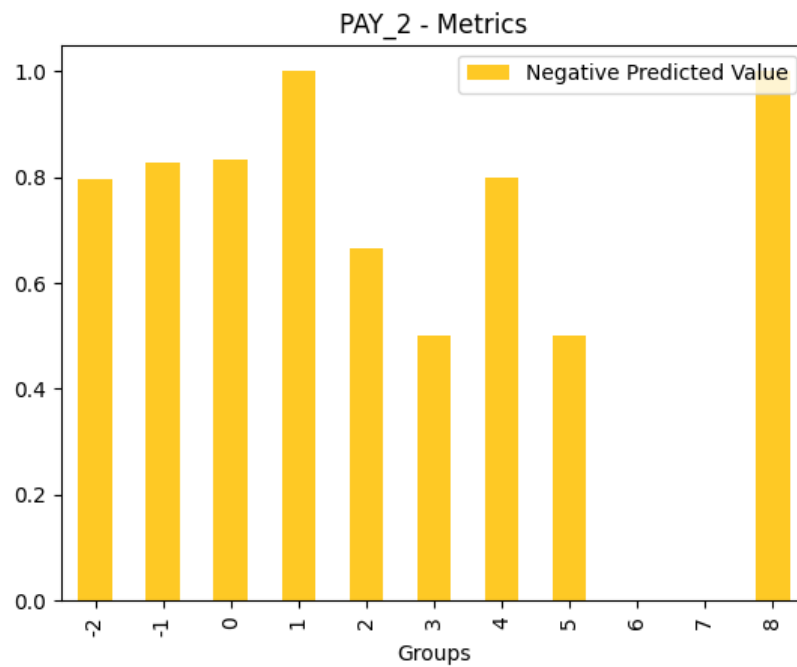


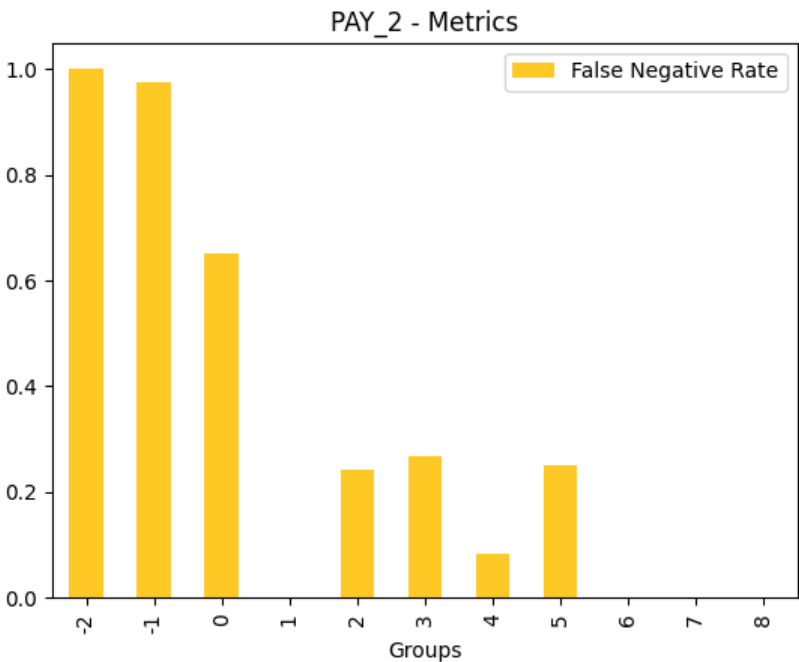
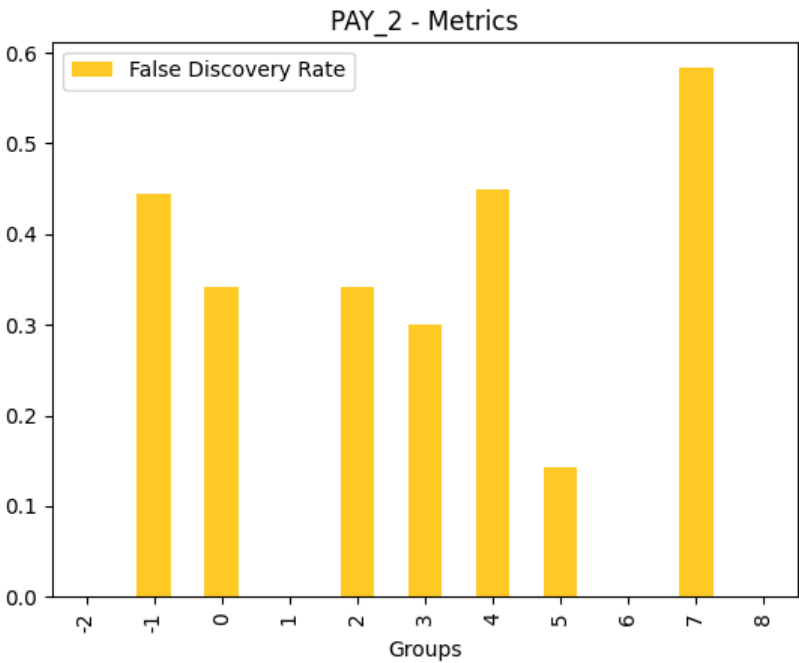
Fairness metrics for the feature: PAY_2

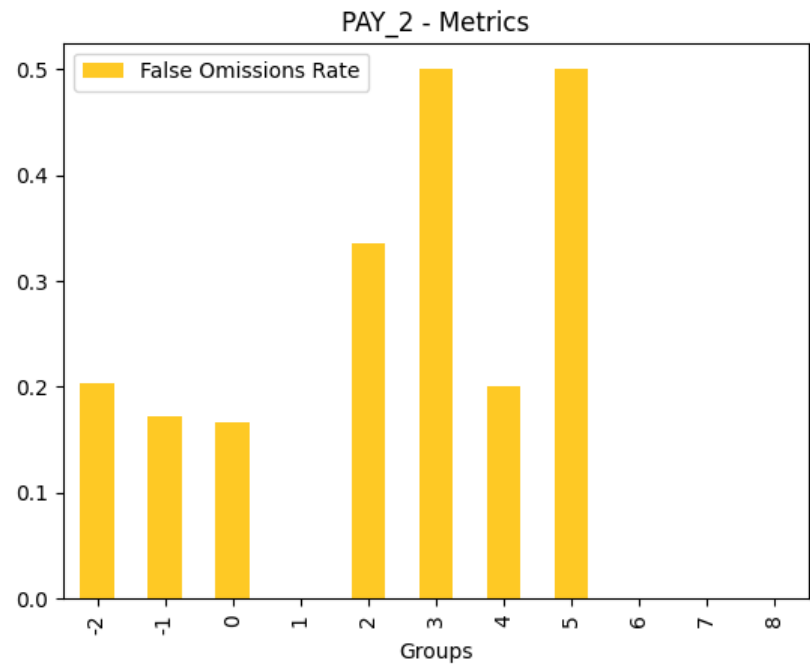




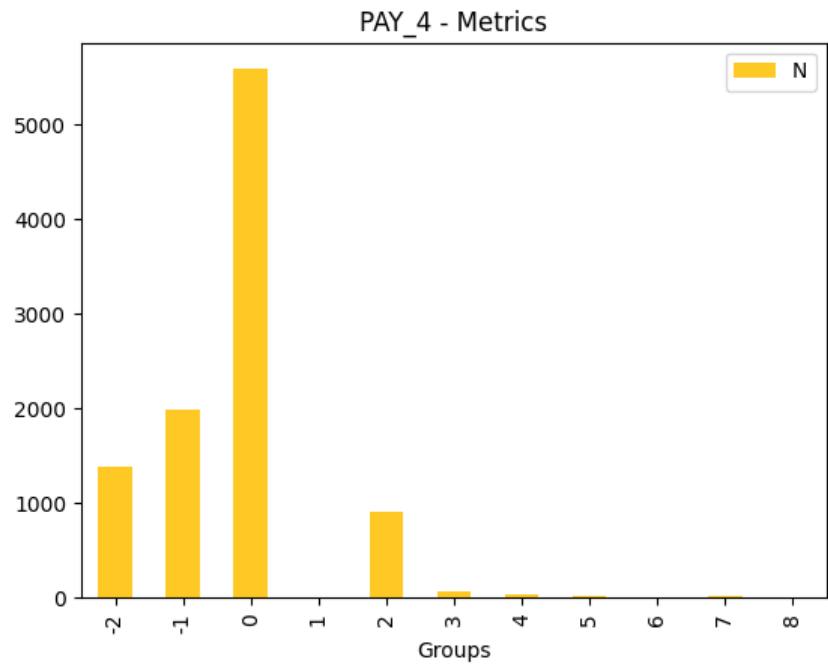


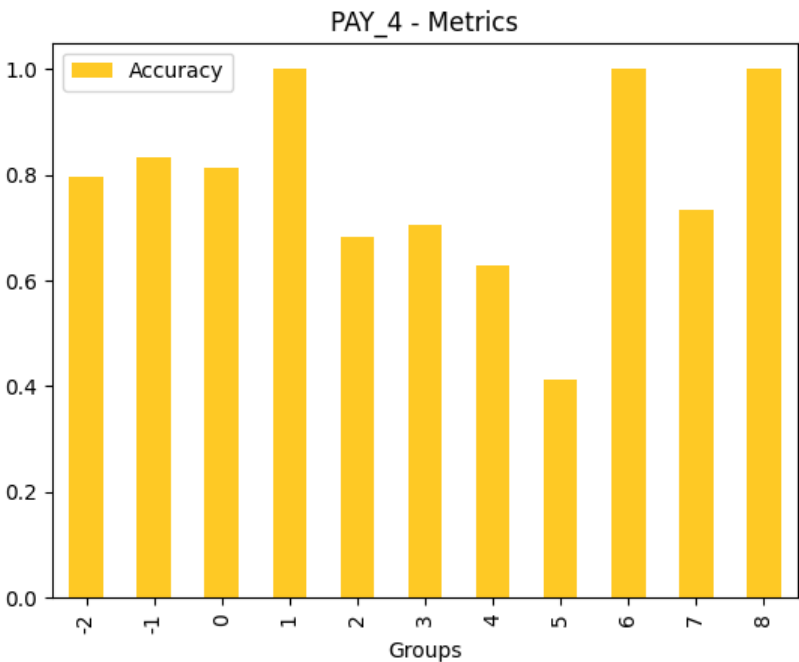
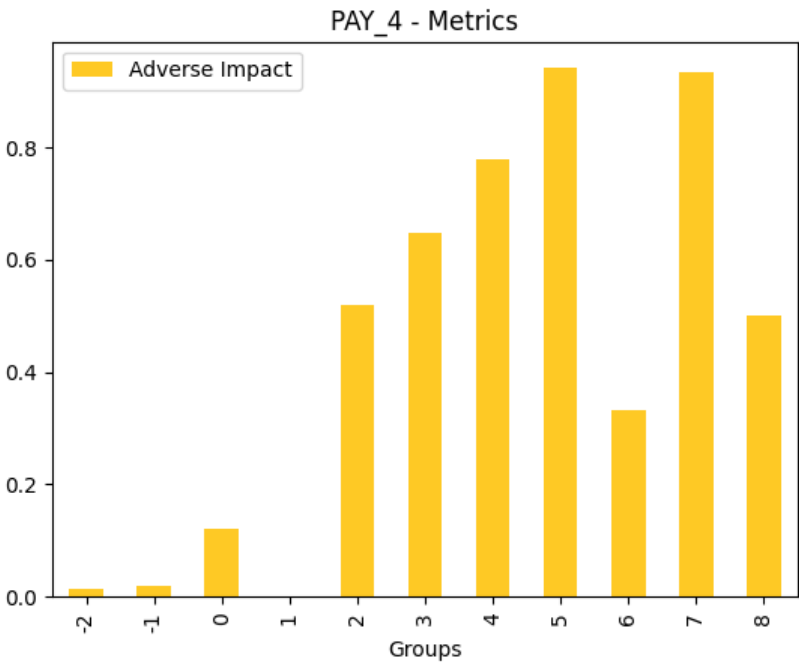


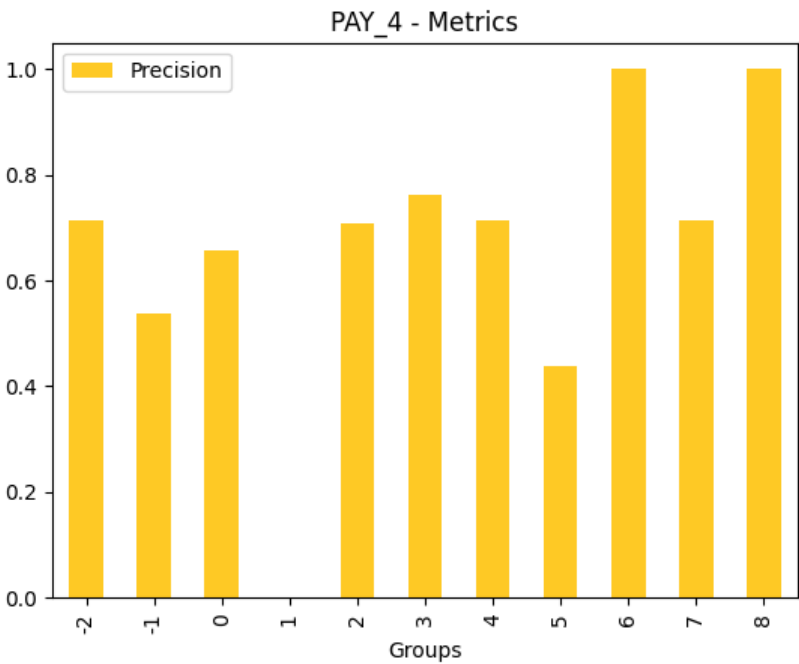
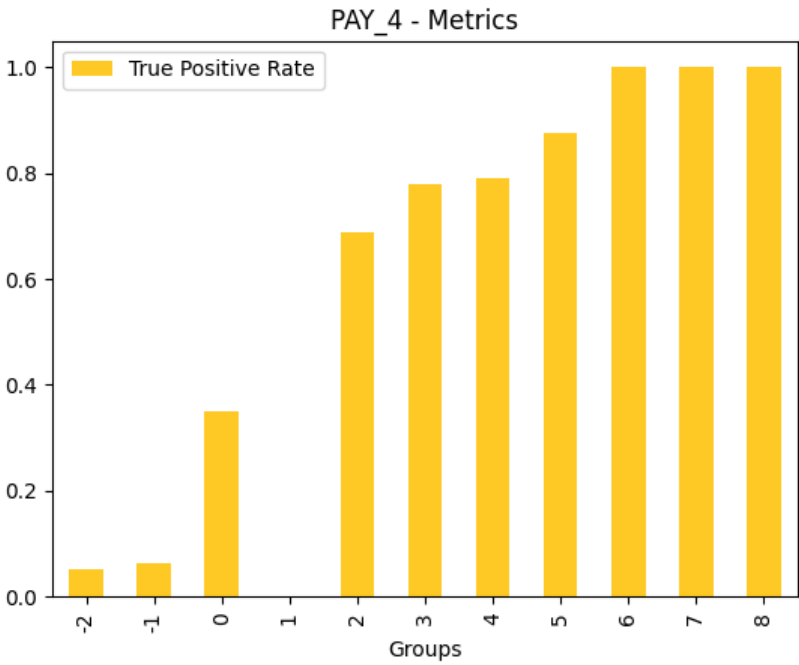


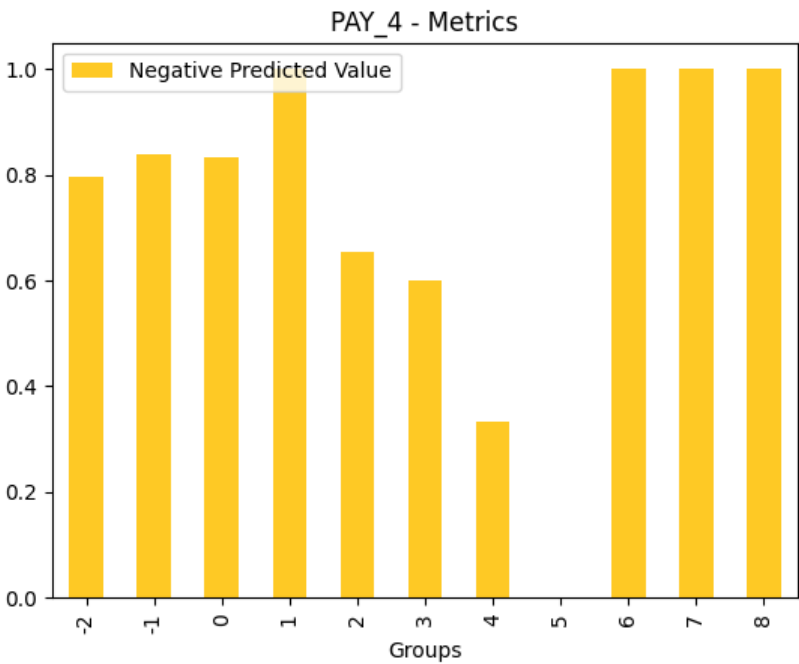
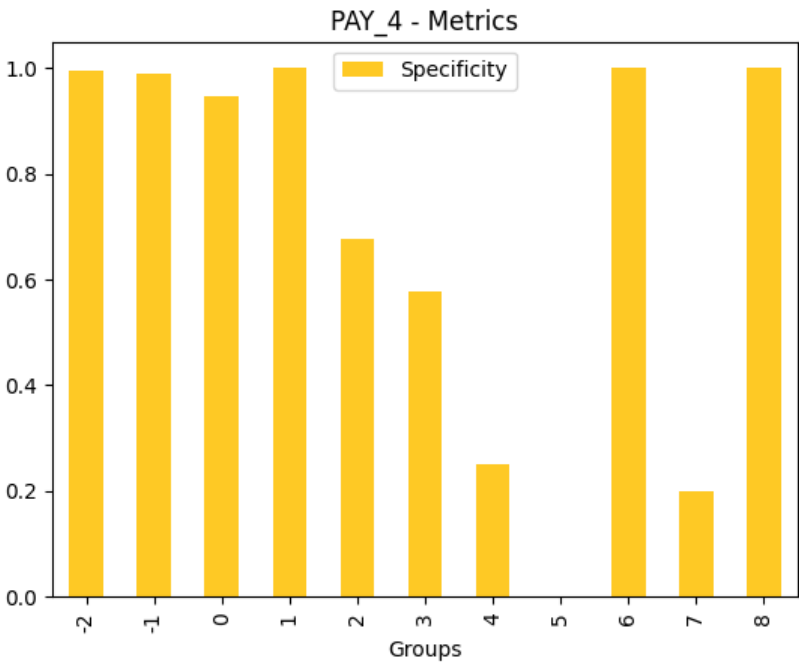


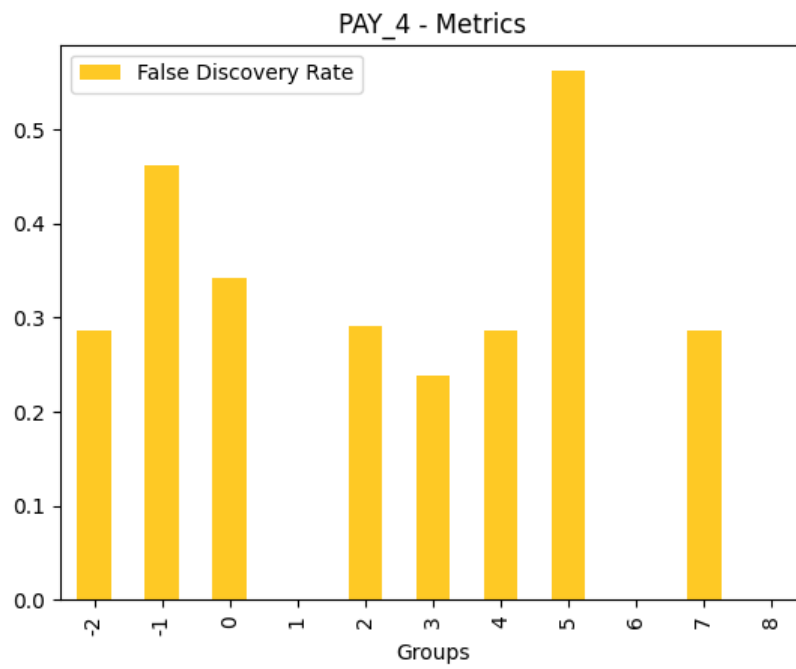
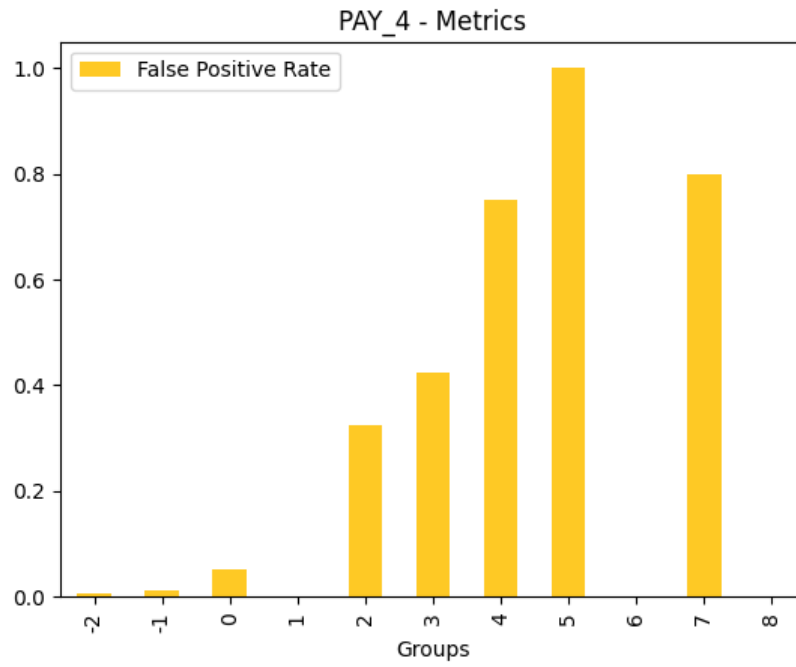
Fairness metrics for the feature: PAY_4

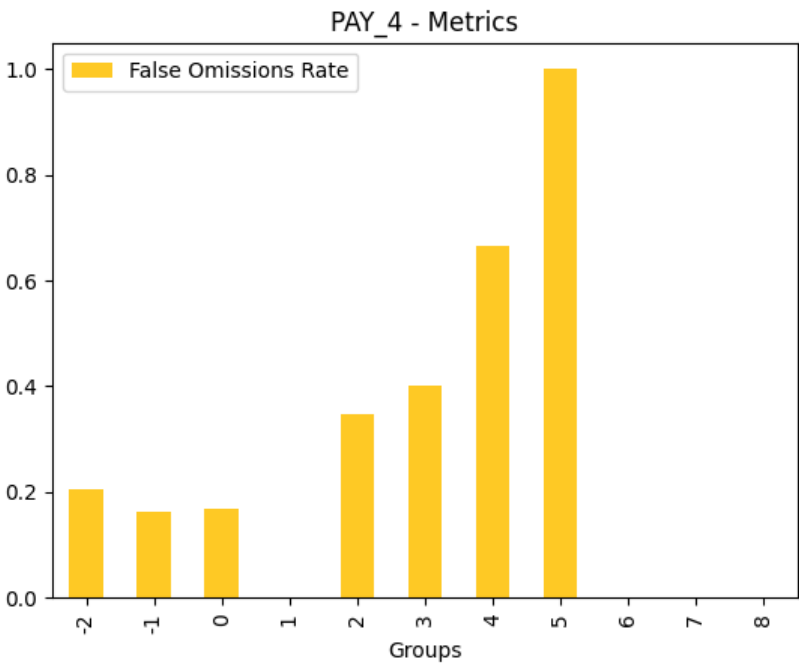
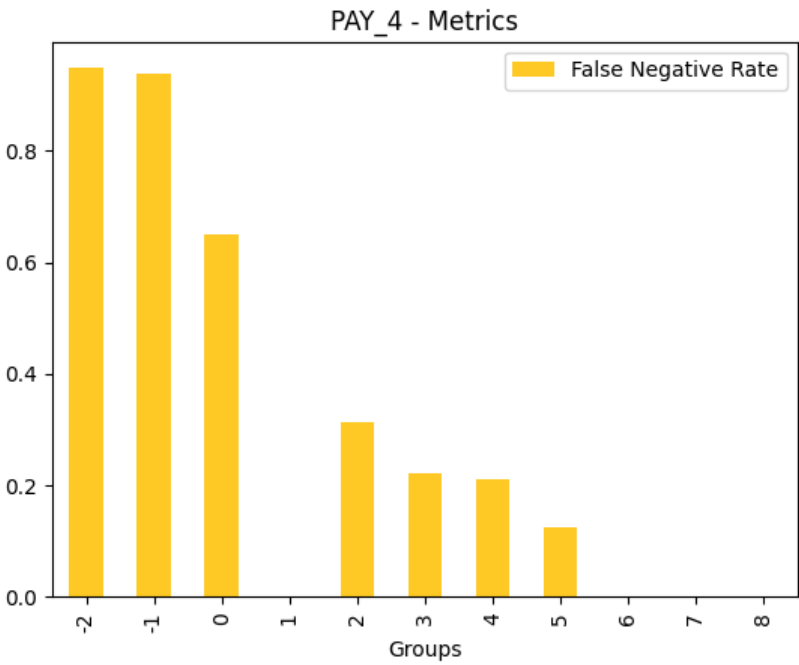




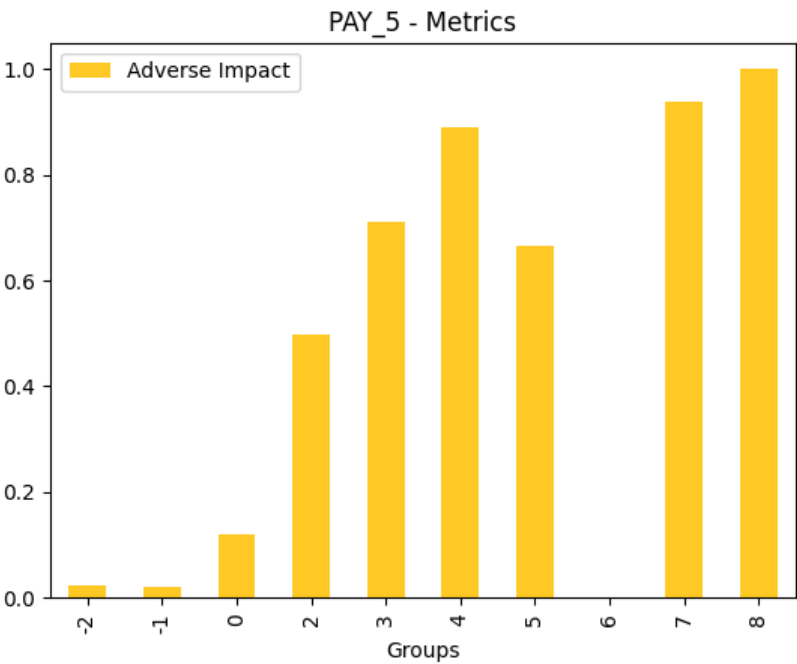
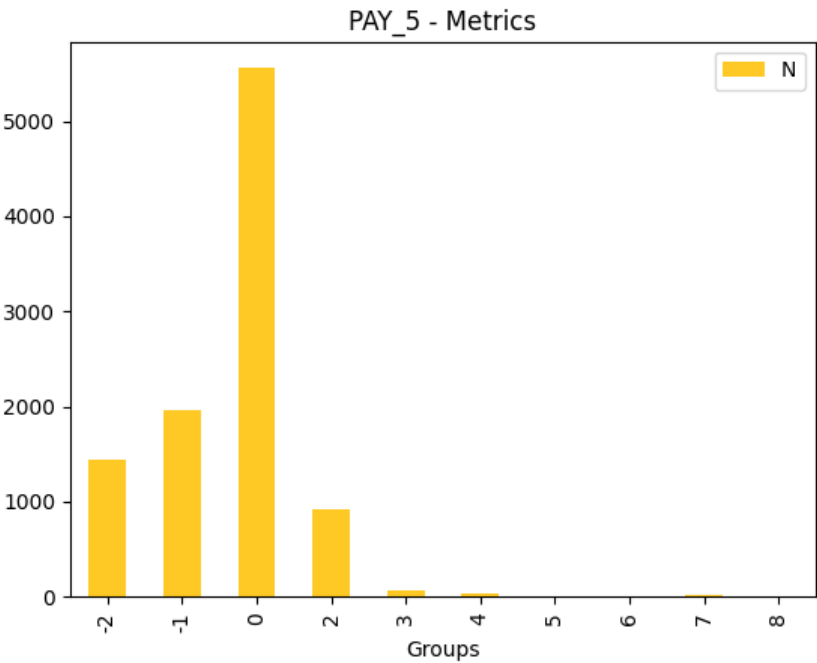


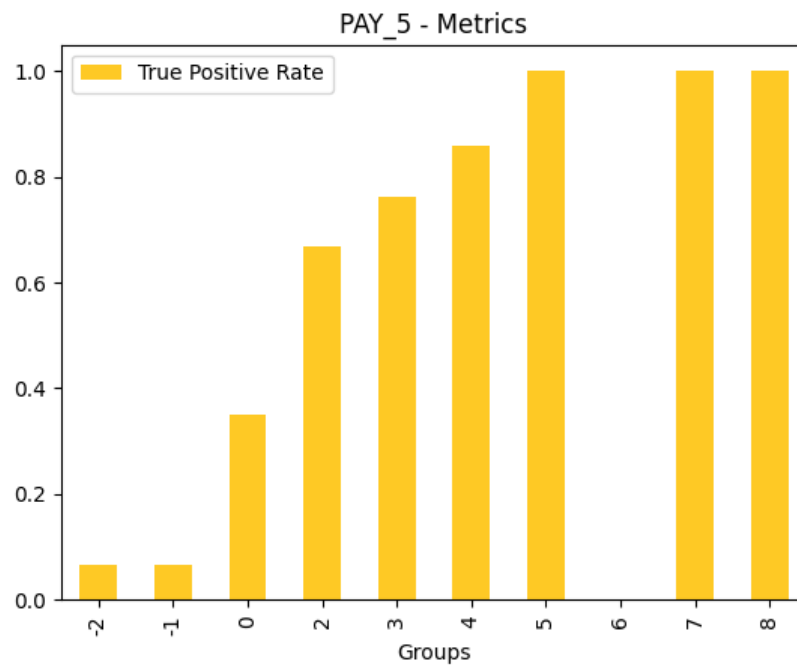
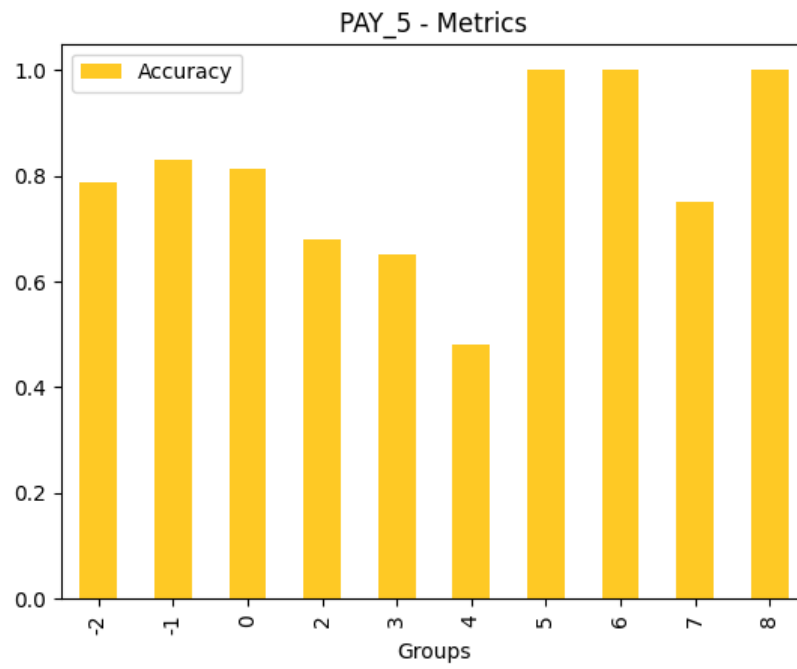


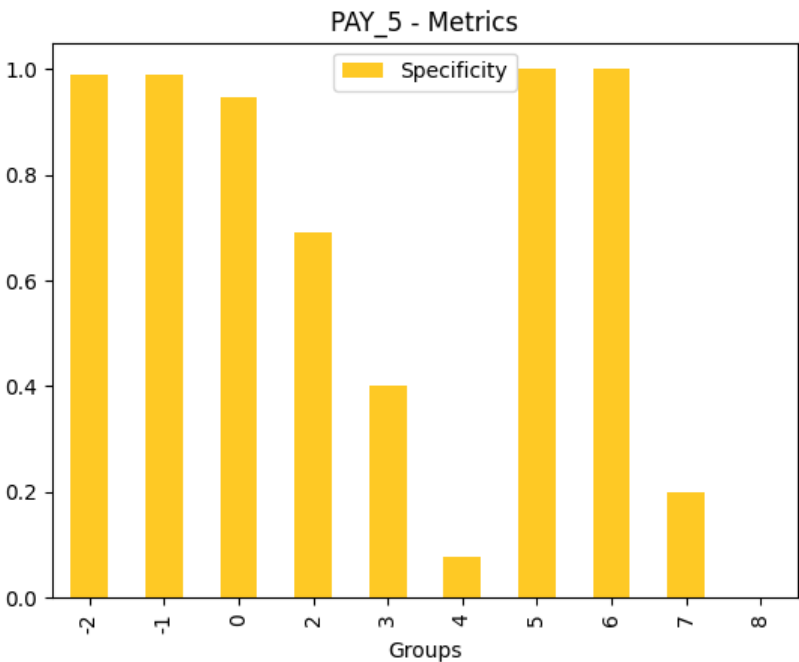
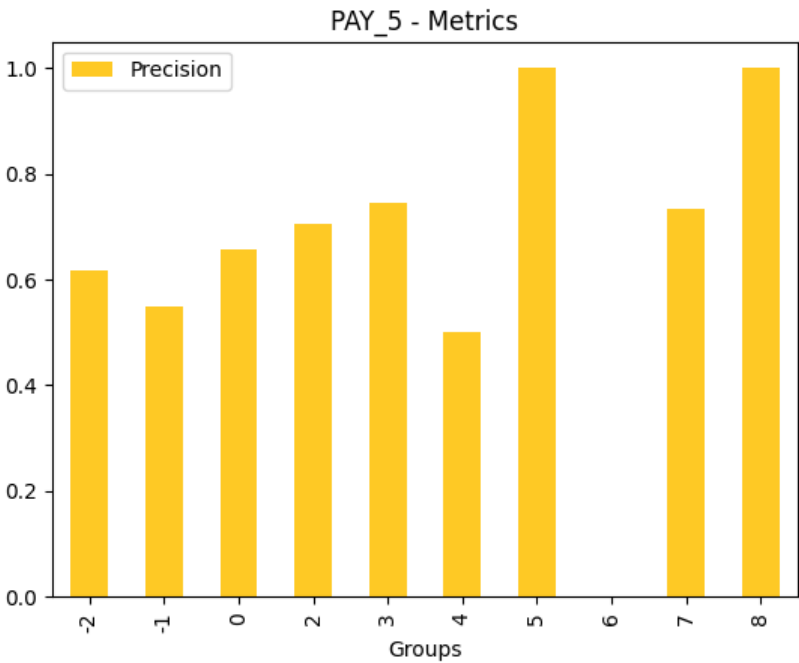


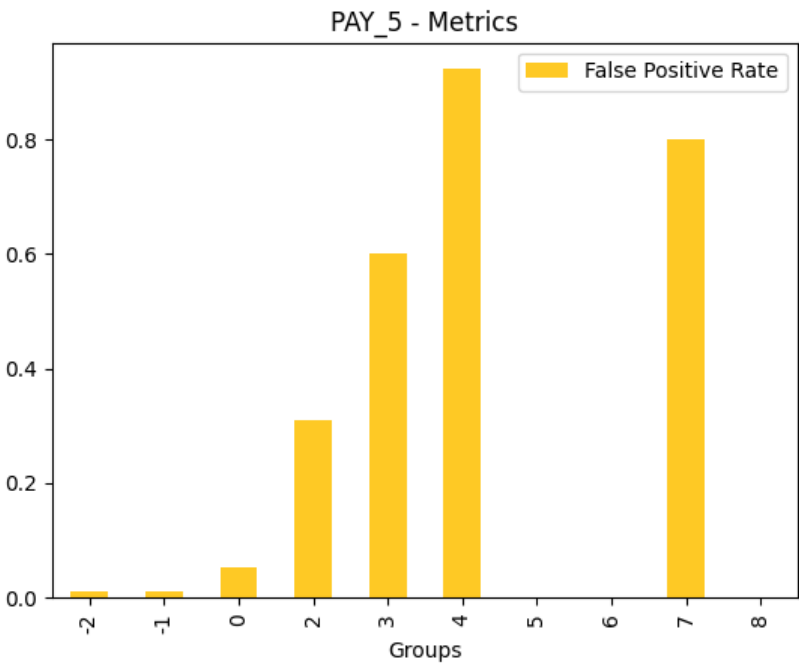
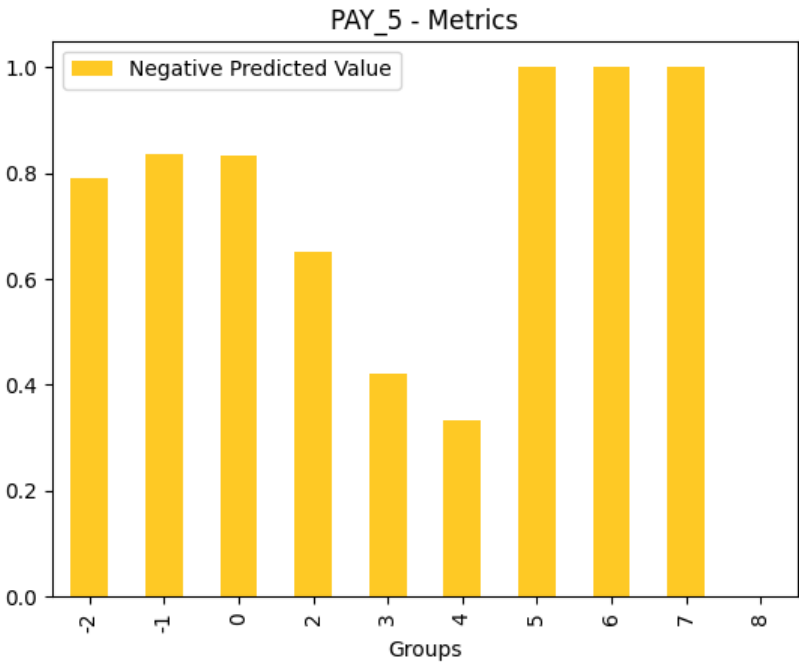


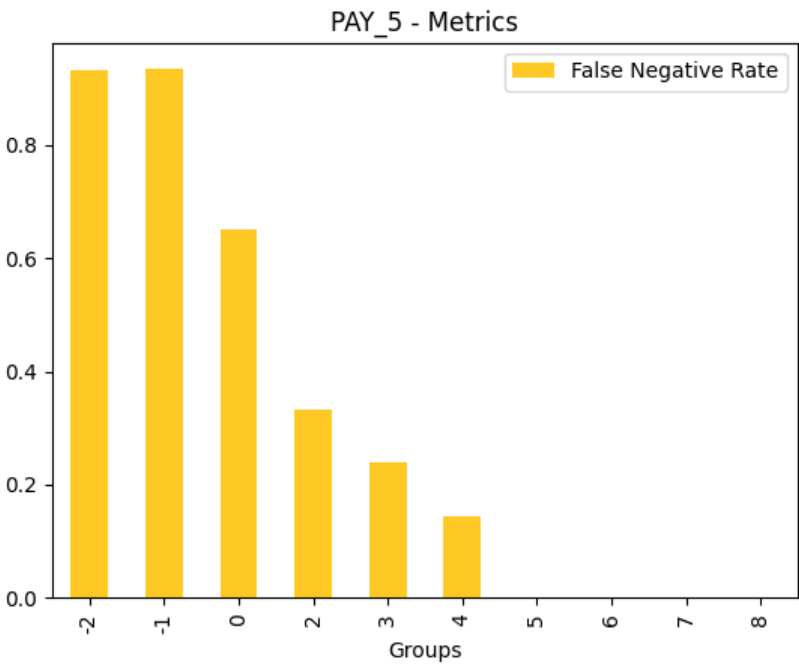
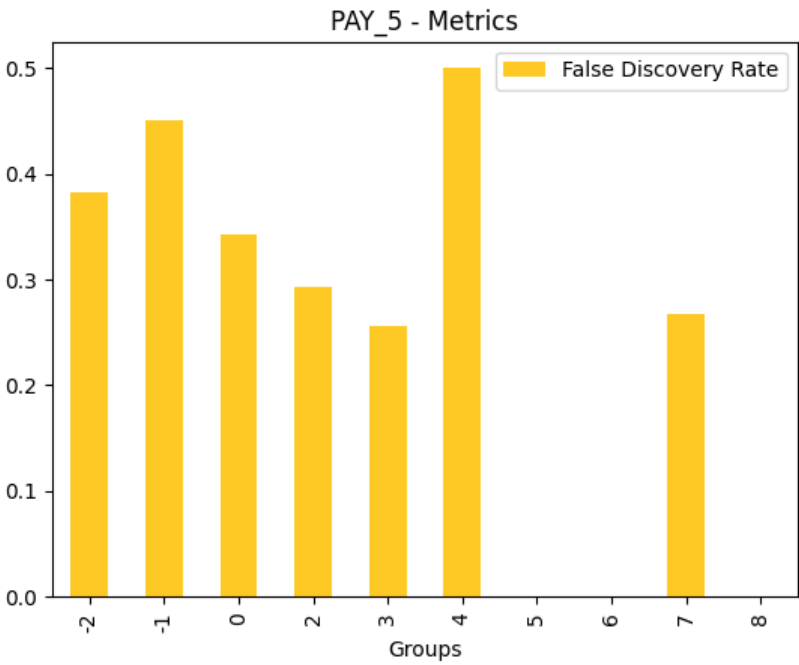
Fairness metrics for the feature: PAY_5

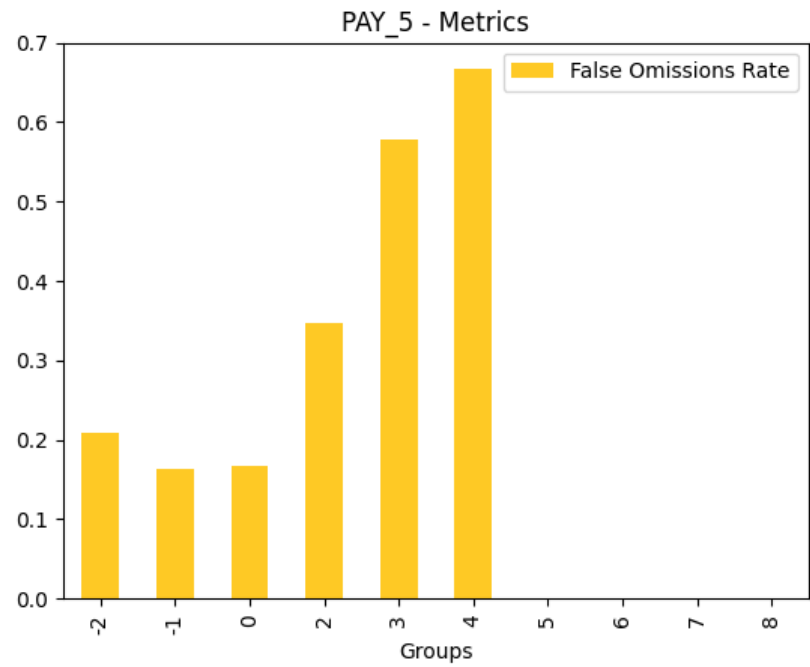




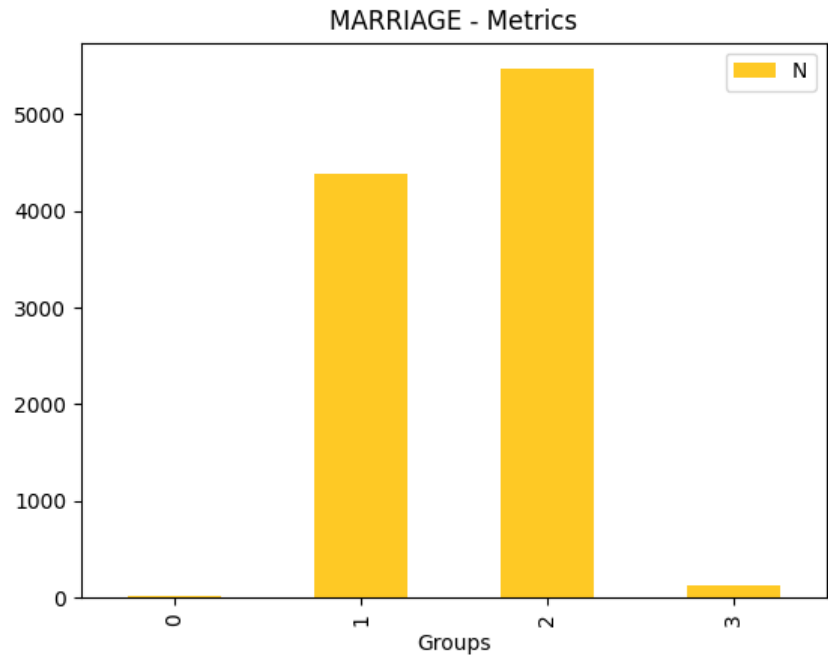


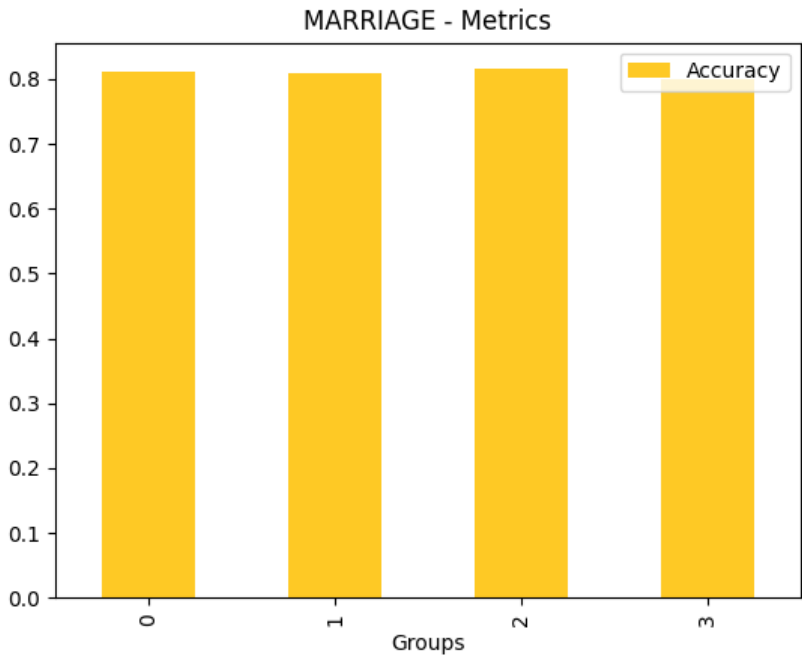
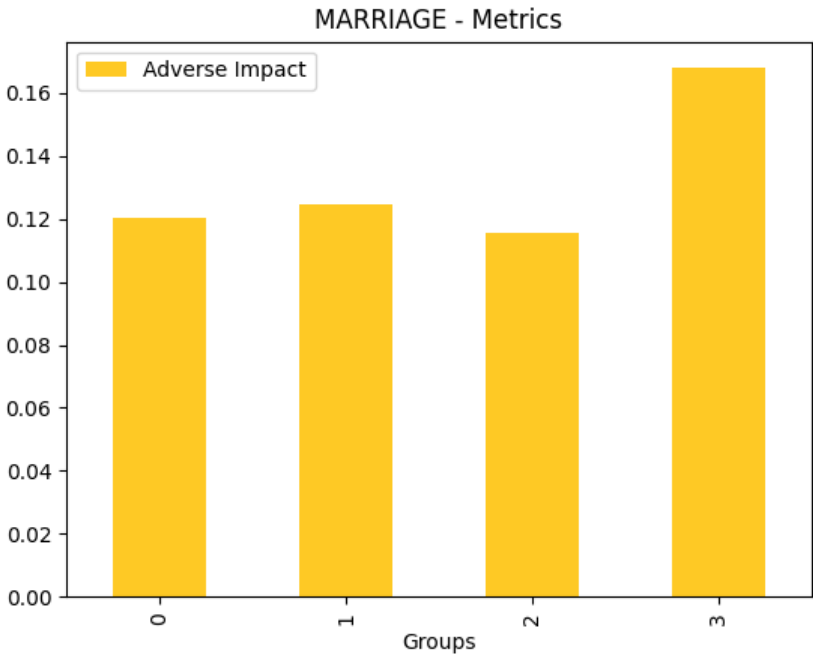


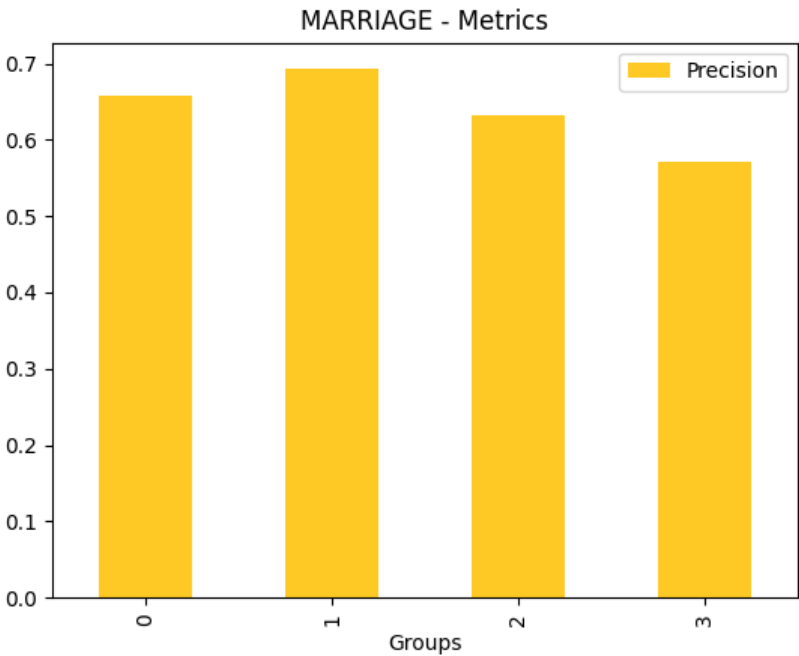
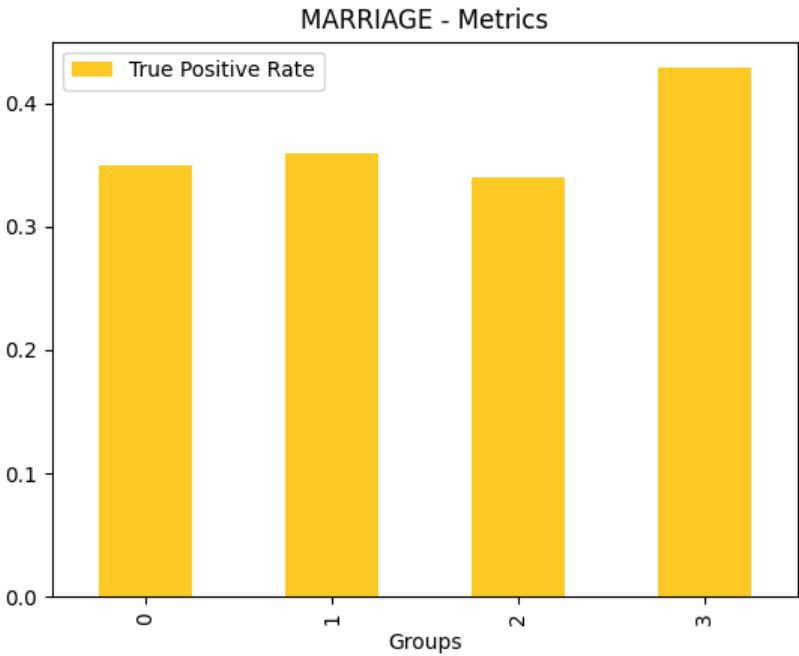


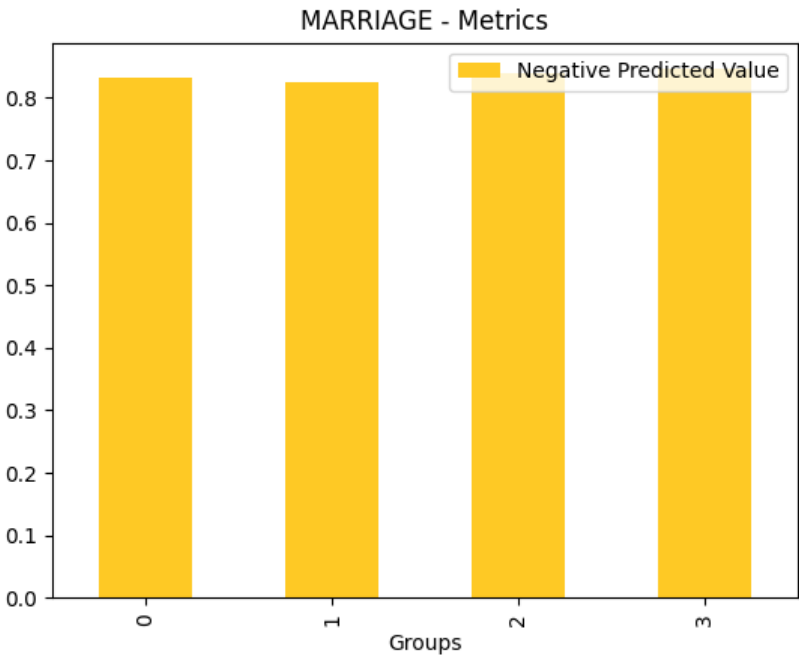
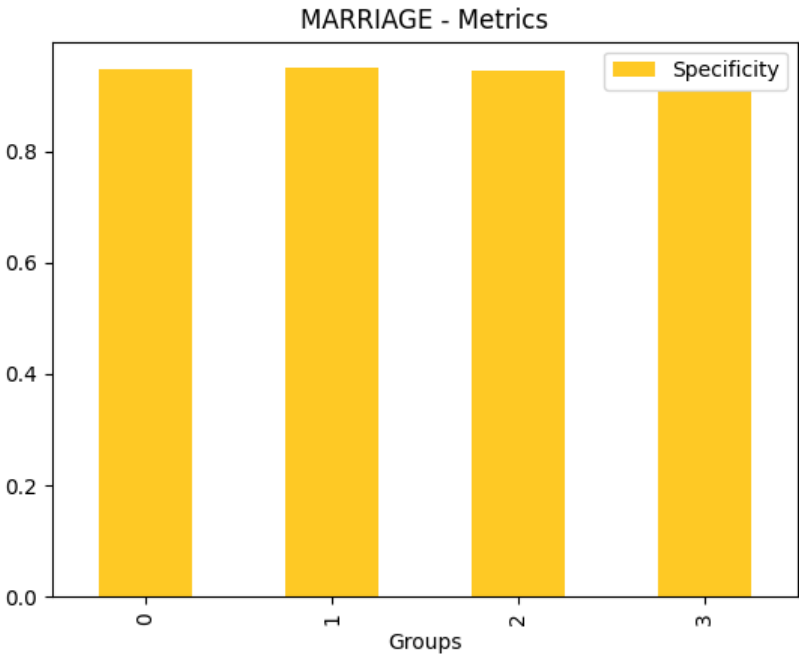


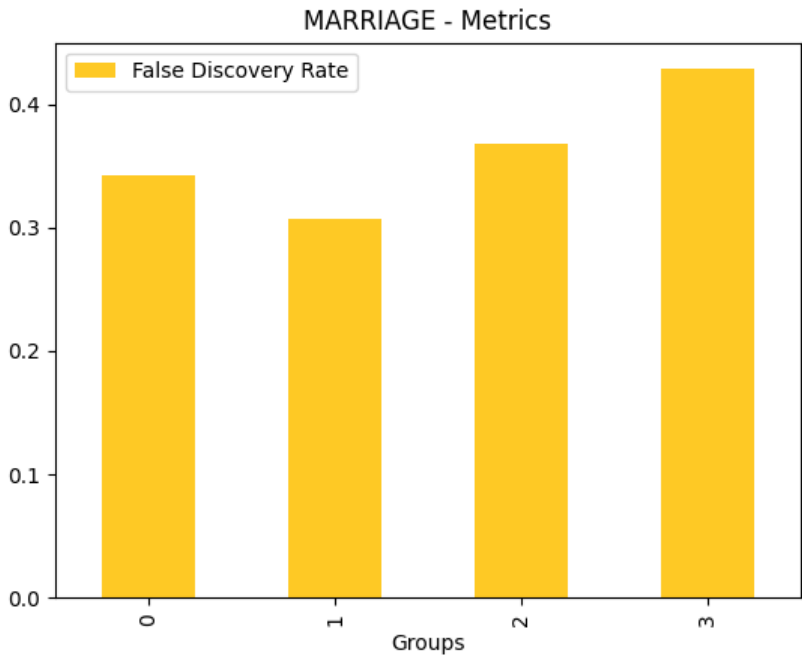
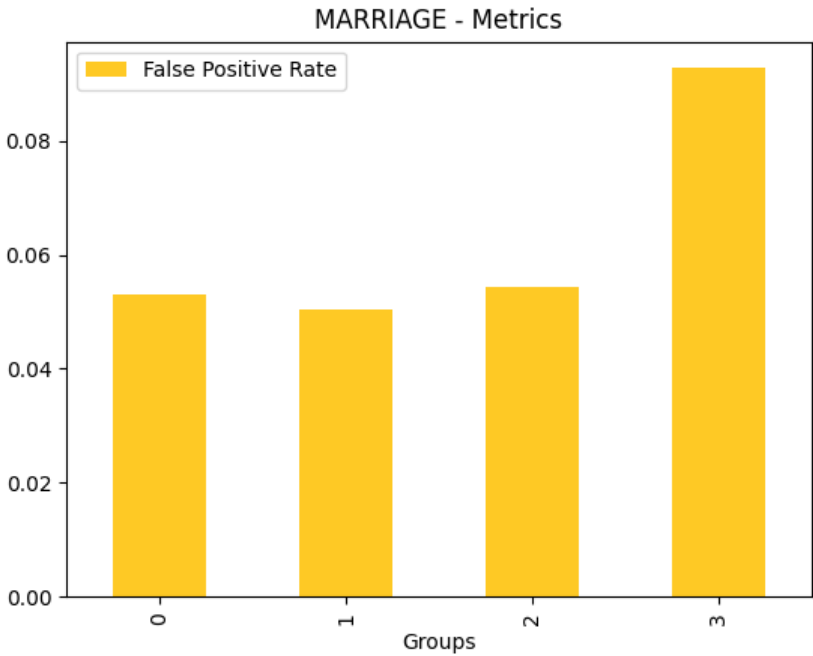
Fairness metrics for the feature: MARRIAGE

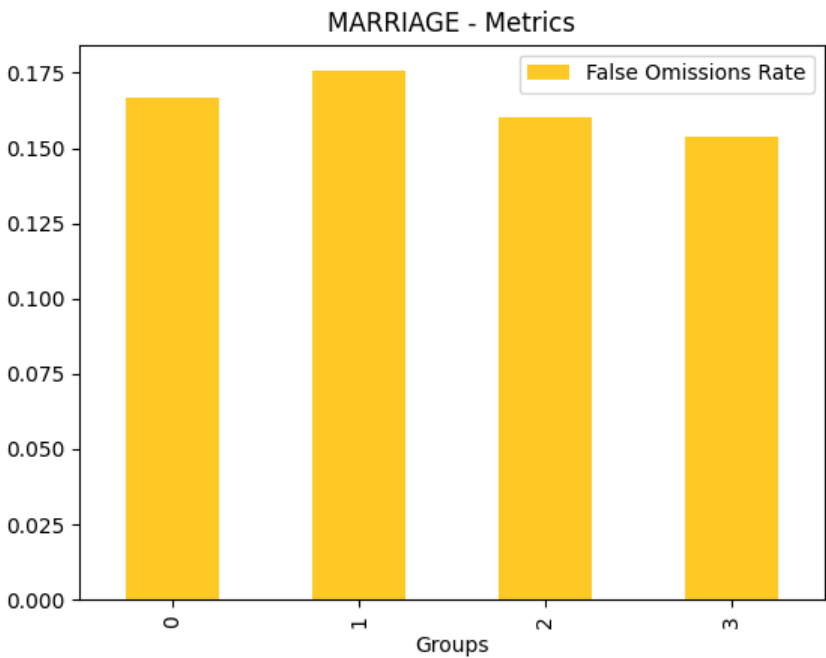
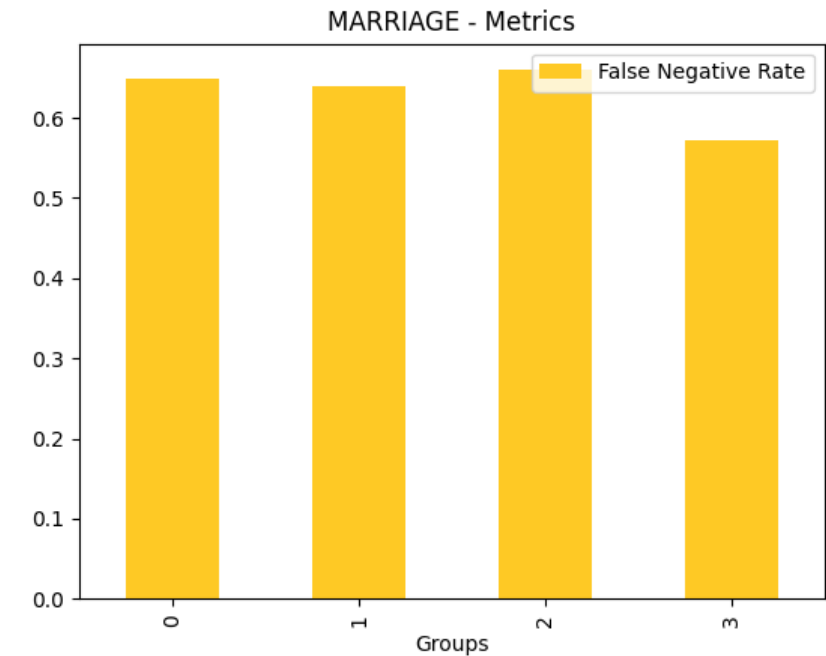




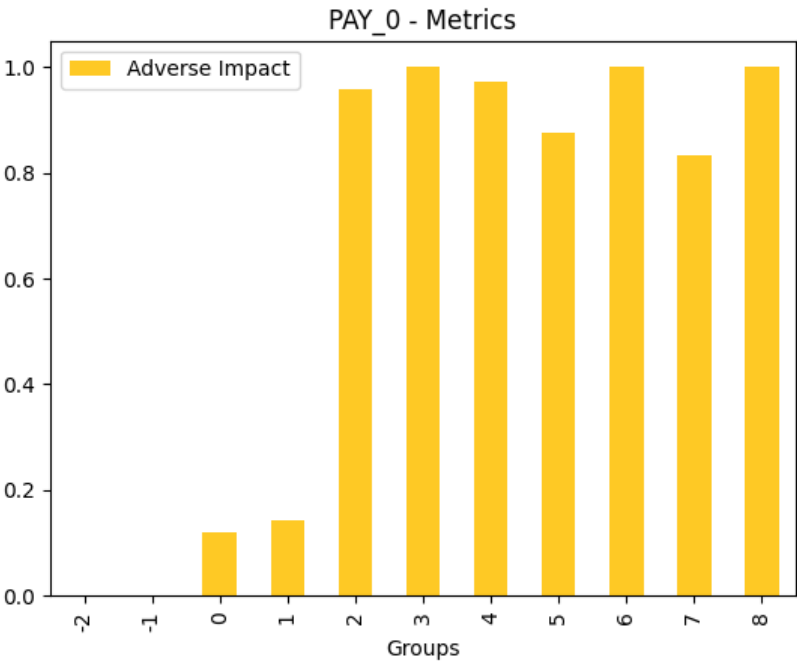
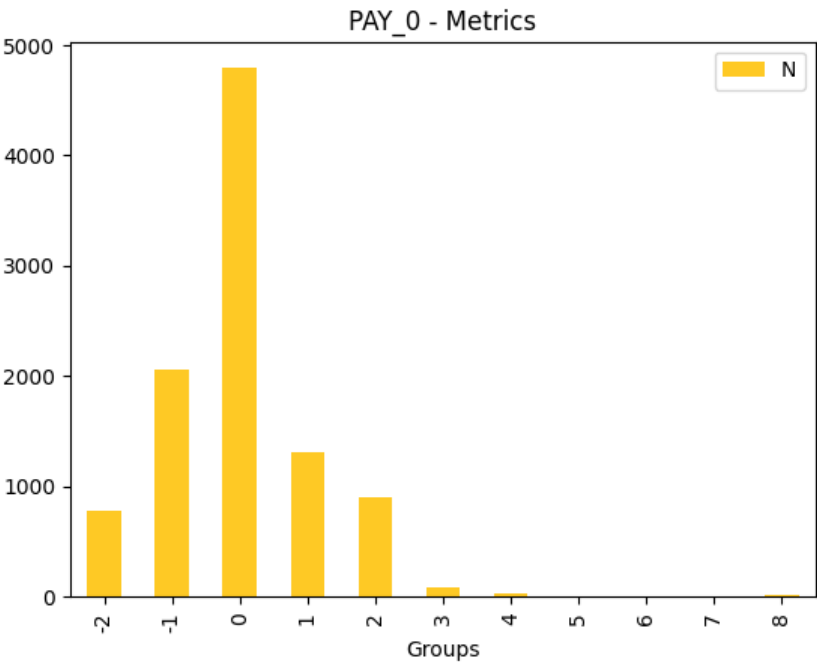


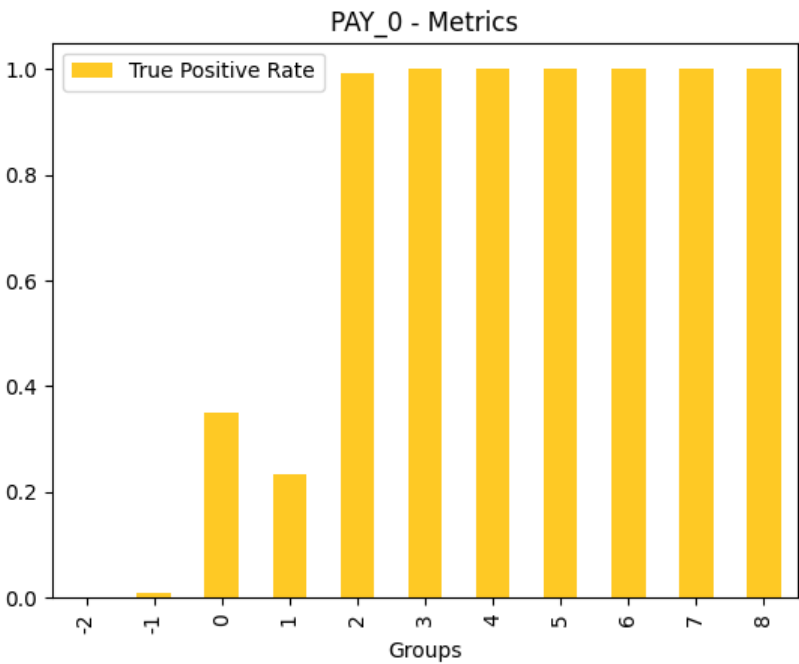
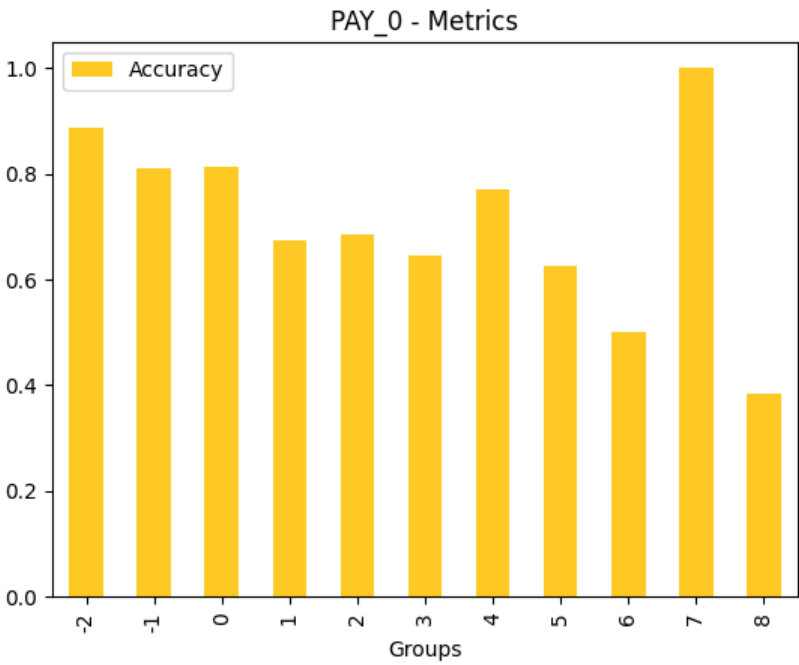


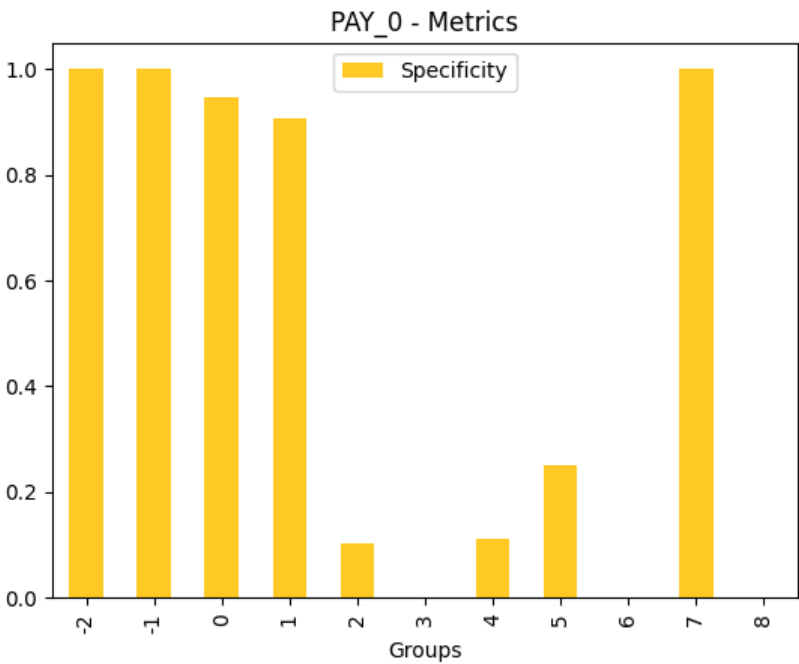
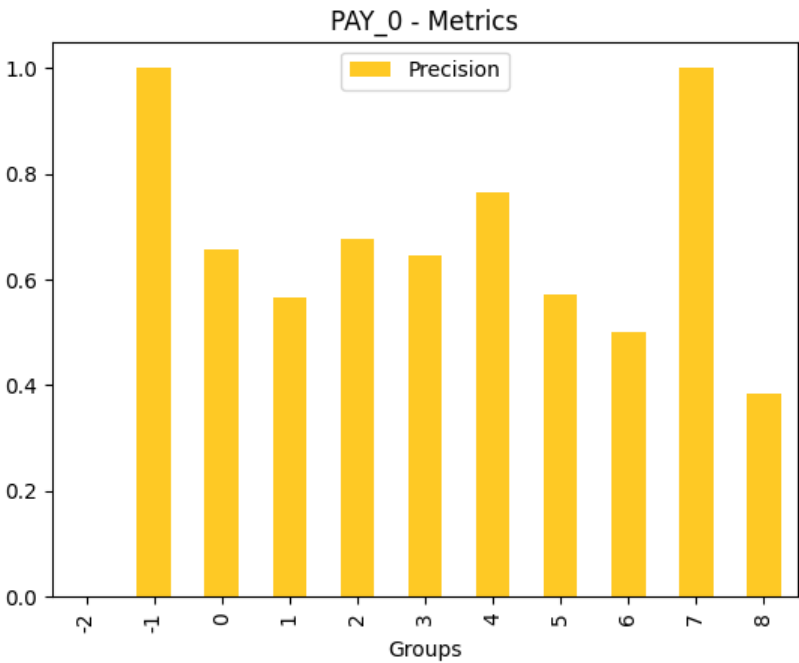


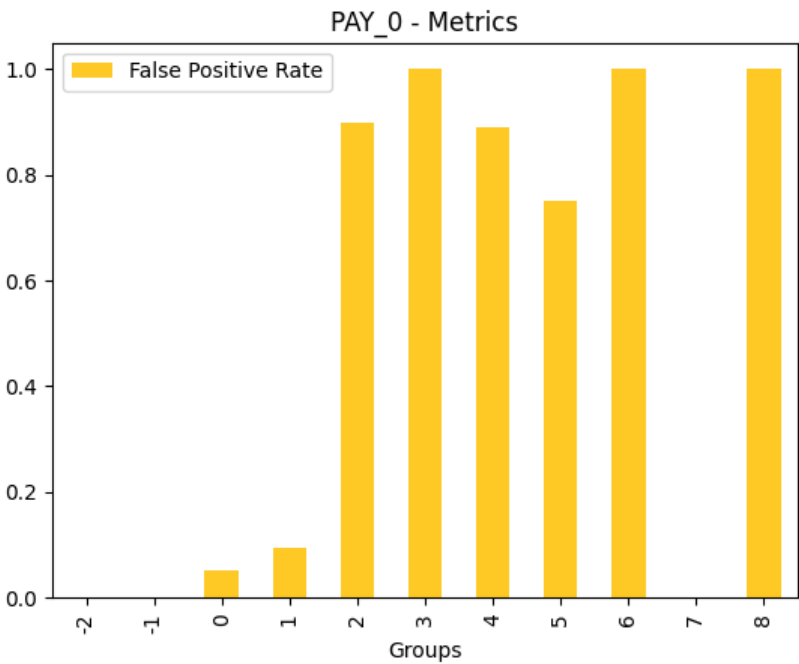
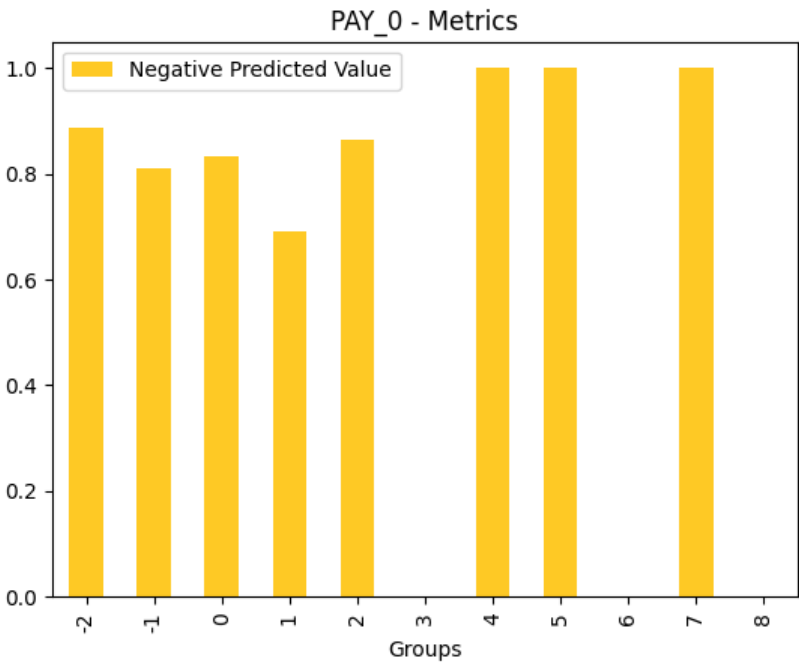


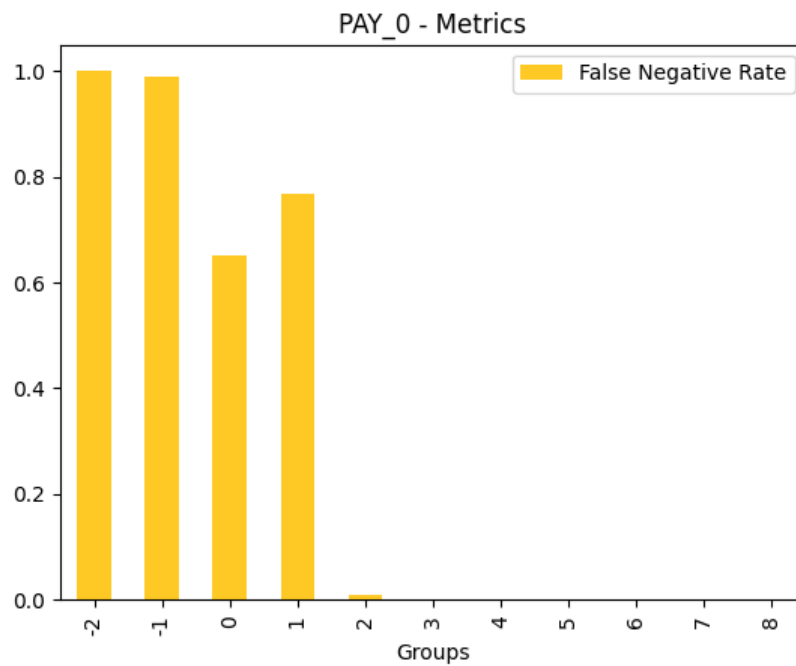
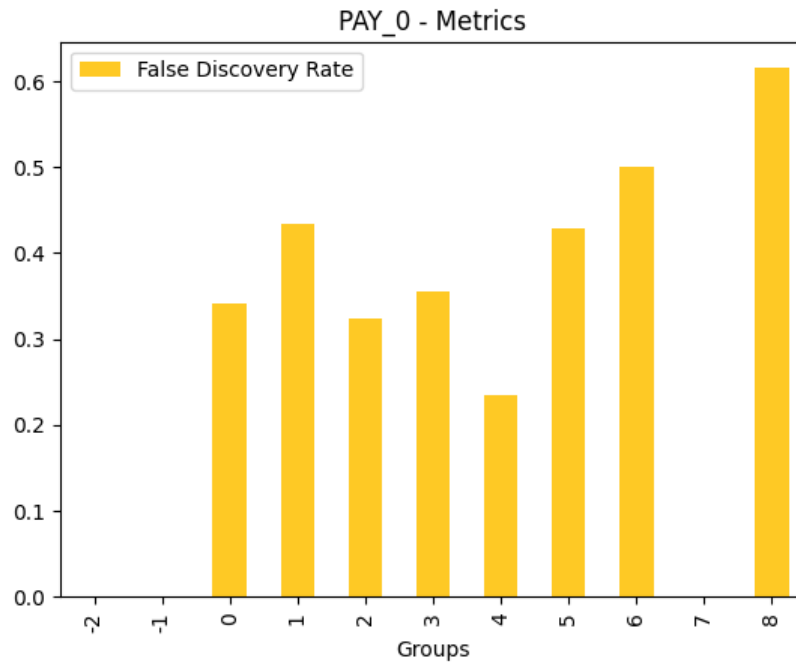
Fairness metrics for the feature: PAY_o

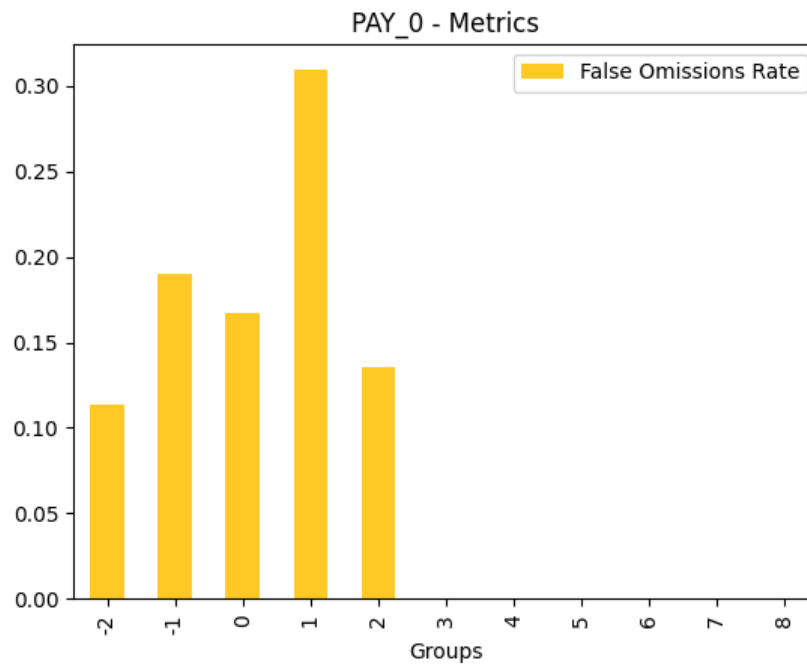












Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types:

- **Disparate Impact Analysis**
global-disparate-impact-analysis
 - [text/plain](#)
- **Disparate Impact Analysis**
global-html-fragment
 - [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
dia_cols	None	List of features for which to compute DIA.	list	None
cut_off	0.0	Cut off.	float	0.0
maximize_metric	F1	Maximize metric.	str	F1
max_cardinality	10	Max cardinality for categorical variables.	int	10
min_cardinality	2	Minimum cardinality for categorical variables.	int	2
num_card	25	Max cardinality for numeric variables to be considered categorical.	int	25

Explainer metadata

The explainer can explain model types:

- regression
- binomial

Explainer keywords:

- **run-by-default**
- **explains-fairness**
- **h2o-sonar**

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:39:23 T+0100**
- Duration: **5.267s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Shapley Values for Original Features of MOJO Models (Naive Method)

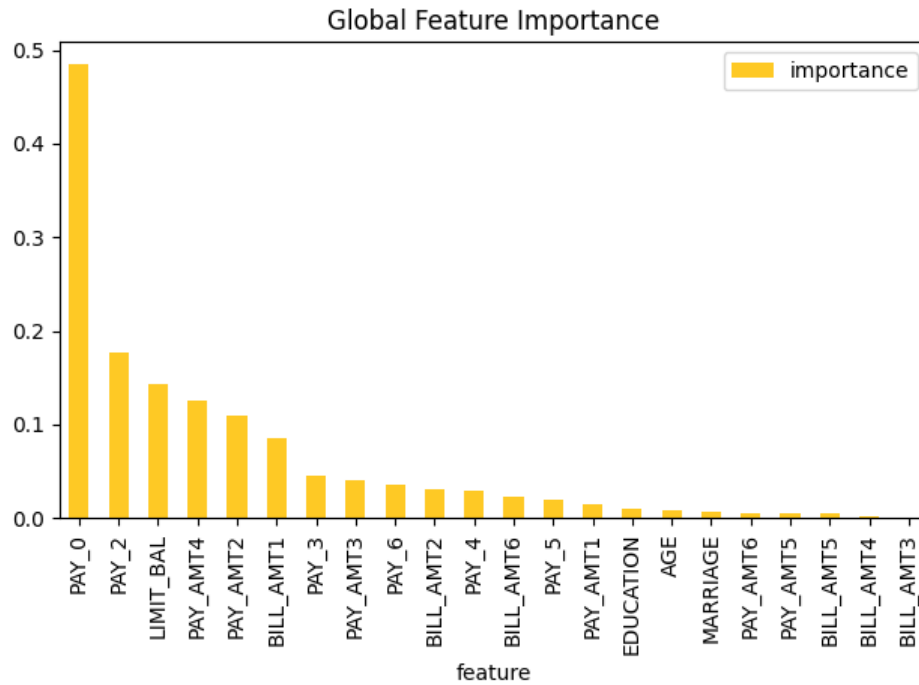
- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

Shapley values for original features of (Driverless AI) MOJO models are approximated from the accompanying Shapley values for transformed features with the Naive Shapley method. This method makes the assumption that input features to a transformer are independent. For example, if the transformed feature, feature1_feature2, has a Shapley value of 0.5, then the Shapley value of the original features feature1 and feature2 will be 0.25 each.

Explanations

Feature importance for the class '1':



The most important original feature of the class 1 is PAY_0 .

Original feature importances for the class '1':

- 1. PAY_0 feature with importance 0.48431425950676193
- 2. PAY_2 feature with importance 0.17728837728463115
- 3. LIMIT_BAL feature with importance 0.14293027323329396
- 4. PAY_AMT4 feature with importance 0.1259882615562616
- 5. PAY_AMT2 feature with importance 0.10983877249736468
- 6. BILL_AMT1 feature with importance 0.08566852904600201
- 7. PAY_3 feature with importance 0.046046926190133244
- 8. PAY_AMT3 feature with importance 0.04056427323384959
- 9. PAY_6 feature with importance 0.035128979444635125
- 10. BILL_AMT2 feature with importance 0.030753264524566466
- 11. PAY_4 feature with importance 0.029223238050425416
- 12. BILL_AMT6 feature with importance 0.023093468860886066
- 13. PAY_5 feature with importance 0.02037994117215276
- 14. PAY_AMT1 feature with importance 0.014774042908039704
- 15. EDUCATION feature with importance 0.009833406838281326
- 16. AGE feature with importance 0.008313157282831355
- 17. MARRIAGE feature with importance 0.007252847269852646
- 18. PAY_AMT6 feature with importance 0.00588242508913081
- 19. PAY_AMT5 feature with importance 0.00541516162084872
- 20. BILL_AMT5 feature with importance 0.004646423485466219
- 21. BILL_AMT4 feature with importance 0.0014406579161764855
- 22. BILL_AMT3 feature with importance 0.0008182751896645635

Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by [media types](#) :

- **Shapley Values for Original Features of MOJO Models (Naive Method)**
global-feature-importance
 - [application/vnd.h2oai.json+datatable.jay](#)

- [application/vnd.h2oai.json+csv](#)
 - [application/json](#)
- Shapley Values for Original Features of MOJO Models (Naive Method)
local-feature-importance
 - [application/vnd.h2oai.json+datatable.jay](#)
- Shapley Values for Original Features of MOJO Models (Naive Method)
global-html-fragment
 - [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
sample_size	100000	Sample size.	int	100000
fast_approx_contribs	True	Speed up predictions with fast contributions predictions approximation.	bool	True
leakage_warning_threshold	0.95	The threshold above which to report a potentially detected feature importance leak problem.	float	0.95

Explainer metadata

The explainer can explain model types:

- regression
- binomial
- multinomial

Explainer keywords:

- run-by-default
- explains-original-feature-importance
- is_fast
- h2o-sonar

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:39:28 T+0100**
- Duration: **1:39:55**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Shapley Values for Transformed Features of MOJO Models

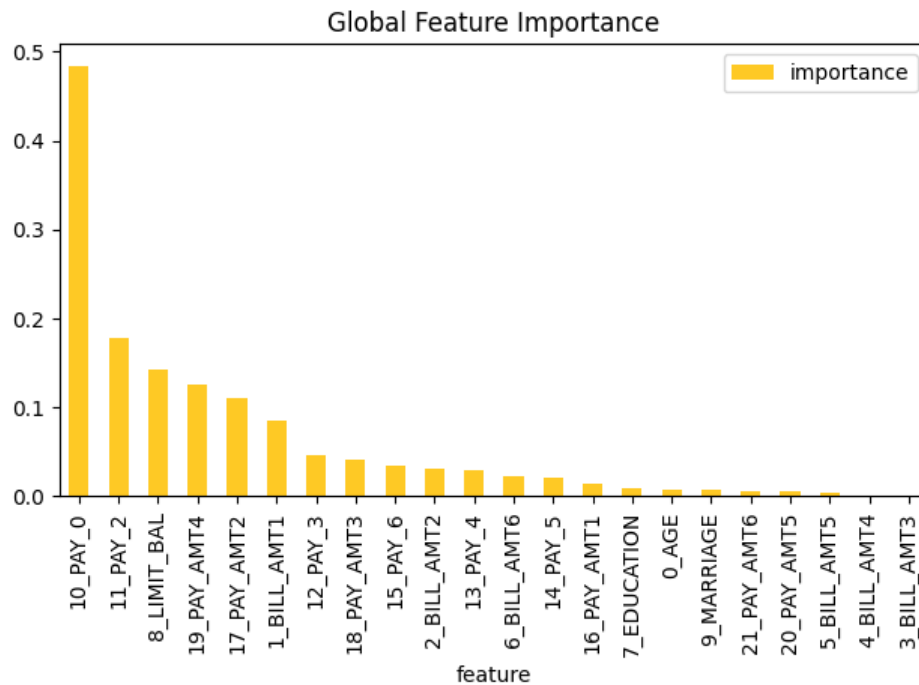
- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

Shapley explanations are a technique with credible theoretical support that presents consistent global and local variable contributions. Local numeric Shapley values are calculated by tracing single rows of data through a trained tree ensemble and aggregating the contribution of each input variable as the row of data moves through the trained ensemble. For regression tasks Shapley values sum to the prediction of the (Driverless AI) MOJO model. For classification problems, Shapley values sum to the prediction of the MOJO model before applying the link function. Global Shapley values are the average of the absolute local Shapley values over every row of a data set.

Explanations

Feature importance for the class '1':



The most important transformed feature of the class 1 is 10_PAY_0 .

Transformed feature importances for the class '1':

- 1. 10_PAY_0 feature with importance 0.48431425950676193
- 2. 11_PAY_2 feature with importance 0.17728837728463115
- 3. 8_LIMIT_BAL feature with importance 0.14293027323329396
- 4. 19_PAY_AMT4 feature with importance 0.1259882615562616
- 5. 17_PAY_AMT2 feature with importance 0.10983877249736468
- 6. 1_BILL_AMT1 feature with importance 0.08566852904600201
- 7. 12_PAY_3 feature with importance 0.04604692619013323
- 8. 18_PAY_AMT3 feature with importance 0.04056427323384959
- 9. 15_PAY_6 feature with importance 0.035128979444635125
- 10. 2_BILL_AMT2 feature with importance 0.03075326452456647

- 11. 13_PAY_4 feature with importance 0.02922323805042542
- 12. 6_BILL_AMT6 feature with importance 0.023093468860886063
- 13. 14_PAY_5 feature with importance 0.02037994117215276
- 14. 16_PAY_AMT1 feature with importance 0.014774042908039705
- 15. 7_EDUCATION feature with importance 0.009833406838281326
- 16. 0_AGE feature with importance 0.008313157282831355
- 17. 9_MARRIAGE feature with importance 0.007252847269852645
- 18. 21_PAY_AMT6 feature with importance 0.00588242508913081
- 19. 20_PAY_AMT5 feature with importance 0.00541516162084872
- 20. 5_BILL_AMT5 feature with importance 0.004646423485466219
- 21. 4_BILL_AMT4 feature with importance 0.0014406579161764855
- 22. 3_BILL_AMT3 feature with importance 0.0008182751896645635

Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types:

- **Shapley Values for Transformed Features of MOJO Models**

global-feature-importance

- [application/vnd.h2oai.json+datatable.jay](#)
- [application/vnd.h2oai.json+csv](#)
- [application/json](#)

- **Shapley Values for Transformed Features of MOJO Models**

local-feature-importance

- [application/vnd.h2oai.json+datatable.jay](#)

- **Shapley Values for Transformed Features of MOJO Models**

global-html-fragment

- [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
sample_size	100000	Sample size.	int	100000
calculate_predictions	False	Score dataset and include predictions in the explanation (local explanations speed-up cache).	bool	False
fast_approx_contrib	True	Speed up predictions with fast contributions predictions approximation.	bool	True

Explainer metadata

The explainer can explain model types:

- regression
- binomial
- multinomial

Explainer keywords:

- run-by-default
- explains-transformed-feature-importance
- is_fast
- h2o-sonar

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:39:29 T+0100**
- Duration: **1.346s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Residual Surrogate Decision Tree

- [Model problems](#)
- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Model problems

Explainer identified the following potential problems with the model:

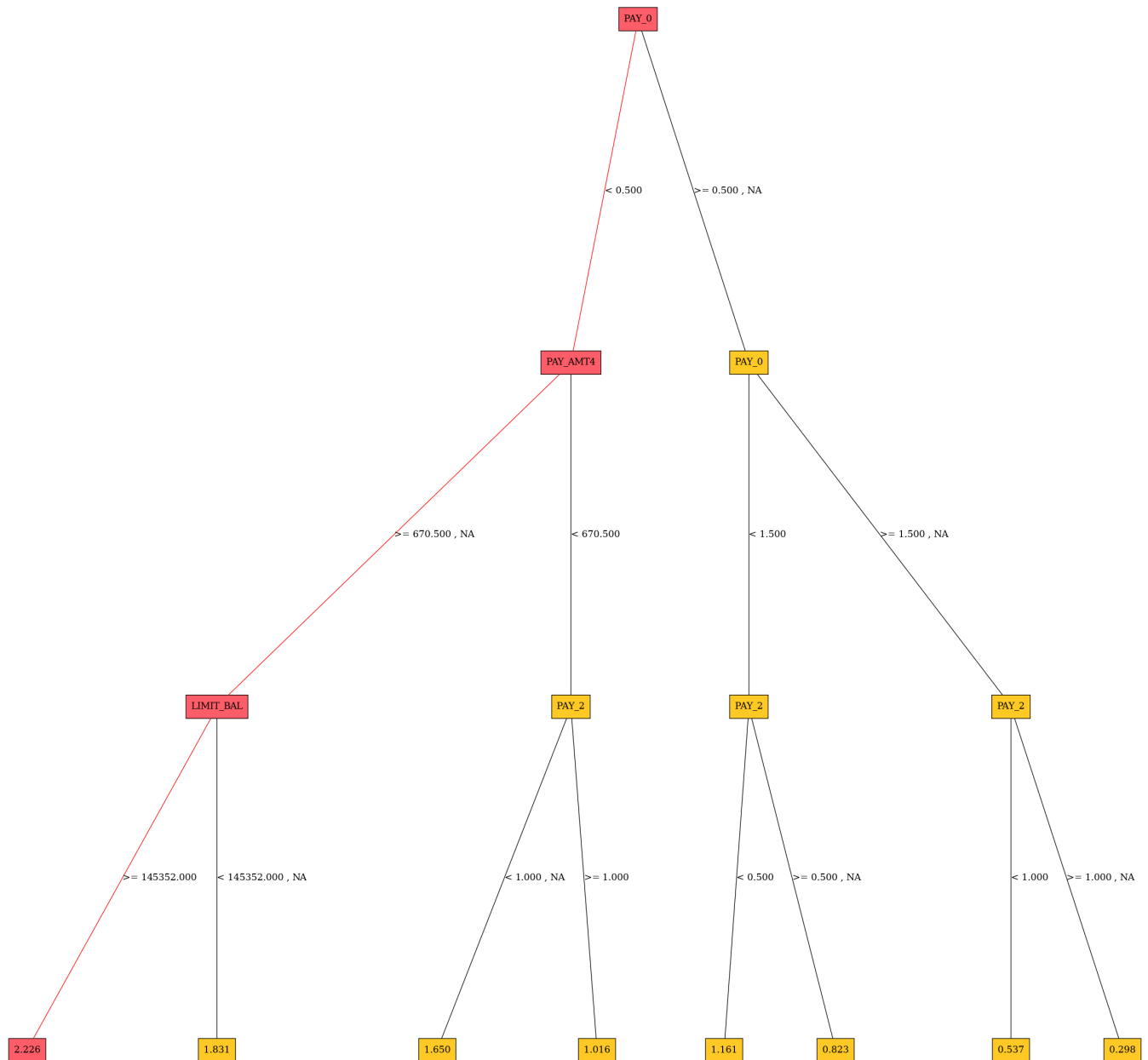
Severity	Type	Problem	Suggested actions	Explainer	Resources
LOW	bias	A path in the residual surrogate decision tree leading to the largest residual (2.226343) may indicate a problem in the model.	Verify that the following surrogate decision tree path does not indicate a model bias or other problem: IF (PAY_o < 0.5) AND (PAY_AMT4 >= 670.5 OR PAY_AMT4 IS N/A) AND (LIMIT_BAL >= 145352.0) THEN AVERAGE VALUE OF TARGET IS 2.226343	Residual Surrogate Decision Tree	GlobalDtExplanation / application/json

Explainer description

The residual surrogate decision tree predicts which paths in the tree (paths explain approximate model behavior) lead to highest or lowest error. The residual surrogate decision tree is created by training a simple decision tree on the residuals of the predictions of the model. Residuals are differences between observed and predicted values which can be used as targets in surrogate models for the purpose of model debugging. The method used to calculate residuals varies depending on the type of problem. For classification problems, logloss residuals are calculated for a specified class (only one residual surrogate decision is created by the explainer and it is built for this class). For regression problems, residuals are determined by calculating the square of the difference between targeted and predicted values.

Explanations

Approximate model behavior for the class '1':



Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types :

- **Residual Decision Tree**
global-decision-tree
 - [application/json](#)
- **Local DT**
local-decision-tree
 - [application/json](#)
- **Surrogate Decision Tree**
global-html-fragment
 - [text/html](#)
- **Residual Decision tree surrogate rules ZIP archive**
global-custom-archive

- [application/zip](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
debug_residuals_class	1	Class for debugging classification model logloss residuals, empty string for debugging regression model residuals.	str	
dt_tree_depth	3	Decision tree depth.	int	3
nfolds	3	Number of CV folds.	int	3
qbin_cols	None	Quantile binning columns.	list	None
qbin_count	0	Quantile bins count.	int	0
categorical_encoding	onehotexplicit	Categorical encoding.	str	onehotexplicit
debug_residuals	True			

Explainer metadata

The explainer can explain model types:

- regression
- binomial
- multinomial

Explainer keywords:

- run-by-default
- requires-h2o3
- explains-model-debugging
- surrogate
- h2o-sonar

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:39:31 T+0100**
- Duration: **5.967s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Surrogate Decision Tree

- [Explainer description](#)

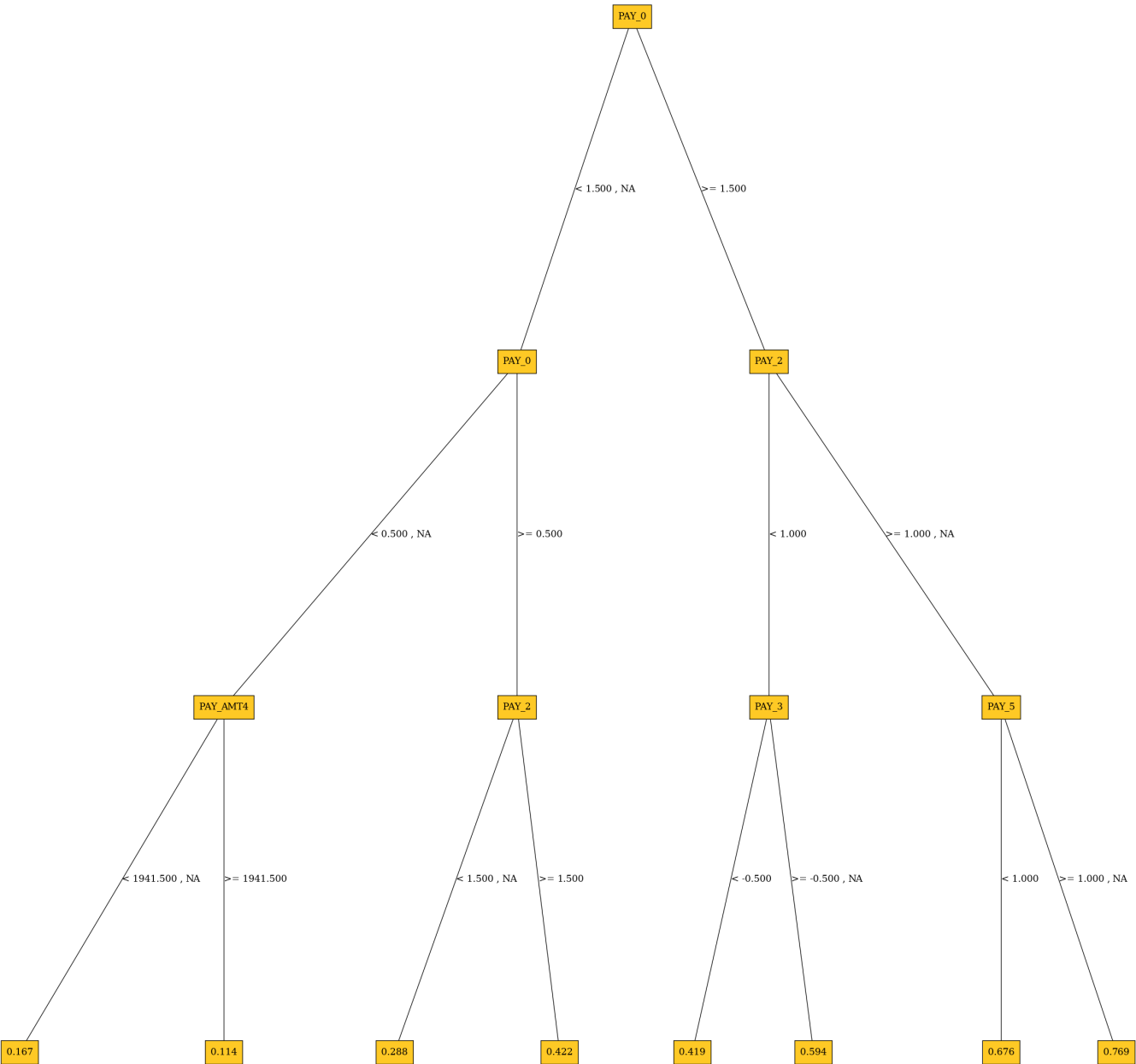
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

The surrogate decision tree is an approximate overall flow chart of the model, created by training a simple decision tree on the original inputs and the predictions of the model.

Explanations

Approximate model behavior for the class '1':



Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types :

- **Decision Tree**
global-decision-tree
 - application/json
- **Local DT**
local-decision-tree
 - application/json
- **Surrogate Decision Tree**
global-html-fragment
 - text/html
- **Decision tree surrogate rules ZIP archive**
global-custom-archive
 - application/zip

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
debug_residuals	False	Debug model residuals.	bool	False
debug_residuals_class		Class for debugging classification model logloss residuals, empty string for debugging regression model residuals.	str	
dt_tree_depth	3	Decision tree depth.	int	3
nfolds	3	Number of CV folds.	int	3
qbin_cols	None	Quantile binning columns.	list	None
qbin_count	0	Quantile bins count.	int	0
categorical_encoding	onehotexplicit	Categorical encoding.	str	onehotexplicit

Explainer metadata

The explainer can explain model types:

- regression
- binomial
- multinomial

Explainer keywords:

- run-by-default
- requires-h2o3
- surrogate
- explains-approximate-behavior
- h2o-sonar

Explainer run

Explainer run details:

- Status code: FINISHED

- Progress: **100%**
- Started: **2023-11-03 11:39:37 T+0100**
- Duration: **1.891s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Shapley Summary Plot for Original Features (Kernel SHAP Method)

- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

Shapley explanations are a technique with credible theoretical support that presents consistent global and local feature contributions.

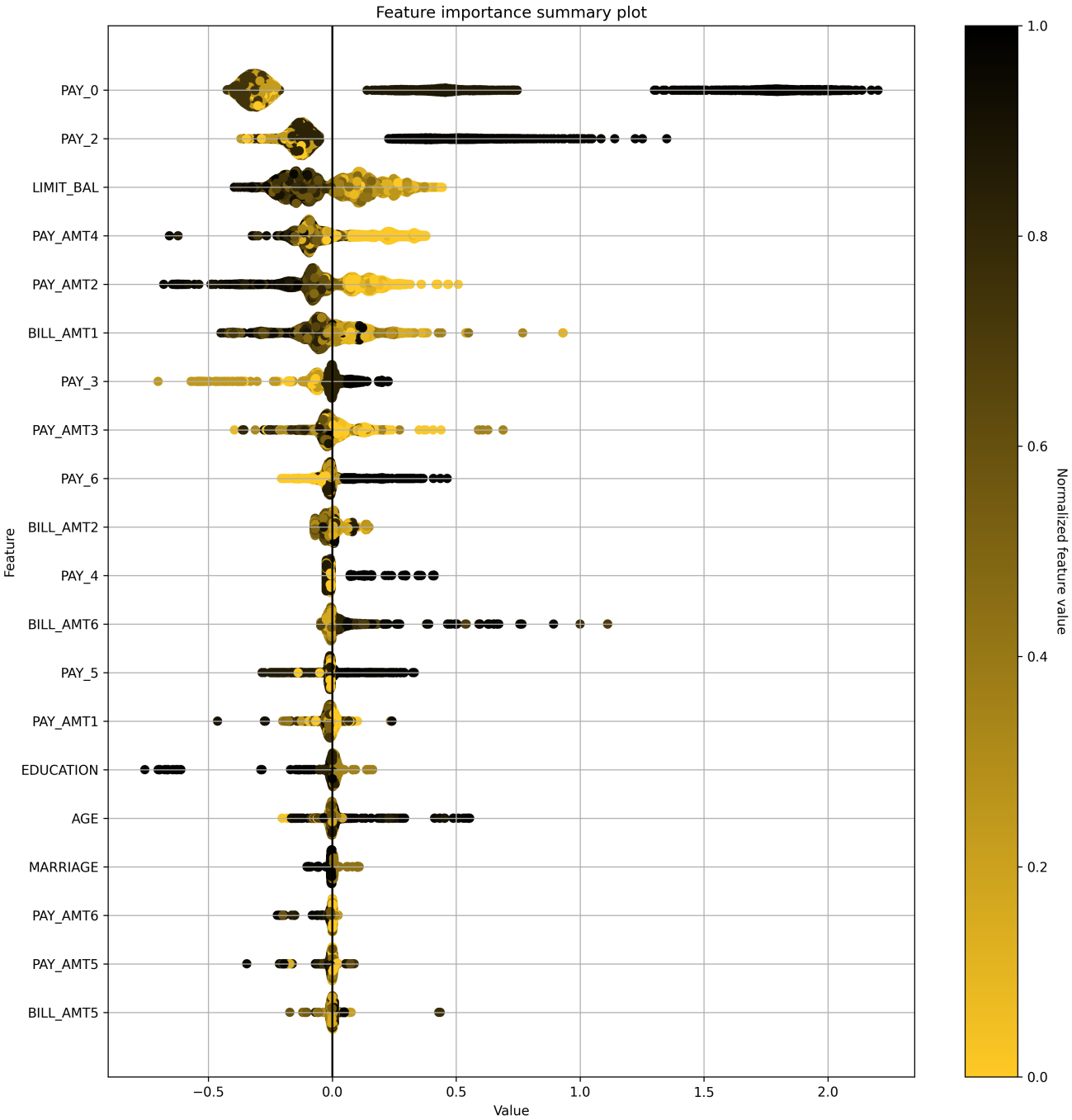
The Shapley Summary Plot shows original features versus their local Shapley values on a sample of the dataset. Feature values are binned by Shapley values and the average normalized feature value for each bin is plotted. The legend corresponds to numeric features and maps to their normalized value - yellow is the lowest value and deep orange is the highest. You can also get a scatter plot of the actual numeric features values versus their corresponding Shapley values. Categorical features are shown in grey and do not provide an actual-value scatter plot.

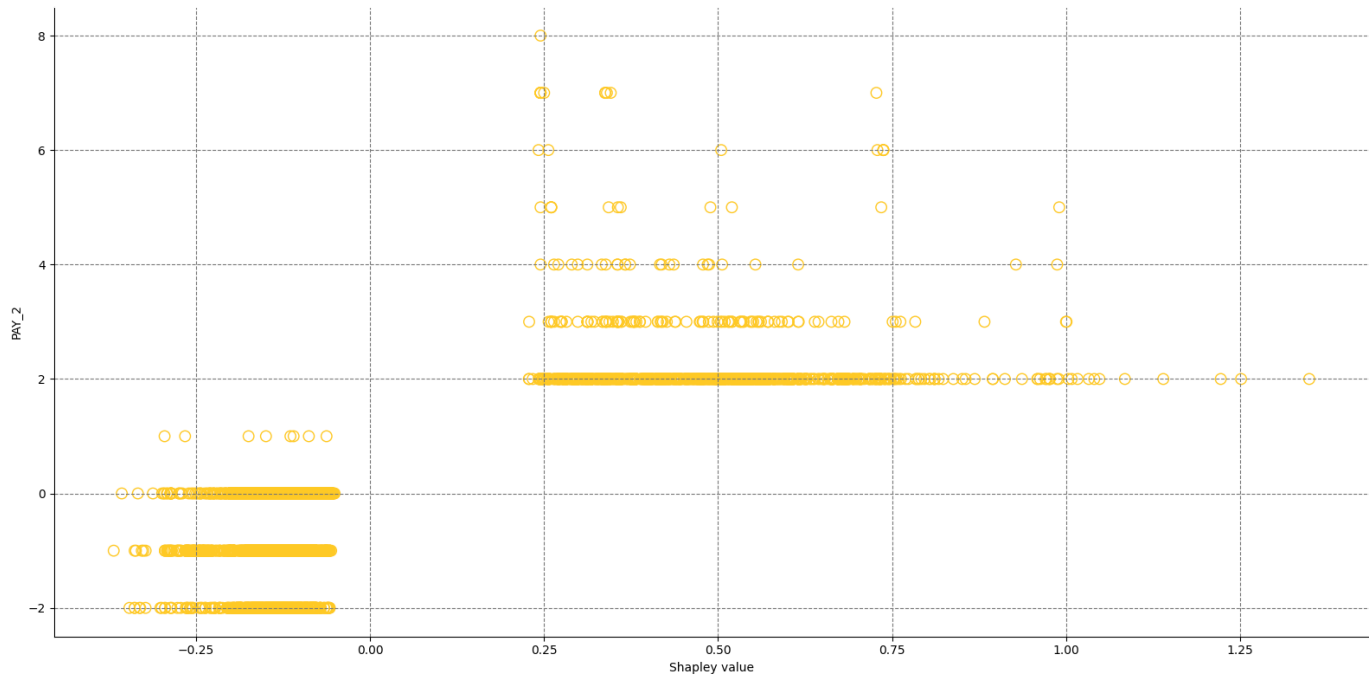
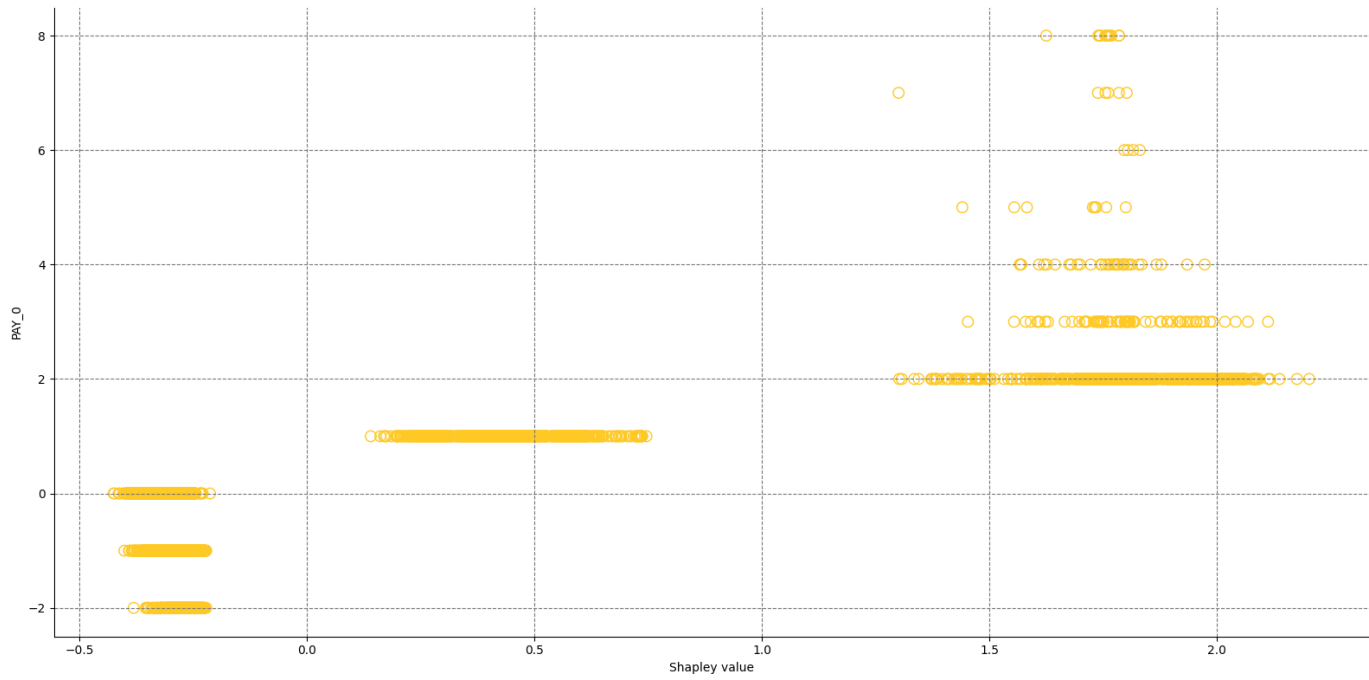
Notes:

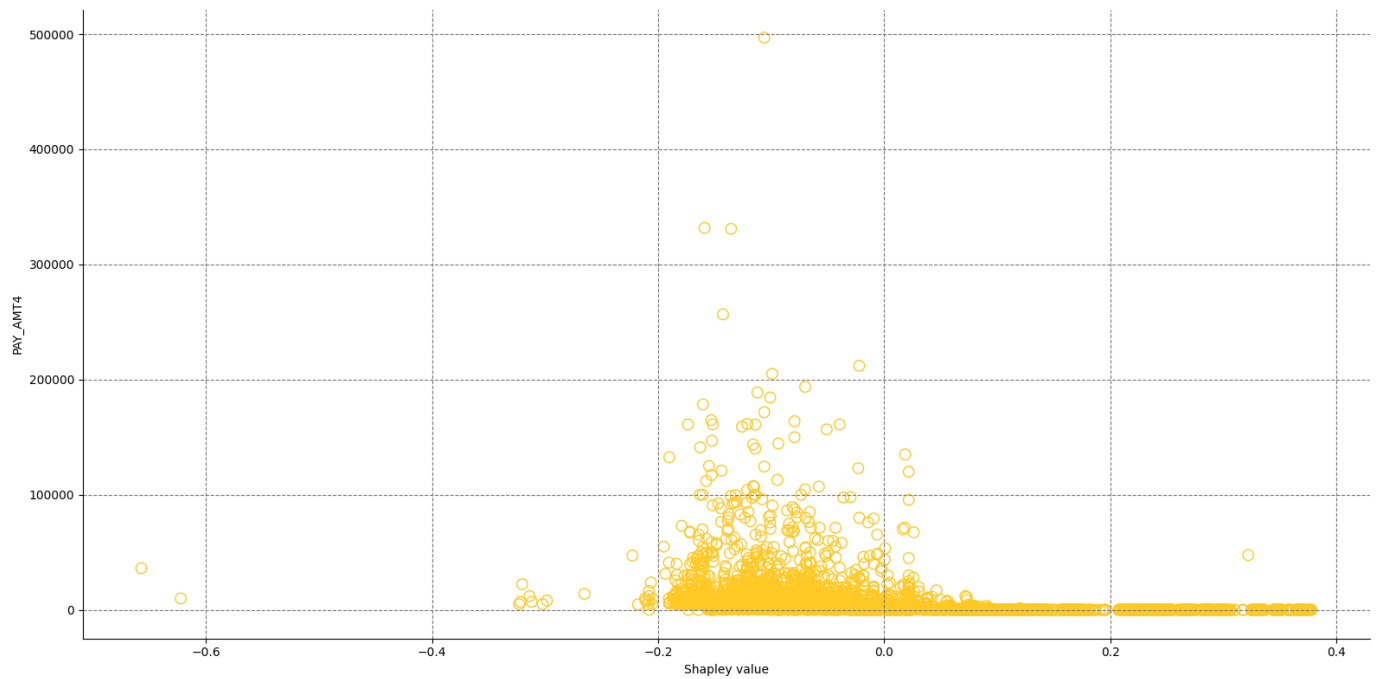
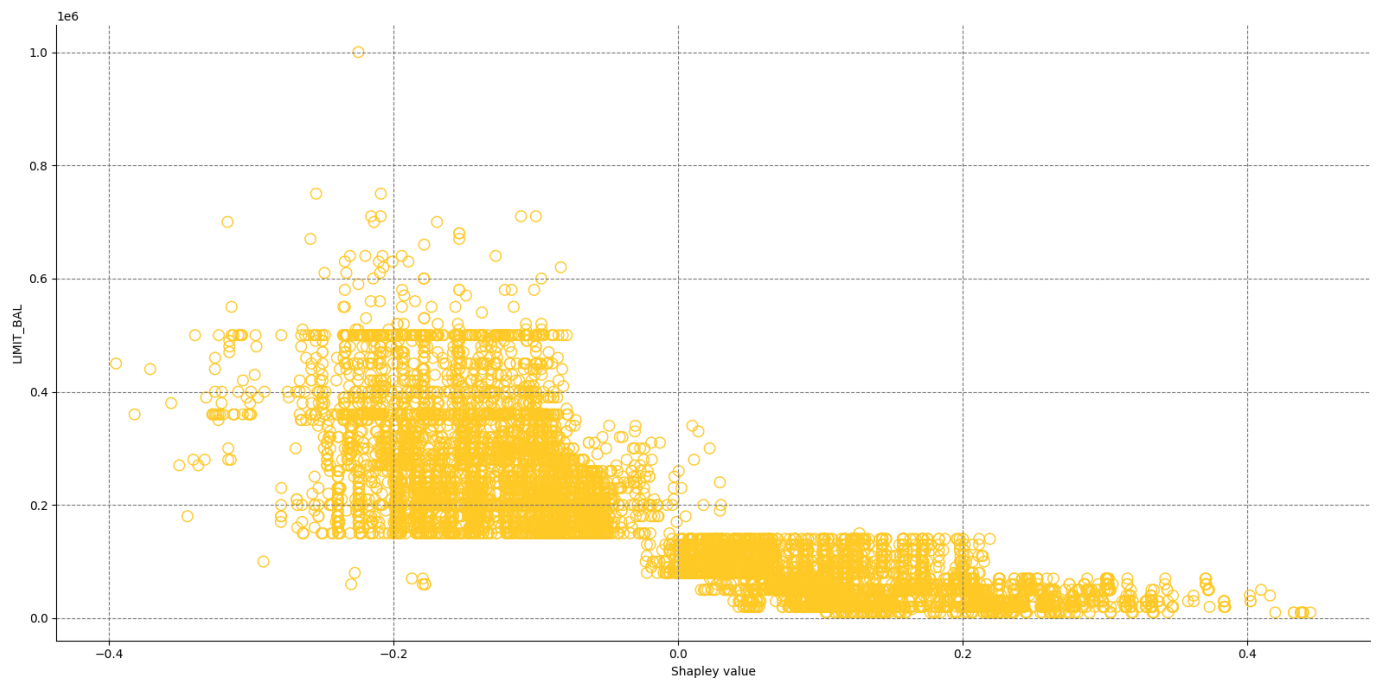
- *The Shapley Summary Plot only shows original features that are used in the model.*
- *The dataset sample size and the number of bins can be updated in the interpretation settings.*

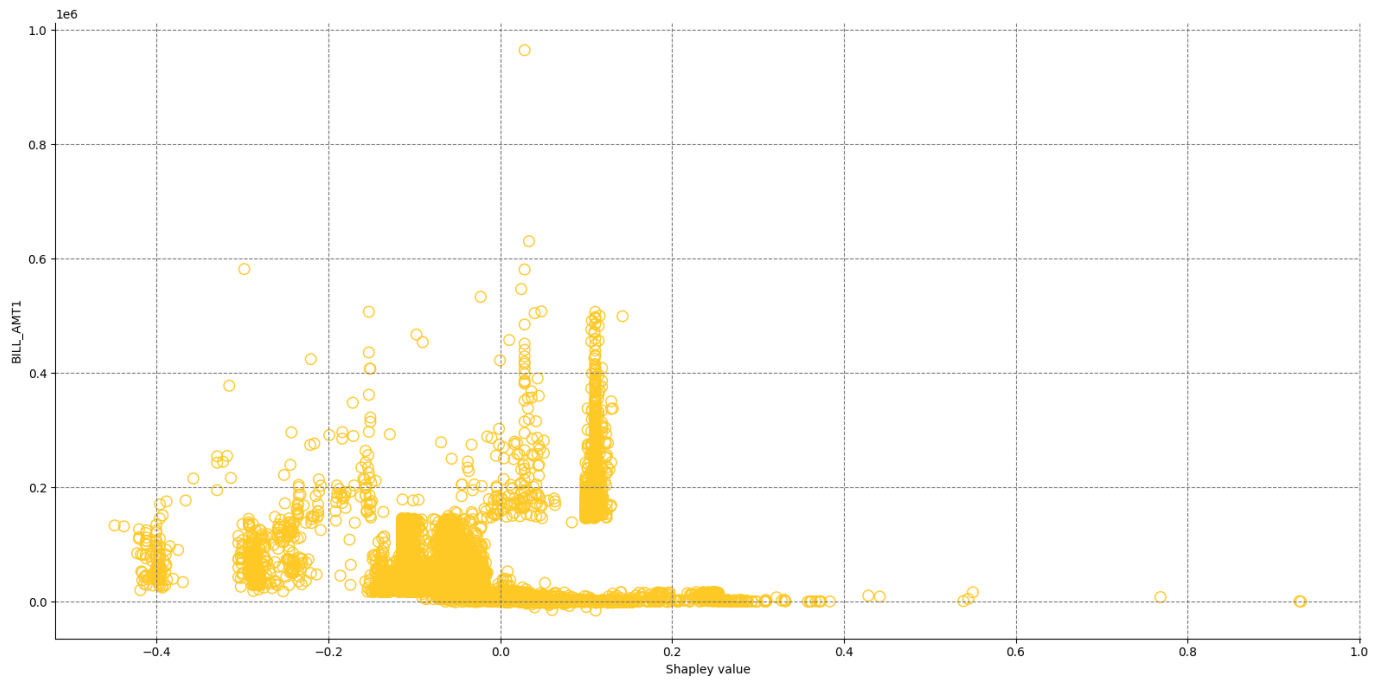
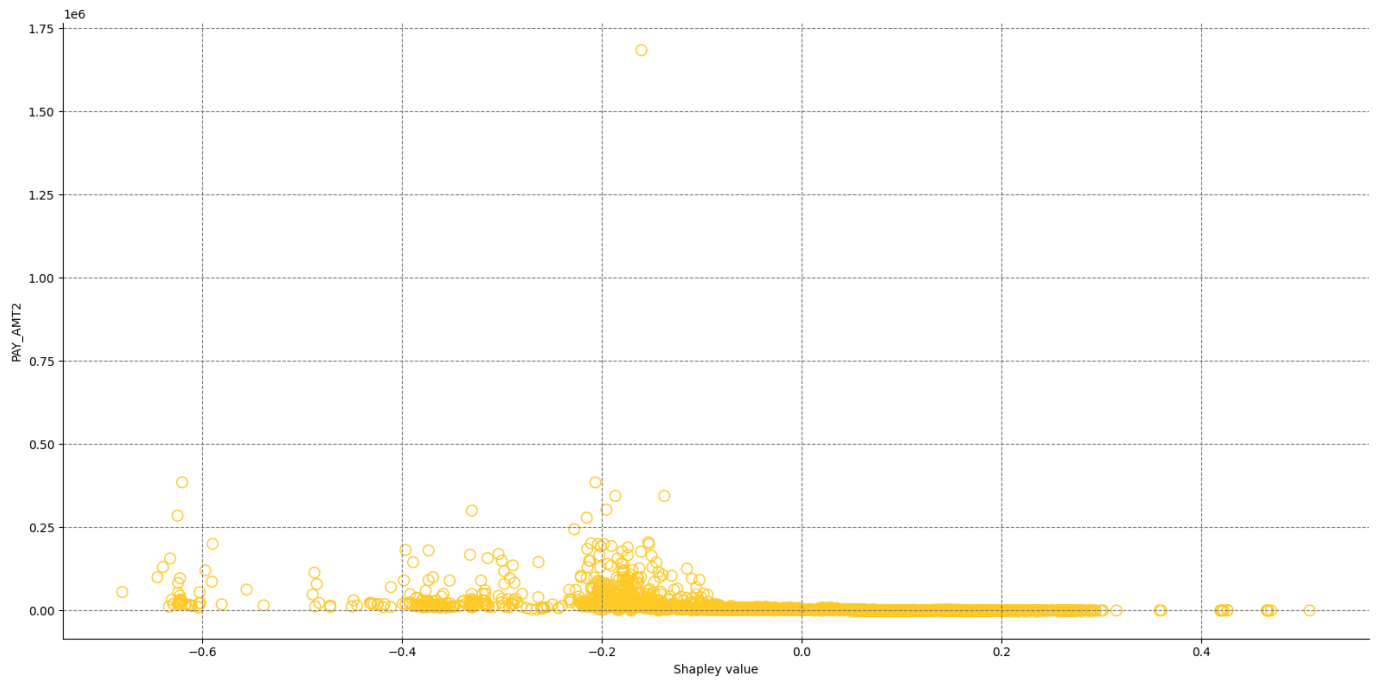
Explanations

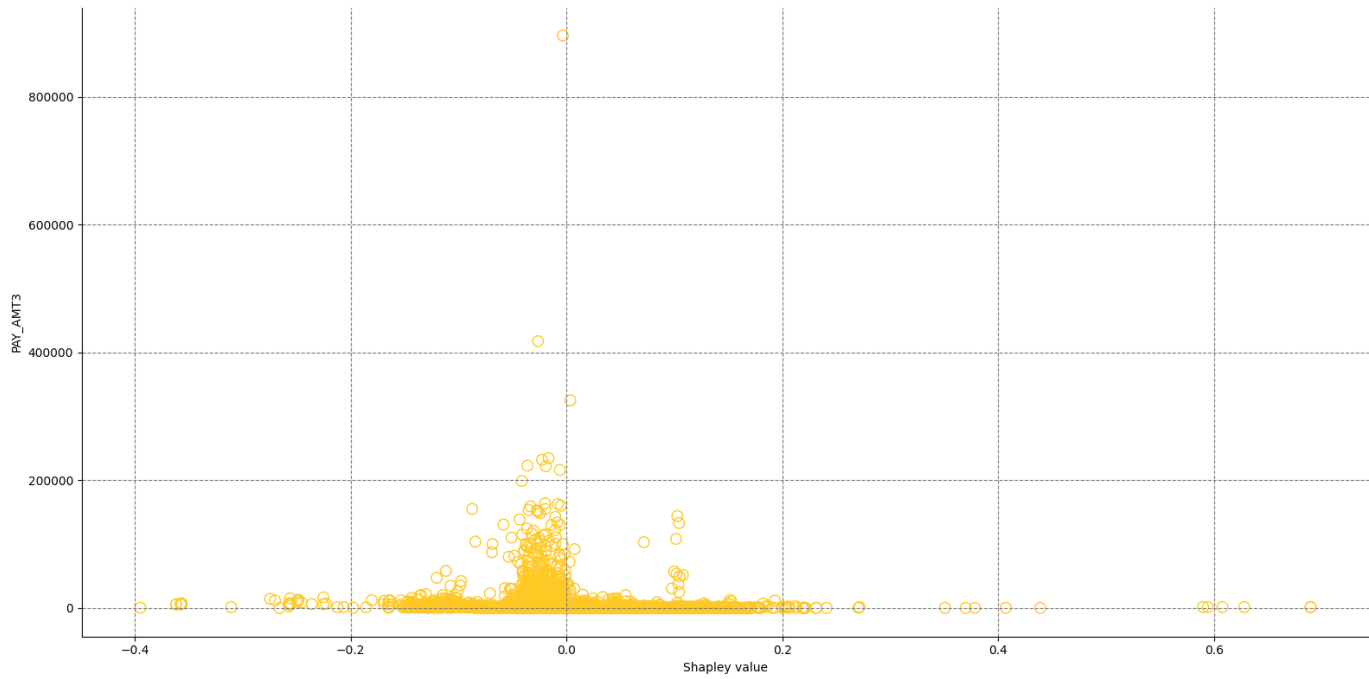
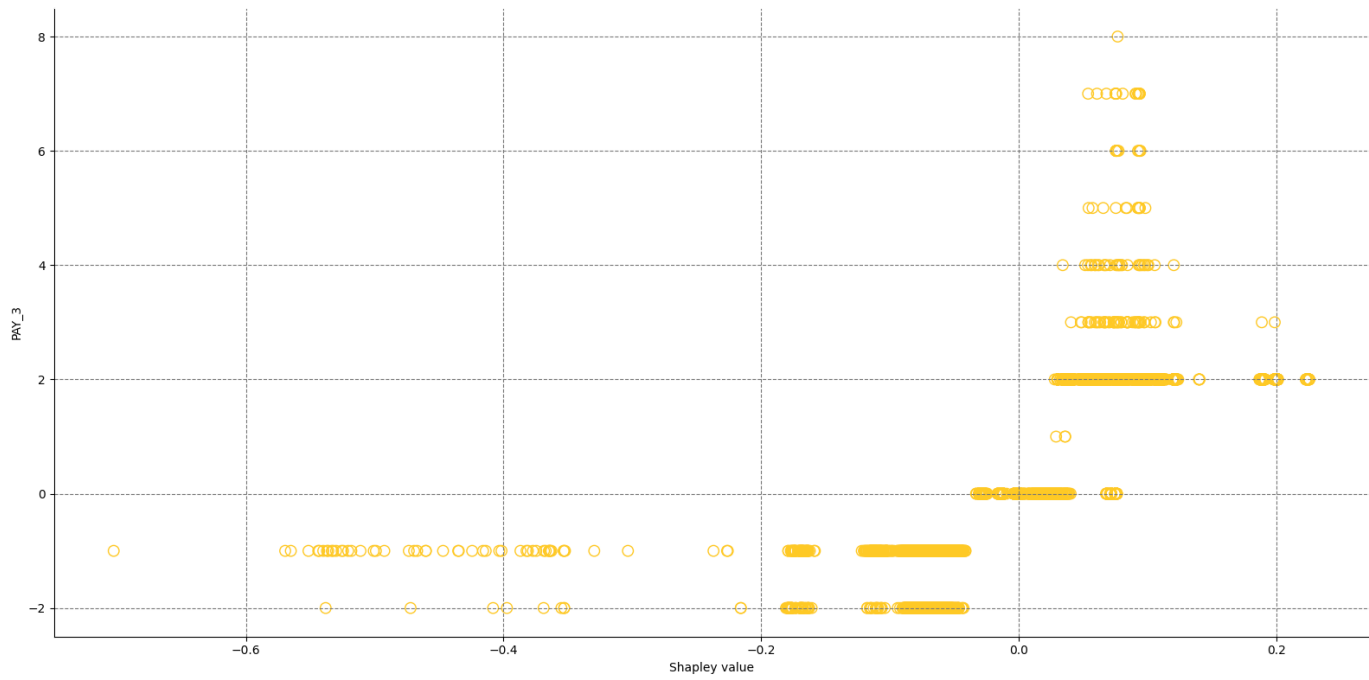
Global Shapley values for original features of class '1':

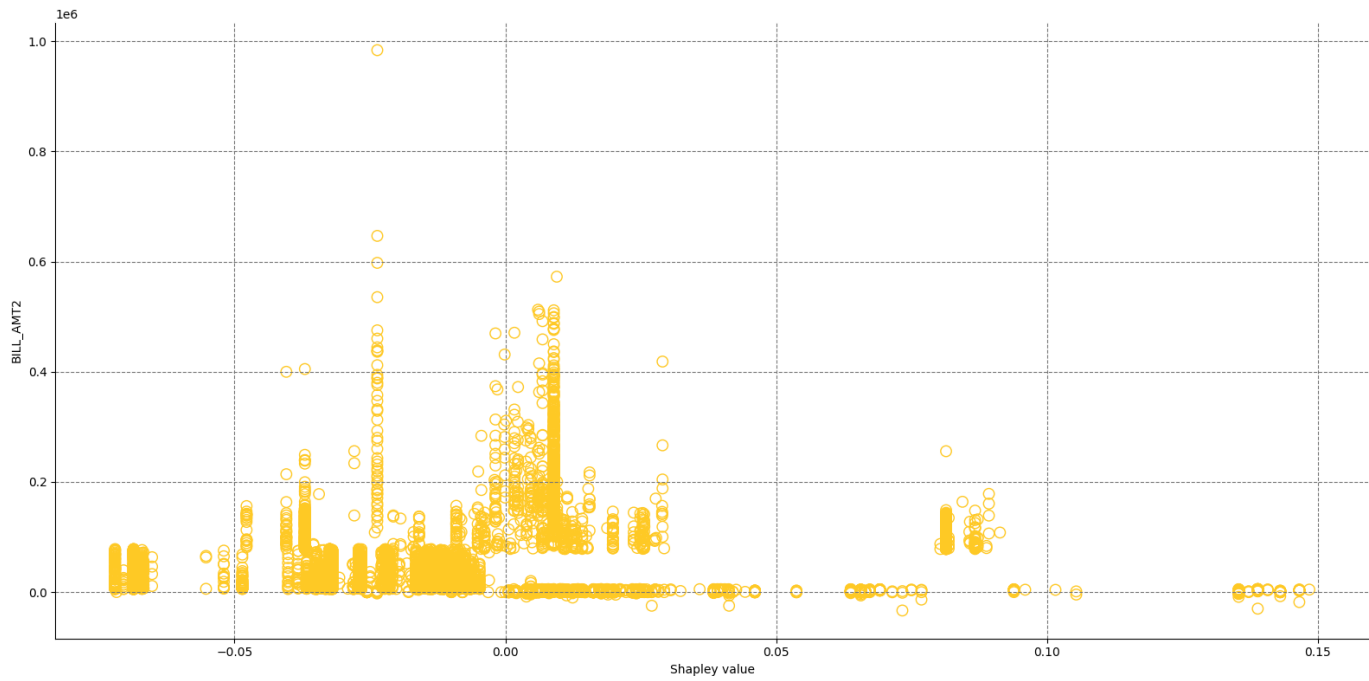
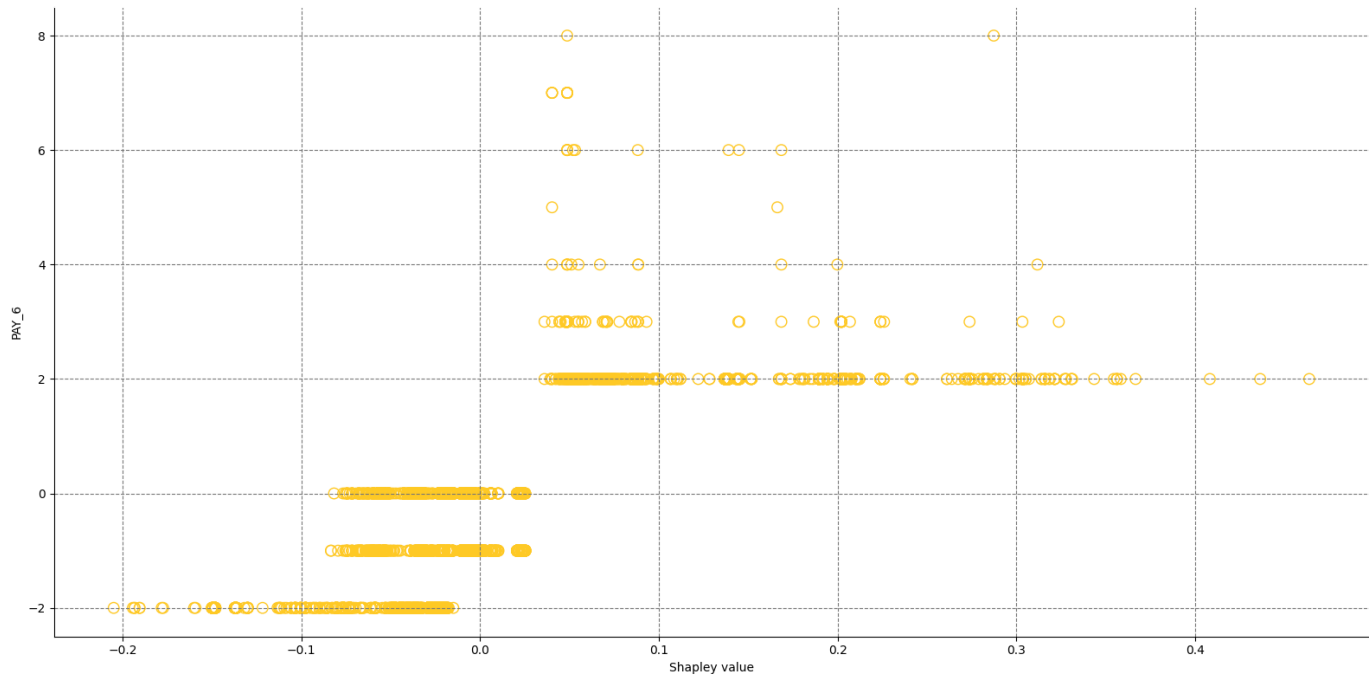


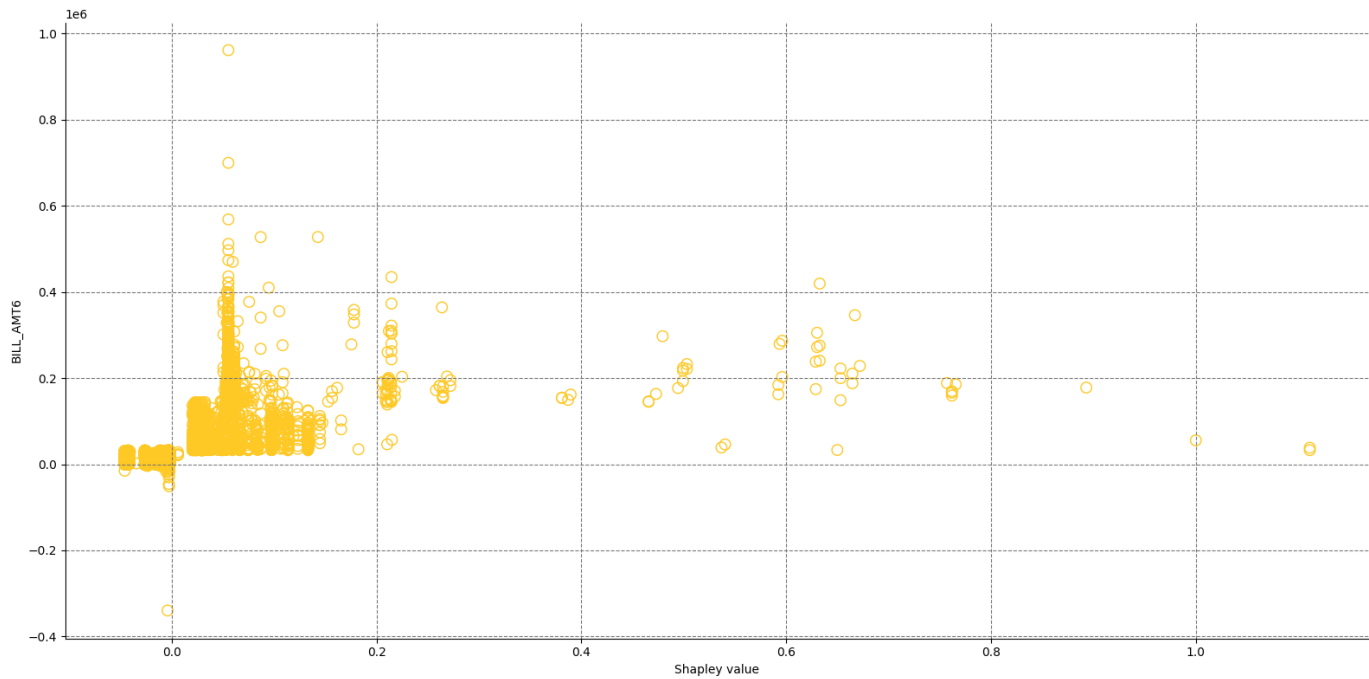
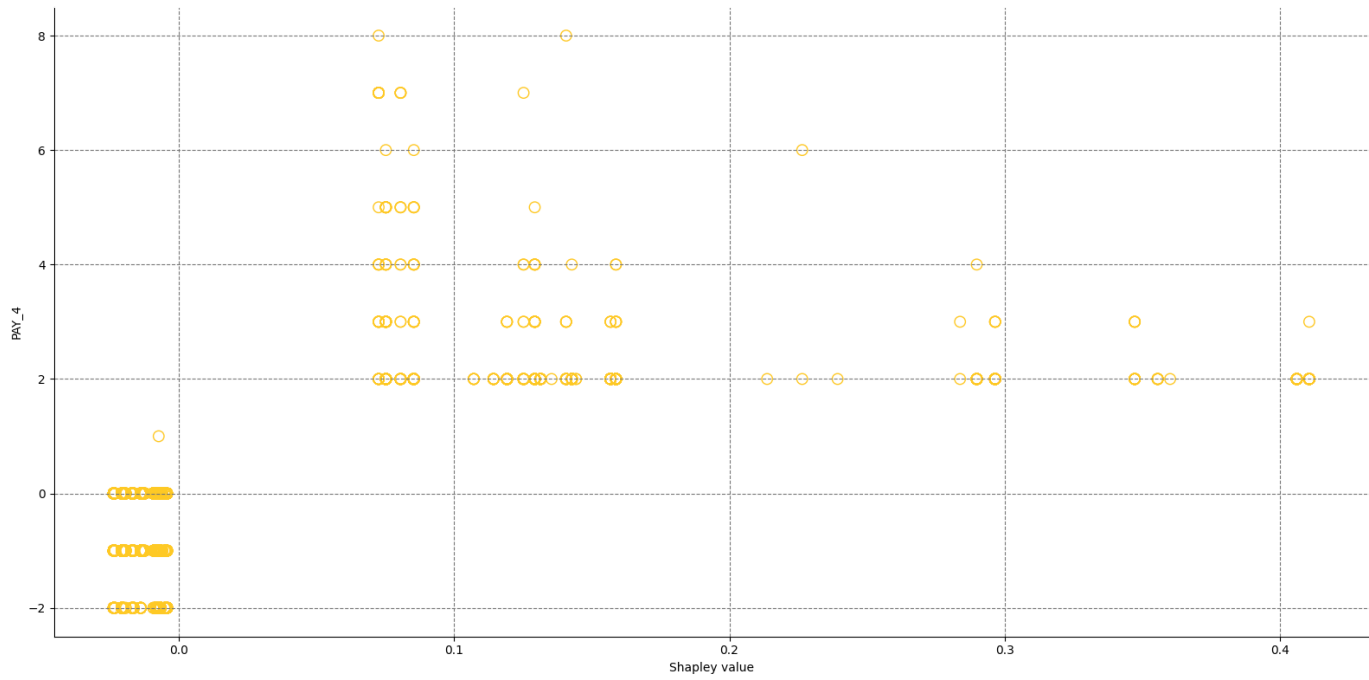


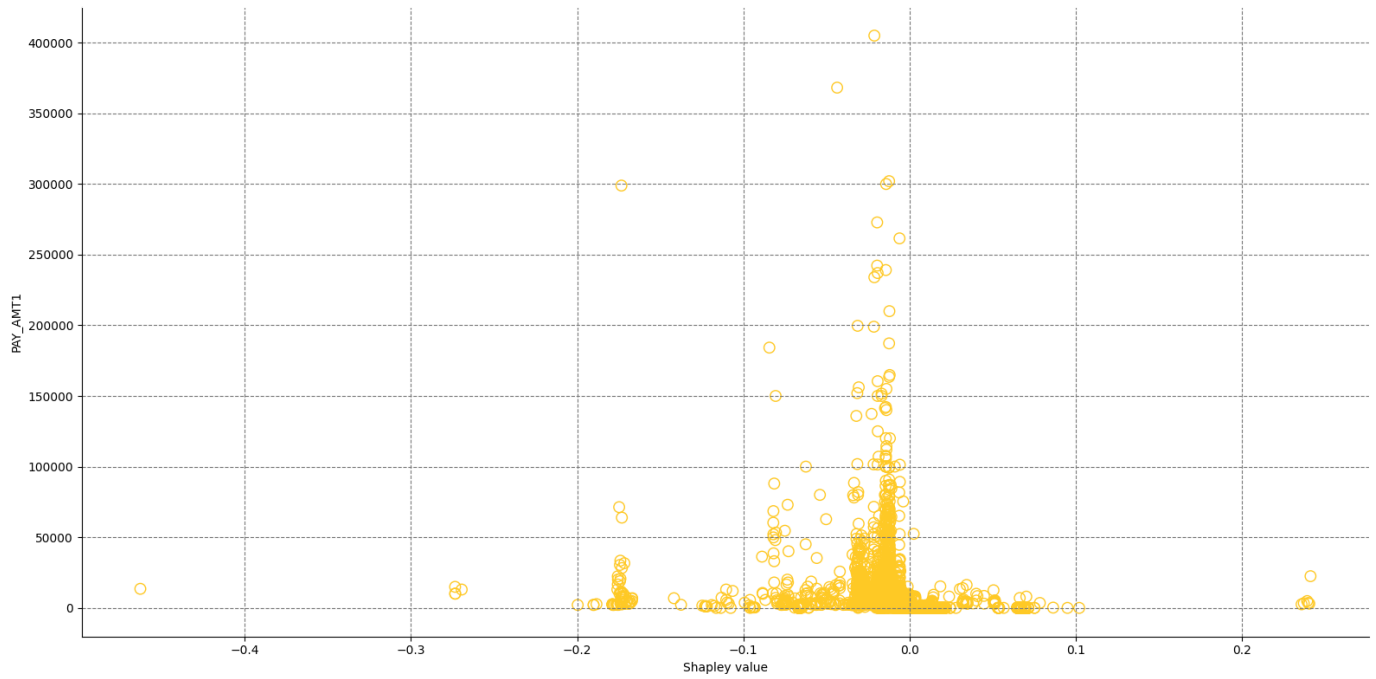
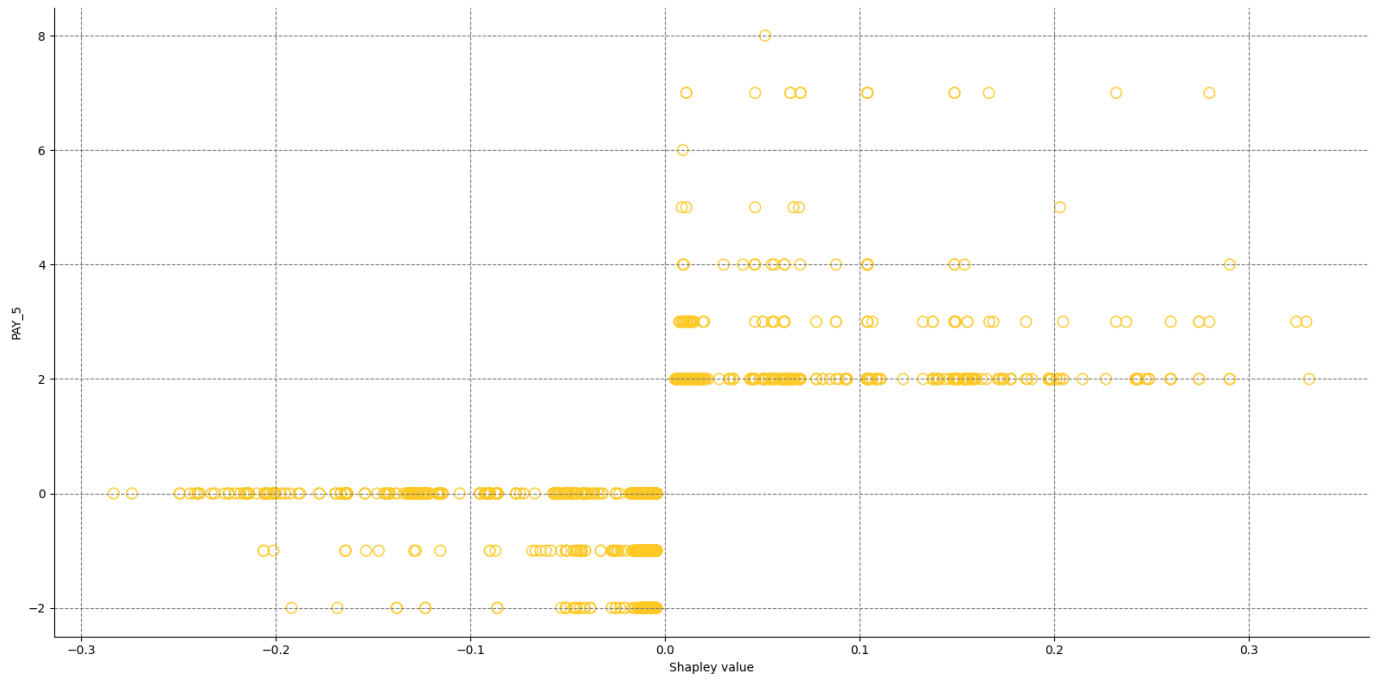


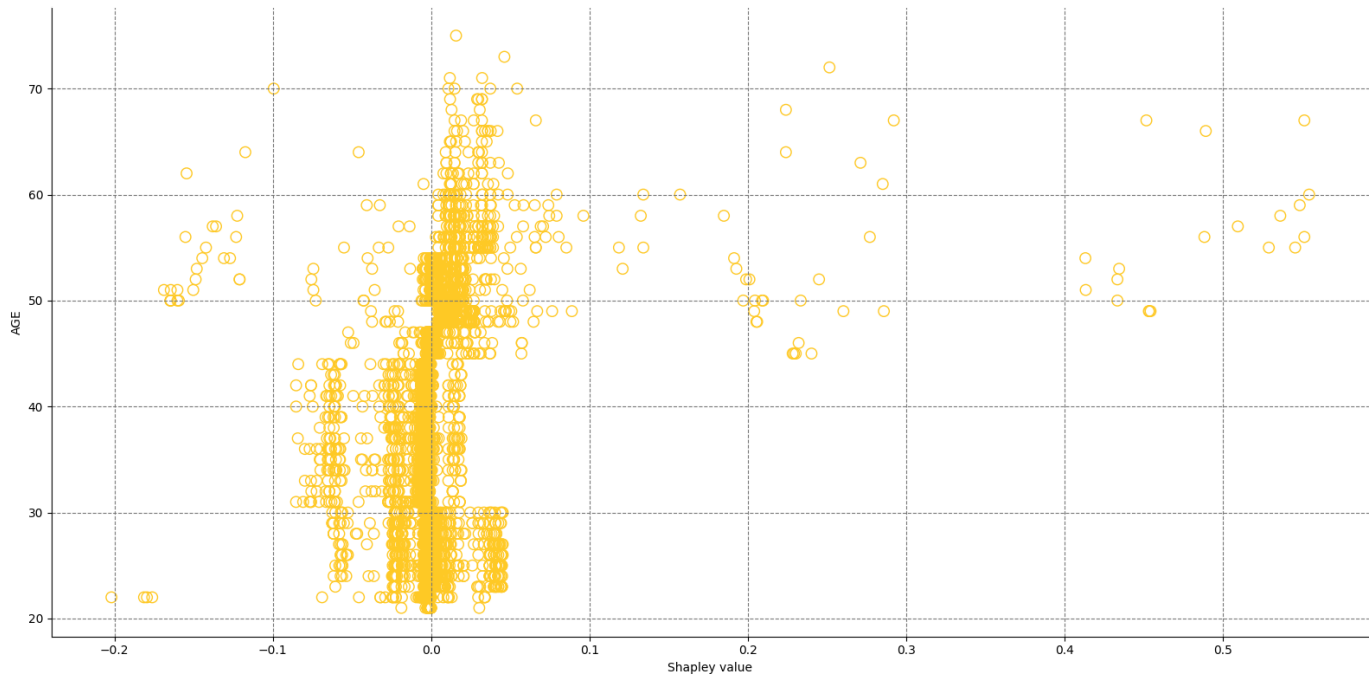
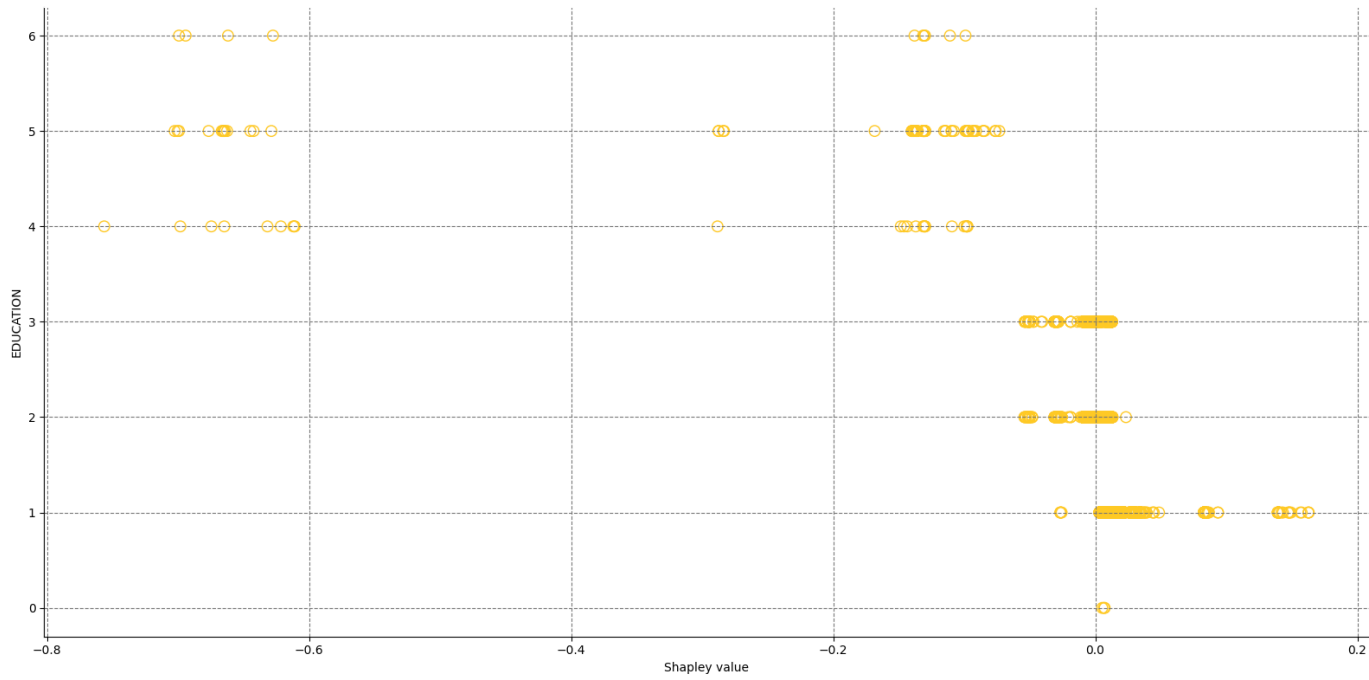


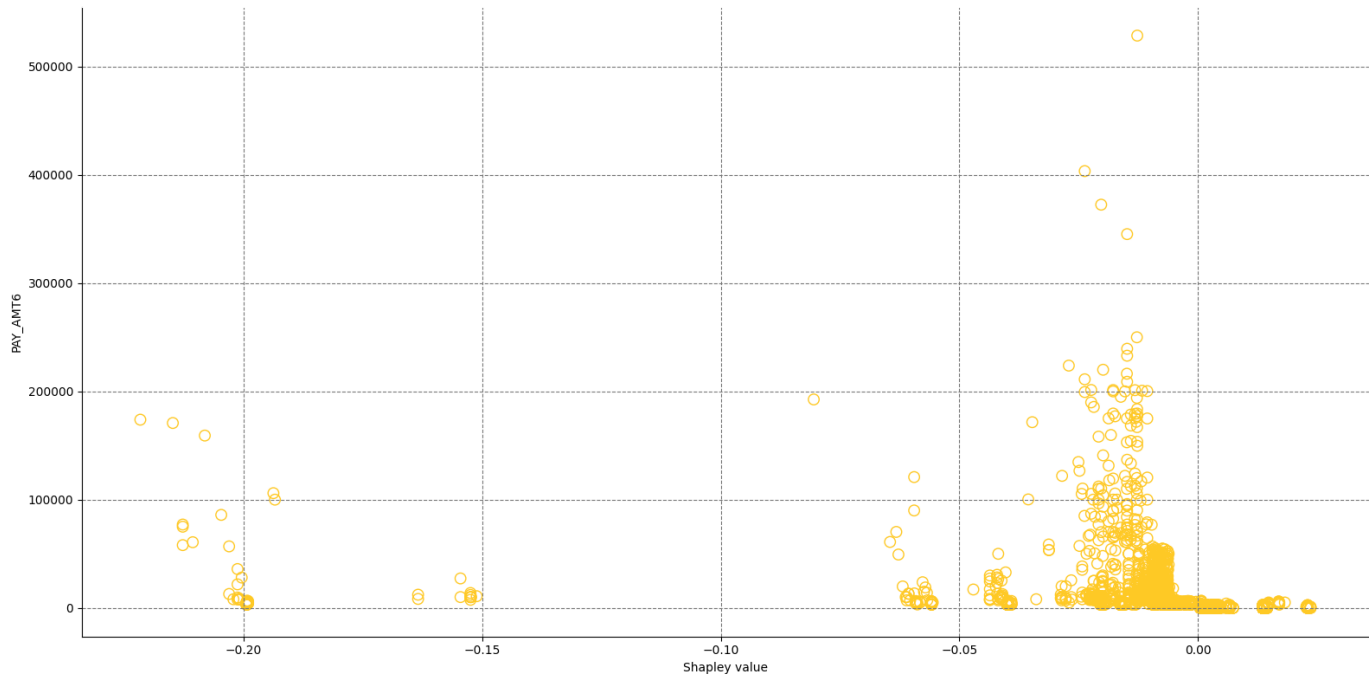
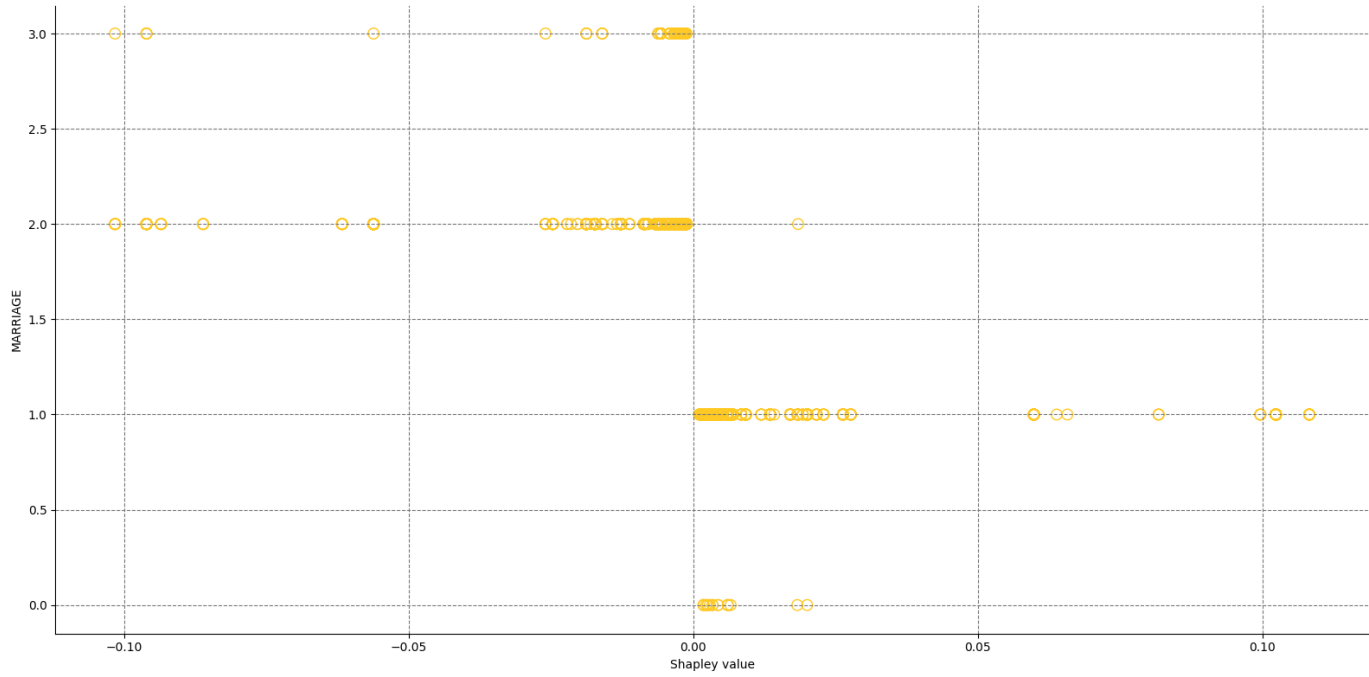


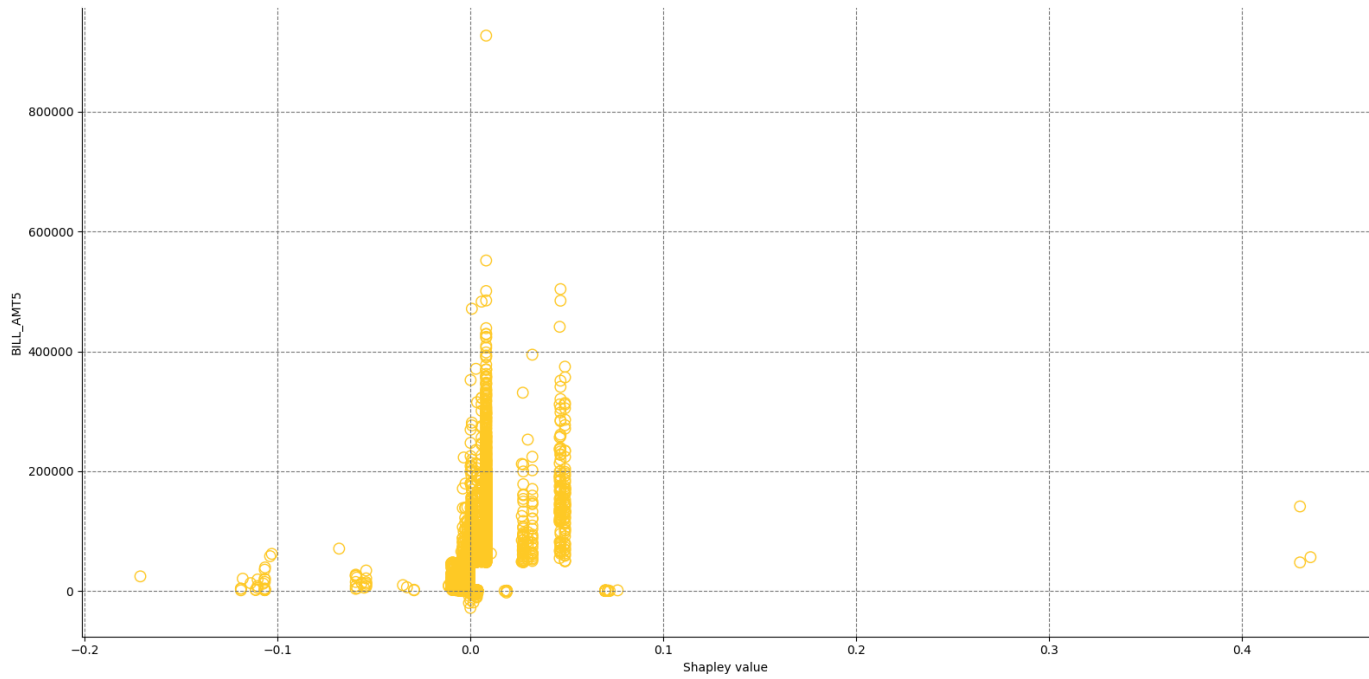
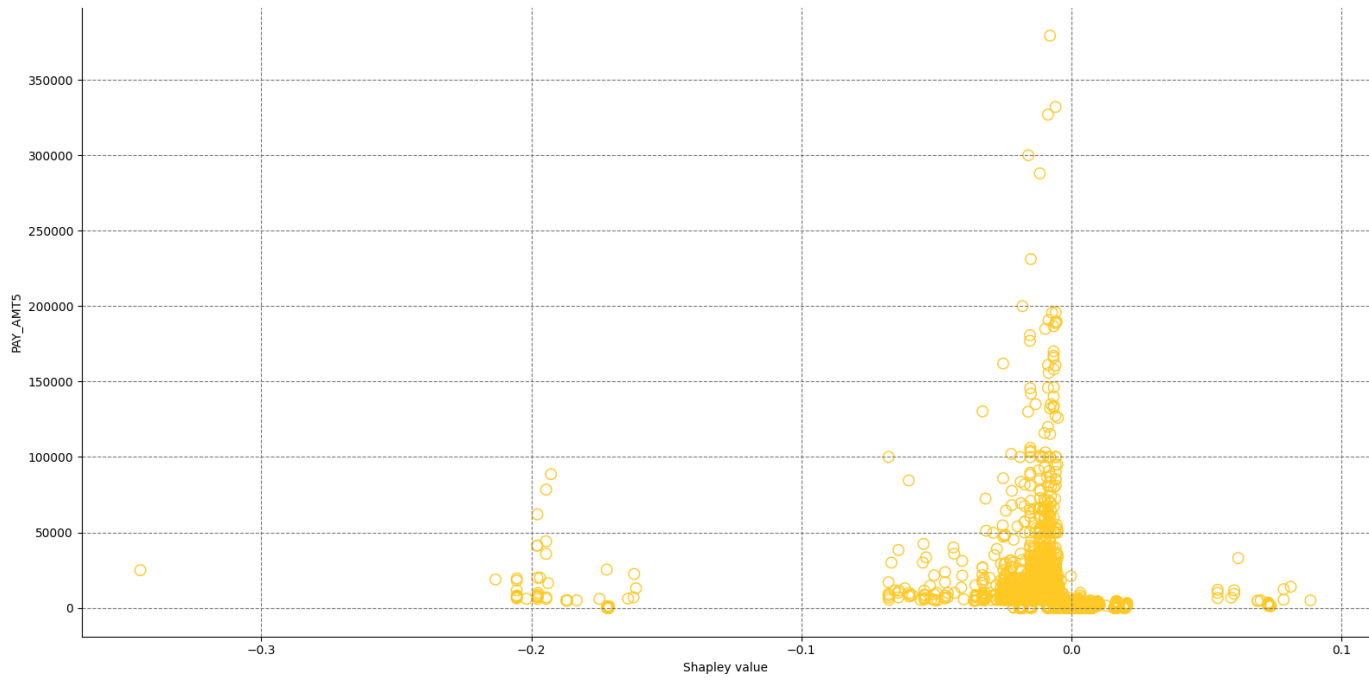


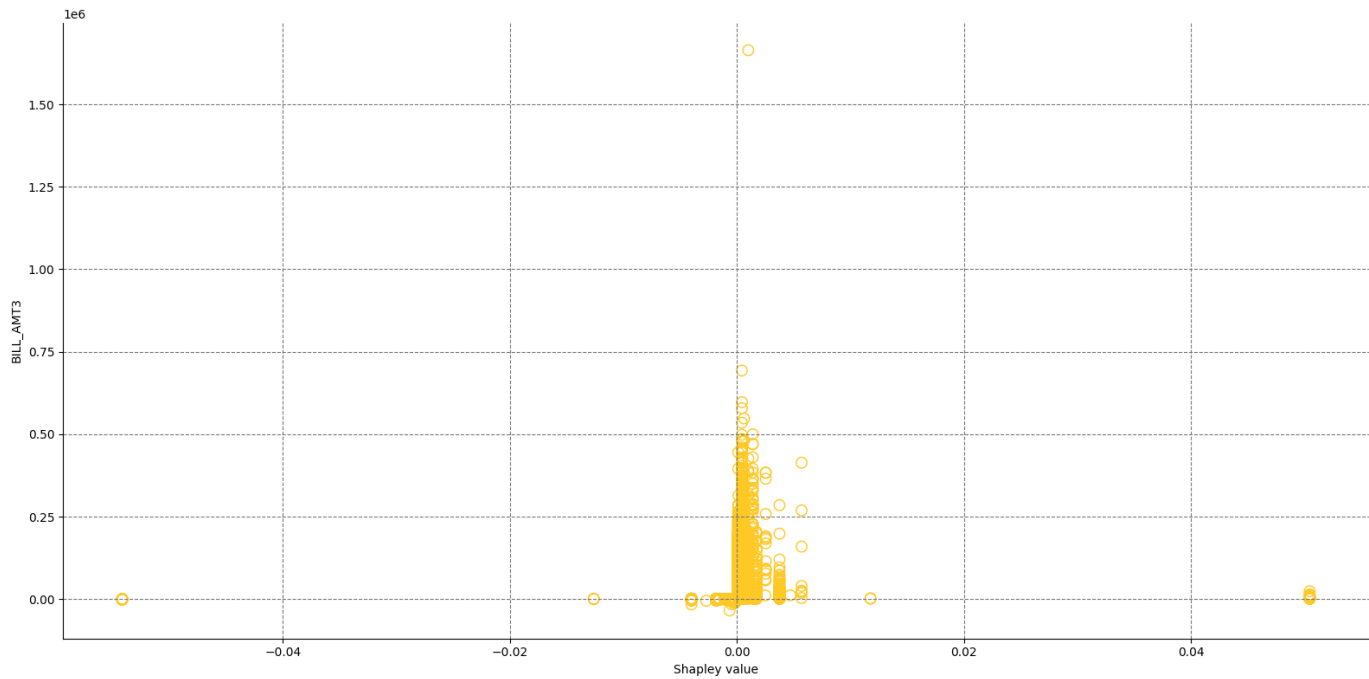
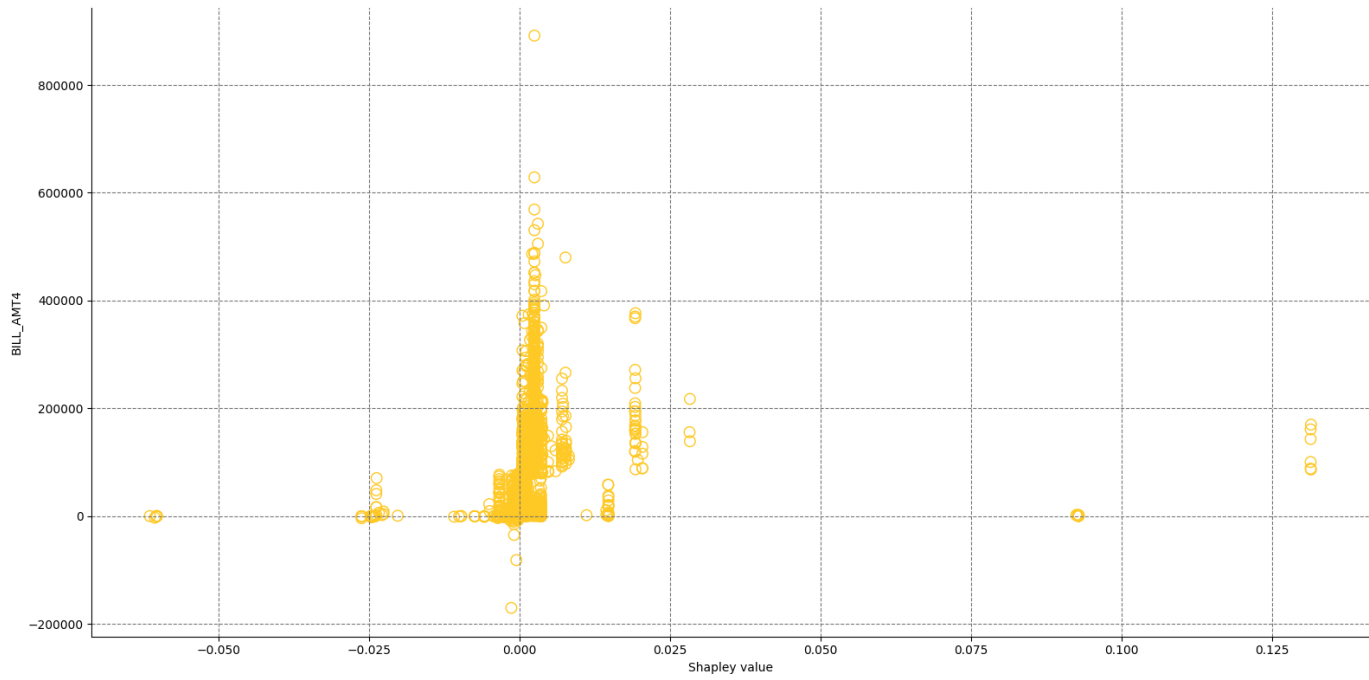


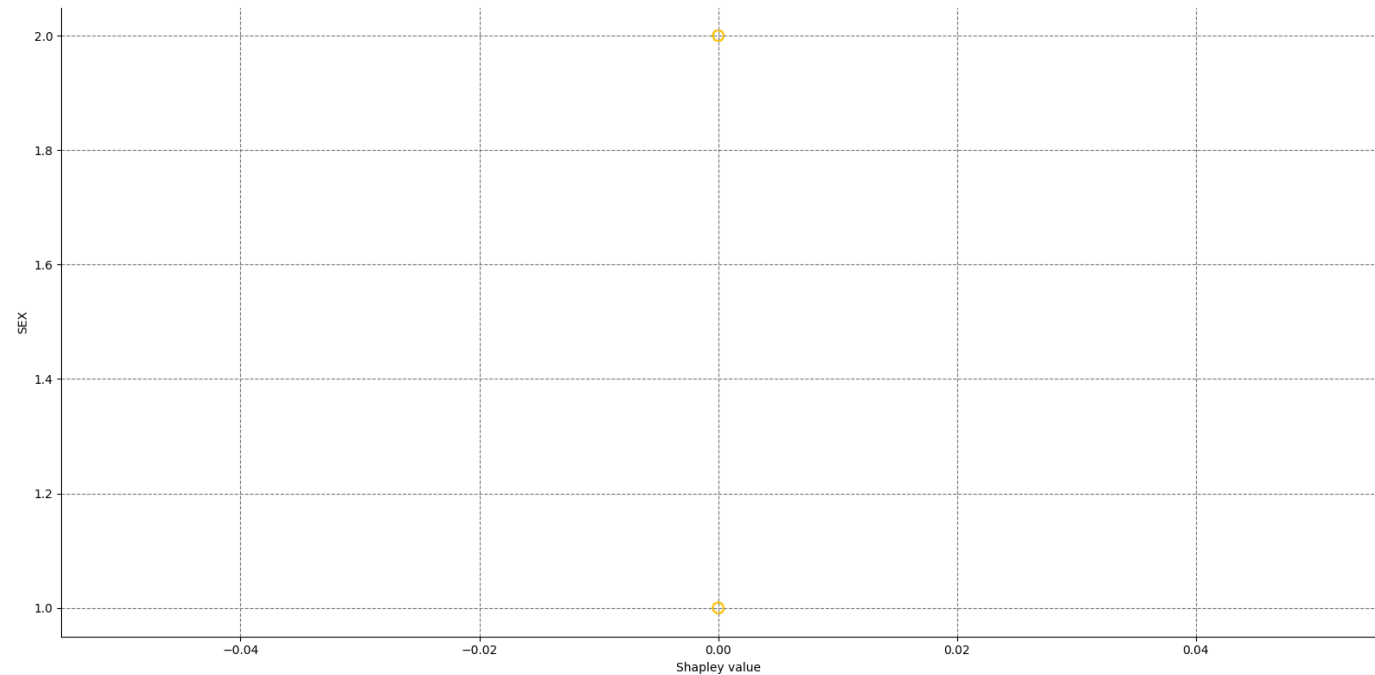












Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types :

- **Shapley Summary Plot for Original Features**
global-summary-feature-importance
 - [application/vnd.h2oai.json+datatable.jay](#)
 - [application/json](#)
 - [text/markdown](#)
- **Shapley Summary Plot for Original Features**
global-html-fragment
 - [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
max_features	50	Maximum number of features to be shown in the plot.	int	50
sample_size	20000	Sample size.	int	20000
x_shapley_resolution	500	x-axis resolution (number of Shapley values bins).	int	500
enable_drilldown_charts	True	Enable creation of per-feature Shapley/feature value scatter plots.	bool	True
fast_approx_contribs	True	Speed up predictions with fast predictions and contributions approximations.	bool	True

Explainer metadata

The explainer can explain model types:

- regression

- binomial
- multinomial

Explainer keywords:

- run-by-default
- explains-feature-behavior
- h2o-sonar

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:39:39 T+0100**
- Duration: **12.327s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Partial Dependence Plot

- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

Partial dependence plot (PDP) portrays the average prediction behavior of the model across the domain of an input variable along with +/- 1 standard deviation bands. Individual Conditional Expectations plot (ICE) displays the prediction behavior for an individual row of data when an input variable is toggled across its domain.

PD binning:

Integer feature:

- bins in **numeric** mode:
 - bins are integers
 - (at most) **grid_resolution** integer values in between minimum and maximum of feature values
 - bin values are created as evenly as possible
 - minimum and maximum is included in bins (if **grid_resolution** is bigger or equal to 2)
- bins in **categorical** mode:
 - bins are integers
 - top **grid_resolution** values from feature values ordered by frequency (int values are converted to strings and most frequent values are used as bins)
- quantile bins in **numeric** mode:
 - bins are integers
 - bin values are created with the aim that there will be the same number of observations per bin

- *q*-quantile used to created *q* bins where *q* is specified by PD parameter
- quantile bins in **categorical** mode:
 - not supported

Float feature:

- bins in **numeric** mode:
 - bins are floats
 - **grid_resolution** float values in between minimum and maximum of feature values
 - bin values are created as evenly as possible
 - minimum and maximum is included in bins (if **grid_resolution** is bigger or equal to 2)
- bins in **categorical** mode:
 - bins are floats
 - top **grid_resolution** values from feature values ordered by frequency (float values are converted to strings and most frequent values are used as bins)
- quantile bins in **numeric** mode:
 - bins are floats
 - bin values are created with the aim that there will be the same number of observations per bin
 - *q*-quantile used to created *q* bins where *q* is specified by PD parameter
- quantile bins in **categorical** mode:
 - not supported

String feature:

- bins in **numeric** mode:
 - not supported
- bins in **categorical** mode:
 - bins are strings
 - top **grid_resolution** values from feature values ordered by frequency
- quantile bins:
 - not supported

Date/datetime feature:

- bins in **numeric** mode:
 - bins are dates
 - **grid_resolution** date values in between minimum and maximum of feature values
 - bin values are created as evenly as possible:
 1. dates are parsed and converted to epoch timestamps i.e integers
 2. bins are created as in case of numeric integer binning
 3. integer bins are converted back to original date format
 - minimum and maximum is included in bins (if **grid_resolution** is bigger or equal to 2)
- bins in **categorical** mode:
 - bins are dates
 - top **grid_resolution** values from feature values ordered by frequency (dates are handled as opaque strings and most frequent values are used as bins)
- quantile bins:
 - not supported

PD out of range binning:**Integer** feature:

- OOR bins in **numeric** mode:
 - OOR bins are integers
 - (at most) **oor_grid_resolution** integer values are added below minimum and above maximum
 - bin values are created by adding/subtracting rounded standard deviation (of feature values) above and below maximum and minimum **oor_grid_resolution** times
 - 1 used if rounded standard deviation would be 0
 - if feature is of unsigned integer type, then bins below 0 are not created

- if rounded standard deviation and/or **oor_grid_resolution** is so high that it would cause lower OOR bins to be negative numbers, then standard deviation of size 1 is tried instead
- OOR bins in **categorical** mode:
 - same as numeric mode

Float feature:

- OOR bins in **numeric** mode:
 - OOR bins are floats
 - **oor_grid_resolution** float values are added below minimum and above maximum
 - bin values are created by adding/subtracting standard deviation (of feature values) above and below maximum and minimum **oor_grid_resolution** times
- OOR bins in **categorical** mode:
 - same as numeric mode

String feature:

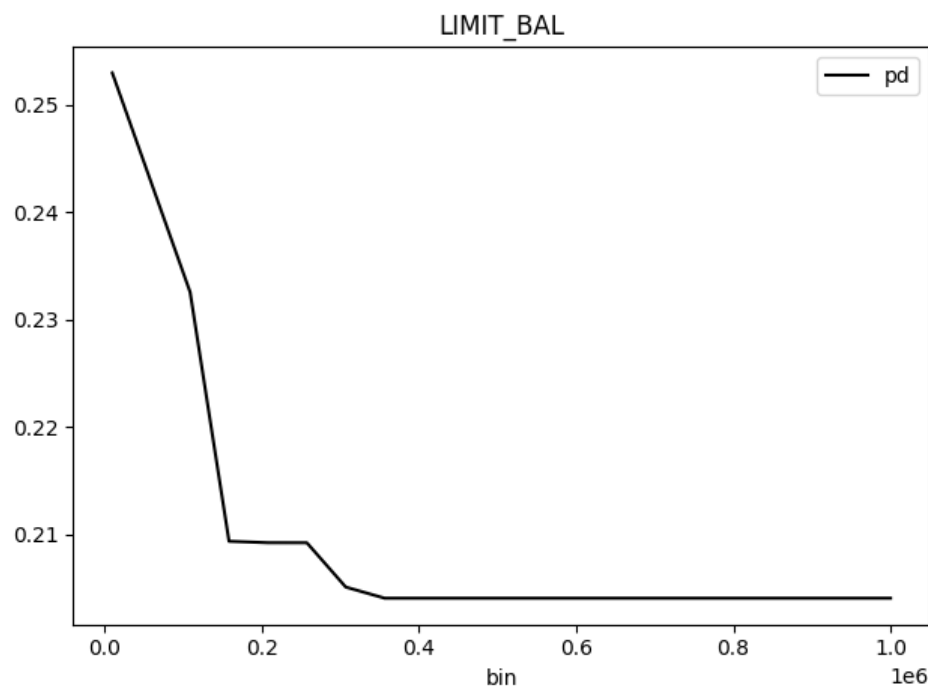
- bins in **numeric** mode:
 - not supported
- bins in **categorical** mode:
 - OOR bins are strings
 - value **UNSEEN** is added as OOR bin

Date feature:

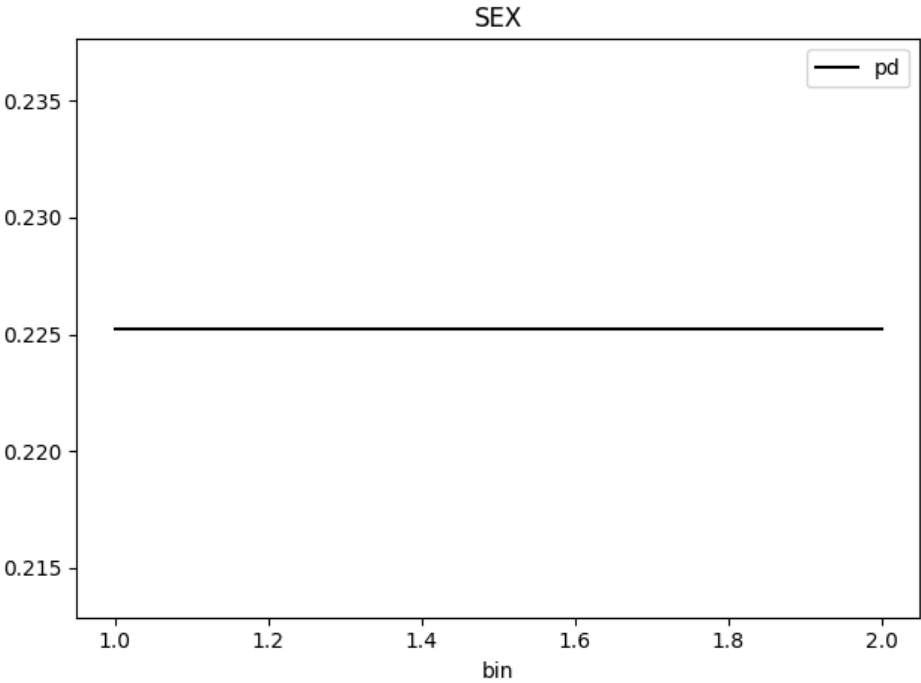
- bins in **numeric** mode:
 - not supported
- bins in **categorical** mode:
 - OOR bins are strings
 - value **UNSEEN** is added as OOR bin

Explanations

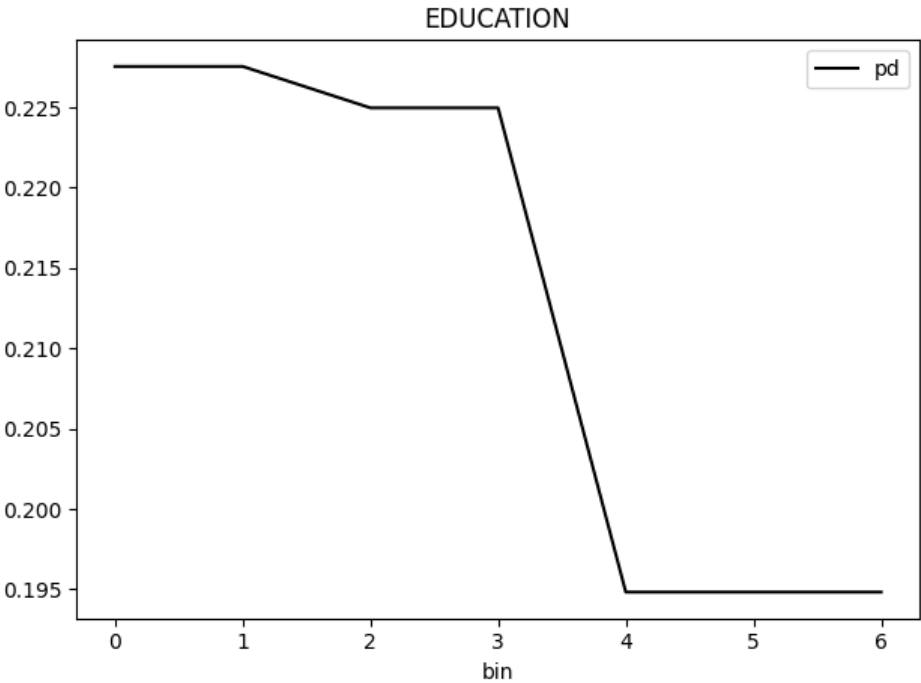
Partial Dependence Plot for the feature 'LIMIT_BAL' and class '1':



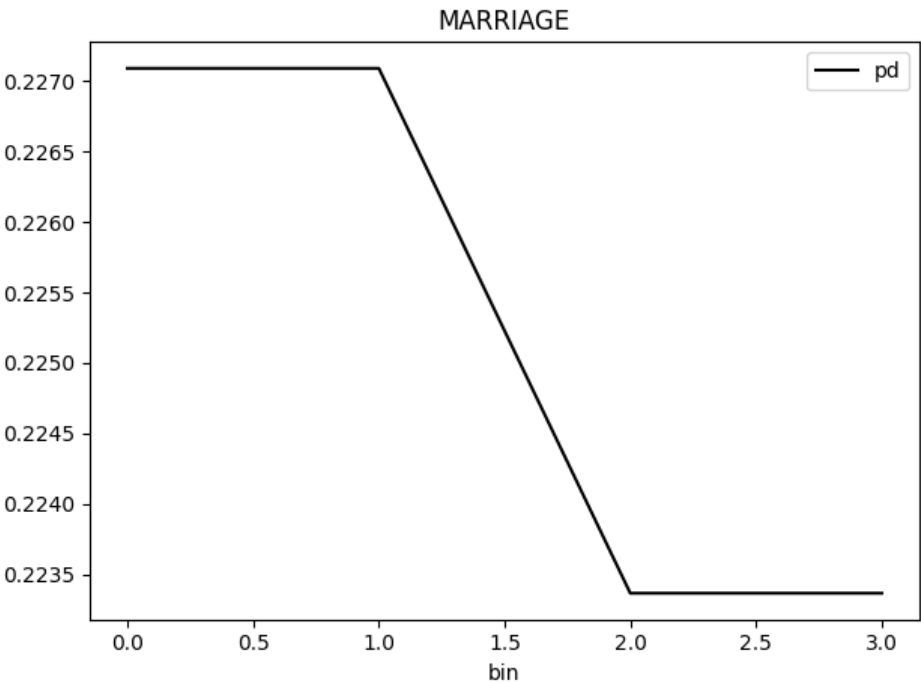
Partial Dependence Plot for the feature 'SEX' and class '1':



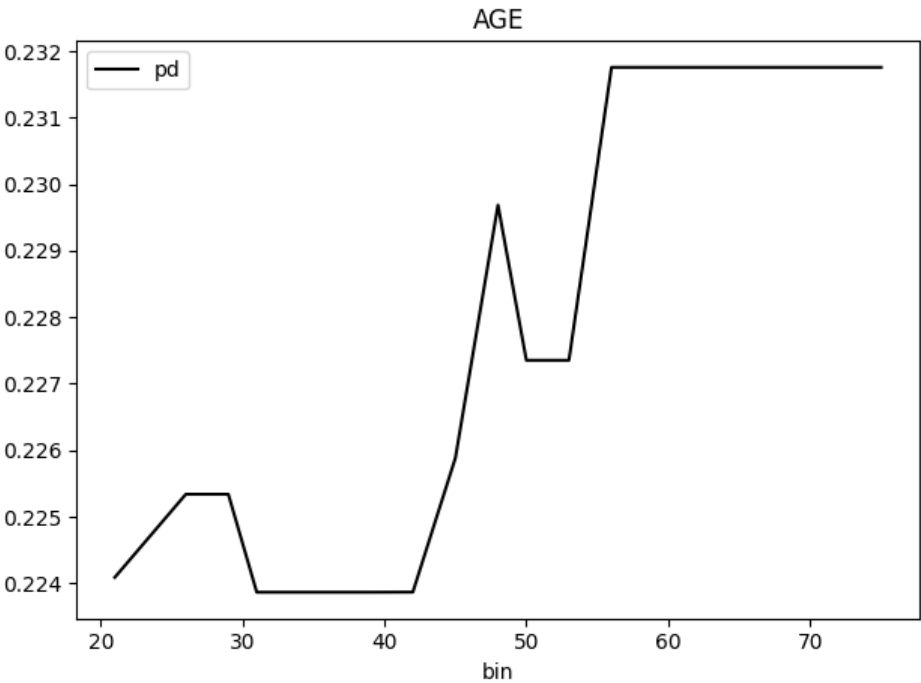
Partial Dependence Plot for the feature 'EDUCATION' and class '1':



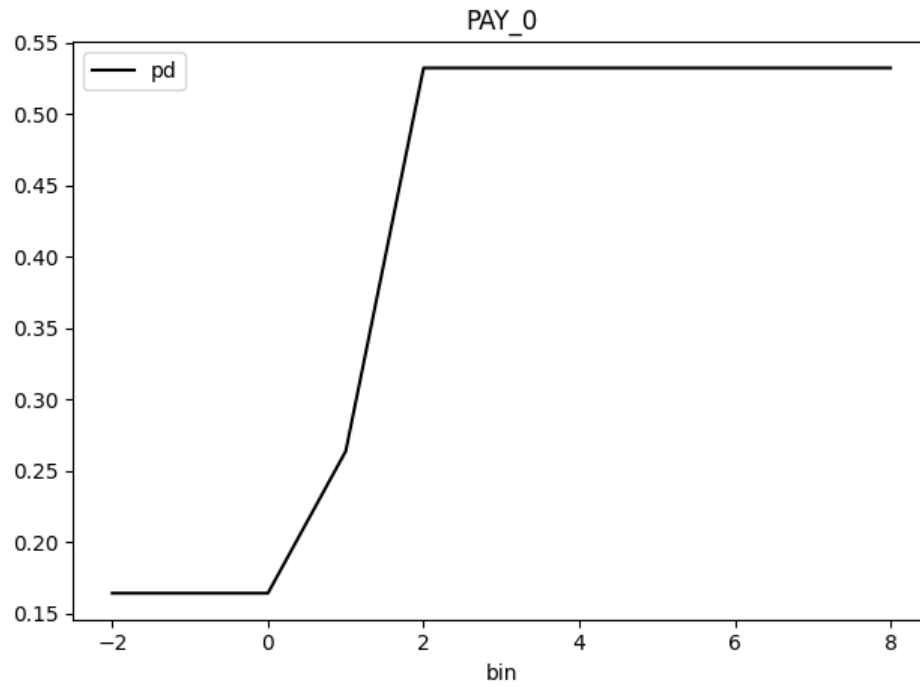
Partial Dependence Plot for the feature 'MARRIAGE' and class '1':



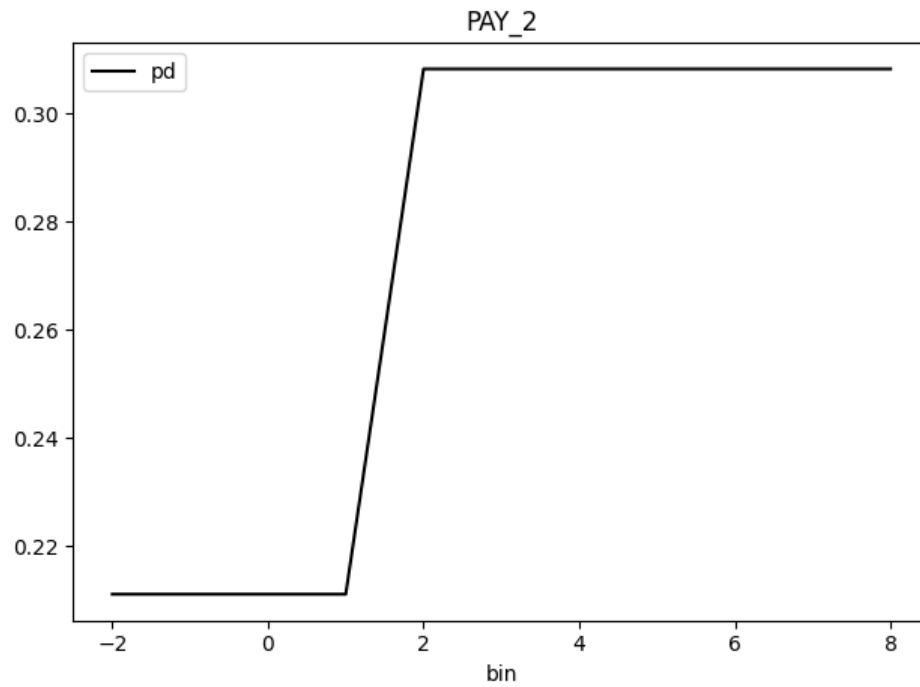
Partial Dependence Plot for the feature 'AGE' and class '1':



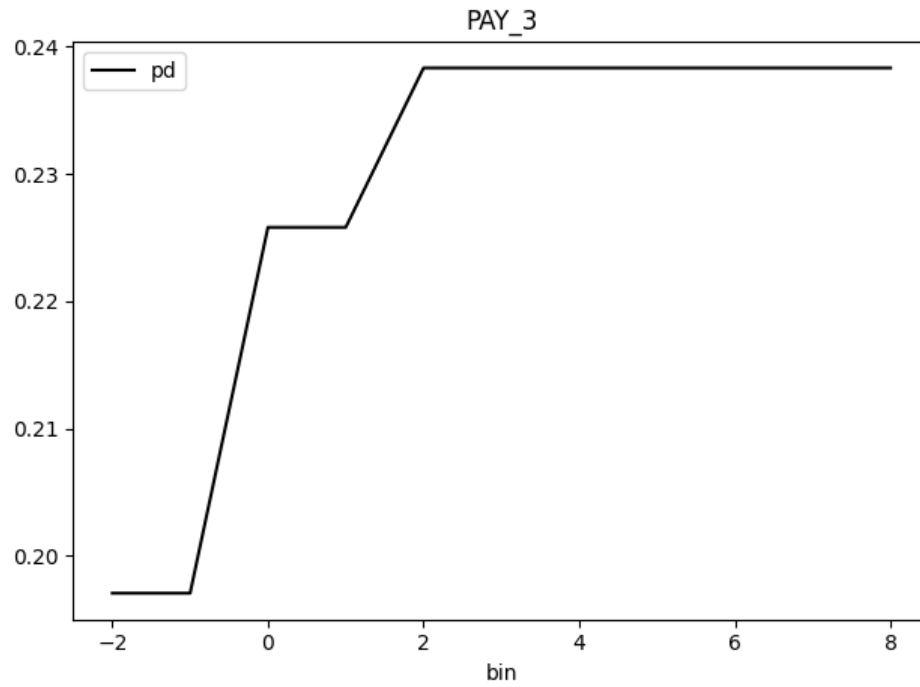
Partial Dependence Plot for the feature 'PAY_o' and class '1':



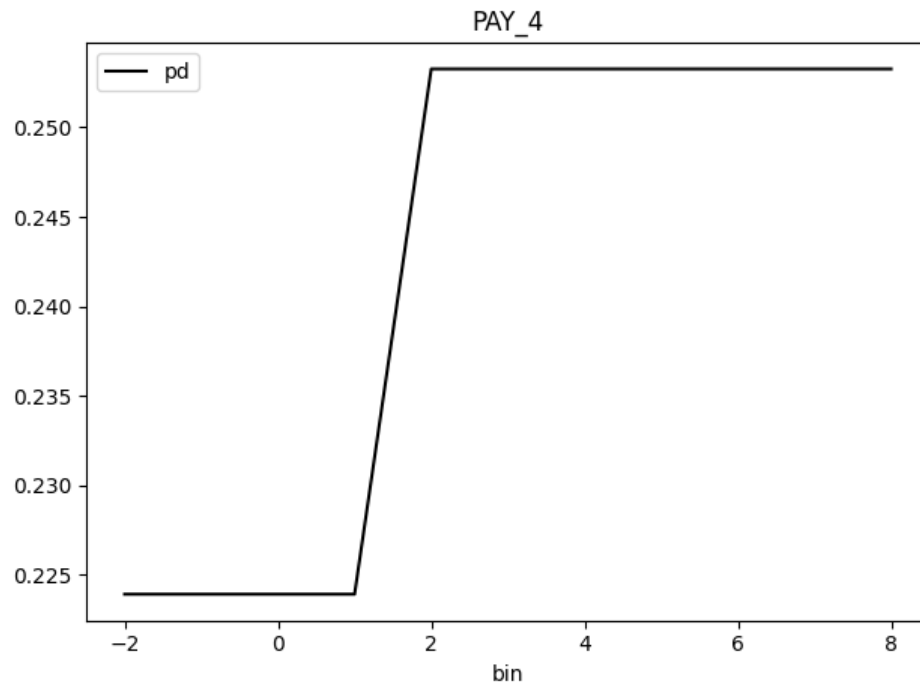
Partial Dependence Plot for the feature 'PAY_2' and class '1':



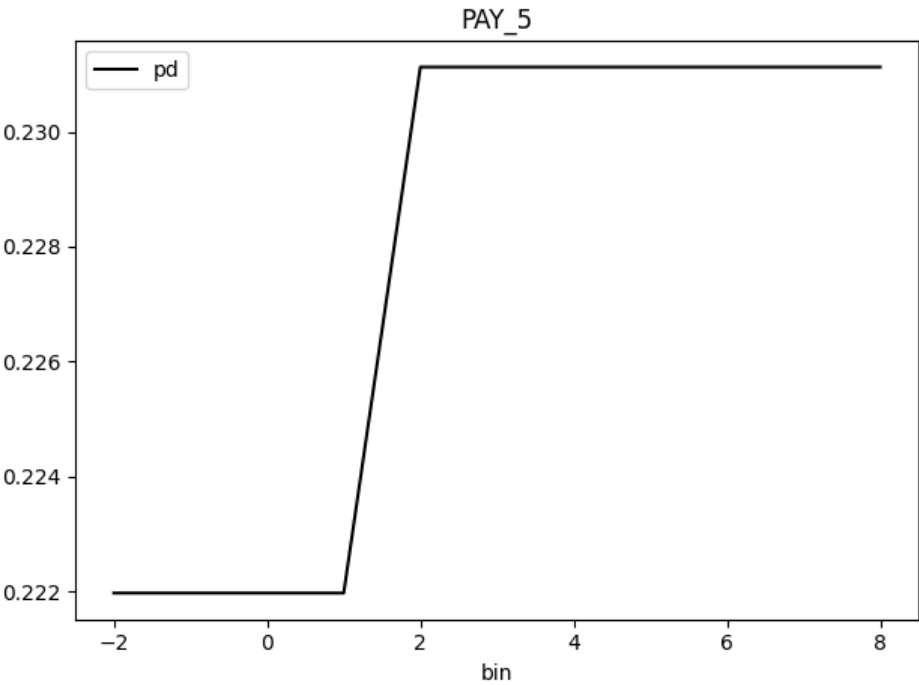
Partial Dependence Plot for the feature 'PAY_3' and class '1':



Partial Dependence Plot for the feature 'PAY_4' and class '1':



Partial Dependence Plot for the feature 'PAY_5' and class '1':



Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types :

- **Partial Dependence Plot (PDP)**
global-partial-dependence
 - [application/json](#)
- **Individual Conditional Expectations (ICE)**
local-individual-conditional-explanation
 - [application/vnd.h2oai.json+datatable.jay](#)
- **Partial Dependence Plot (PDP)**
global-html-fragment
 - [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
sample_size	25000	Sample size for Partial Dependence Plot.	int	25000
max_features	10	Partial Dependence Plot number of features (to see all features used by model set to -1).	int	10
features	None	Partial Dependence Plot feature list.	list	None
oor_grid_resolution	0	Partial Dependence Plot number of out of range bins.	int	0
quantile-bin-grid-resolution	0	Partial Dependence Plot quantile binning (total quantile points used to create bins).	int	0
grid_resolution	20	Partial Dependence Plot observations per bin (number of equally spaced points used to create bins).	int	20

center	False	Center Partial Dependence Plot using ICE centered at 0.	bool	False
sort_bins	True	Ensure bin values sorting.	bool	True
histograms	True	Enable histograms.	bool	True
quantile-bins		Per-feature quantile binning (Example: if choosing features F1 and F2, this parameter is '{"F1": 2,"F2": 5}'. Note, you can set all features to use the same quantile binning with the 'Partial Dependence Plot quantile binning' parameter and then adjust the quantile binning for a subset of PDP features with this parameter).	str	
numcat_num_chart	True	Unique feature values count driven Partial Dependence Plot binning and chart selection.	bool	True
numcat_threshold	11	Threshold for Partial Dependence Plot binning and chart selection (<=threshold categorical, >threshold numeric).	int	11
debug_residuals	False	Debug model residuals.	bool	False

Explainer metadata

The explainer can explain model types:

- regression
- binomial
- multinomial

Explainer keywords:

- run-by-default
- can-add-feature
- explains-feature-behavior
- h2o-sonar

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:39:51 T+0100**
- Duration: **3.015s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Residual Partial Dependence Plot

- [Model problems](#)
- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)

- [Metadata](#)
- [Run](#)
- [Log](#)

Model problems

Explainer identified the following potential problems with the model:

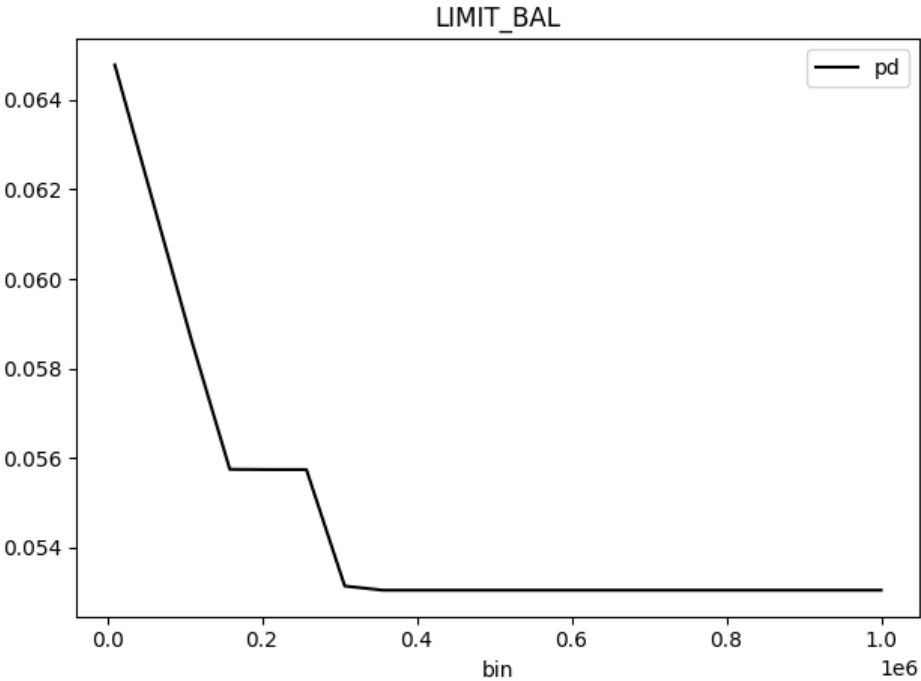
Severity	Type	Problem	Suggested actions	Explainer	Resources
MEDIUM	bias	The residual partial dependence plot of feature 'PAY_o' in class '1' indicates the highest interaction of this feature with the error (residual abs(max-min) = 0.07603767327964306) from all the features used by the model (or features configured for PD calculation)	Verify that feature 'PAY_o' error interaction in class '1' does not indicate a model bias or other problem.	Residual Partial Dependence Plot	PartialDependenceExplanation/application/json

Explainer description

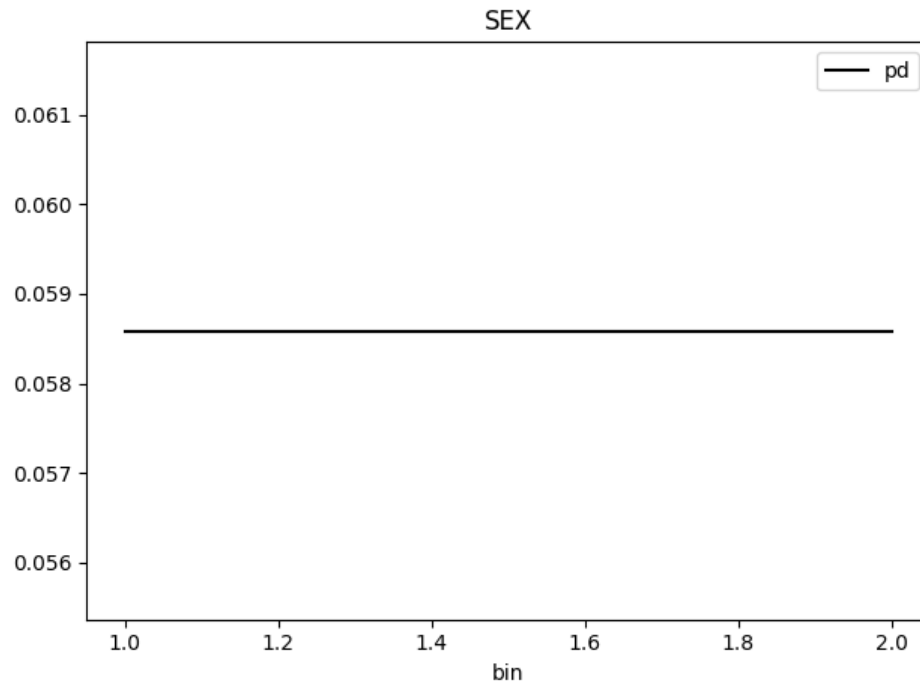
The residual partial dependence plot (PDP) indicates which variables interact most with the error. Residuals are transformed differences between observed and predicted values: the square of the difference between observed and predicted values is used in case of regression problems; $-1 * \log(p)$ is used in case of classification problems. The residual partial dependence is created using normal partial dependence algorithm, while instead of prediction is used the residual. Individual Conditional Expectations plot (ICE) displays the interaction with error for an individual row of data when an input variable is toggled across its domain.

Explanations

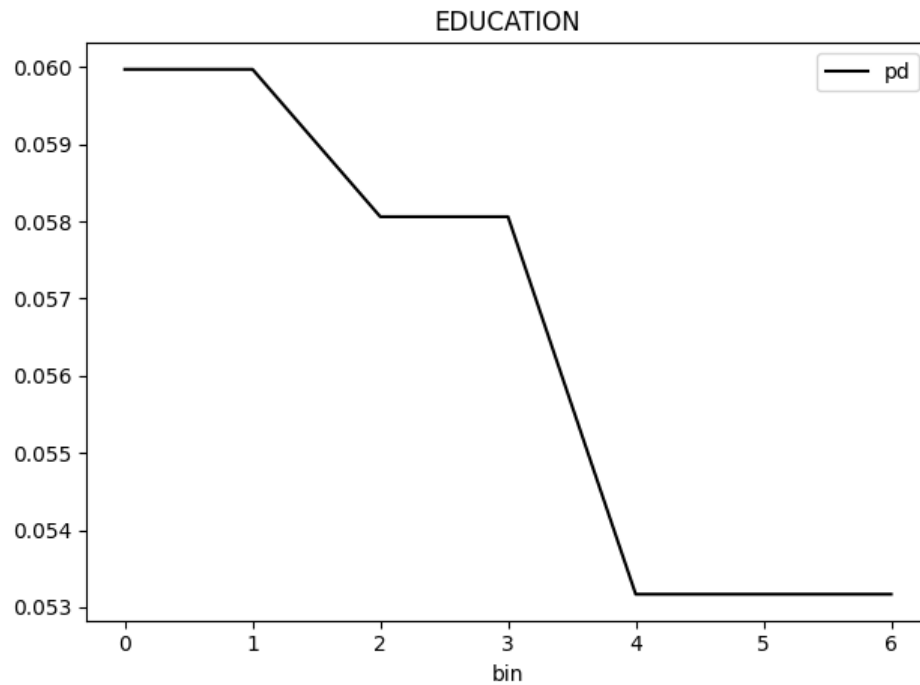
Partial Dependence Plot for the feature 'LIMIT_BAL' and class '1':



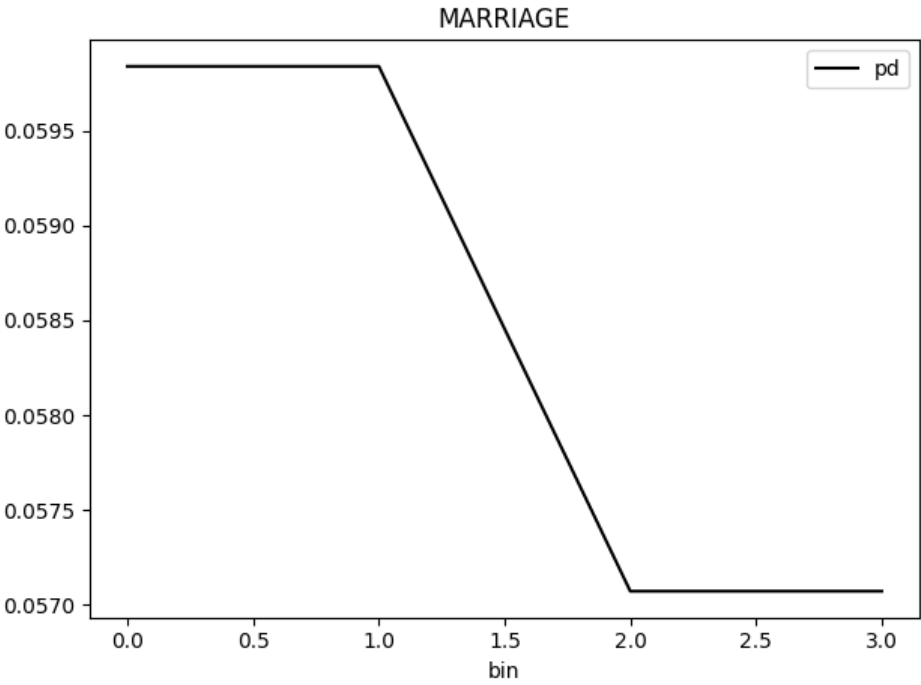
Partial Dependence Plot for the feature 'SEX' and class '1':



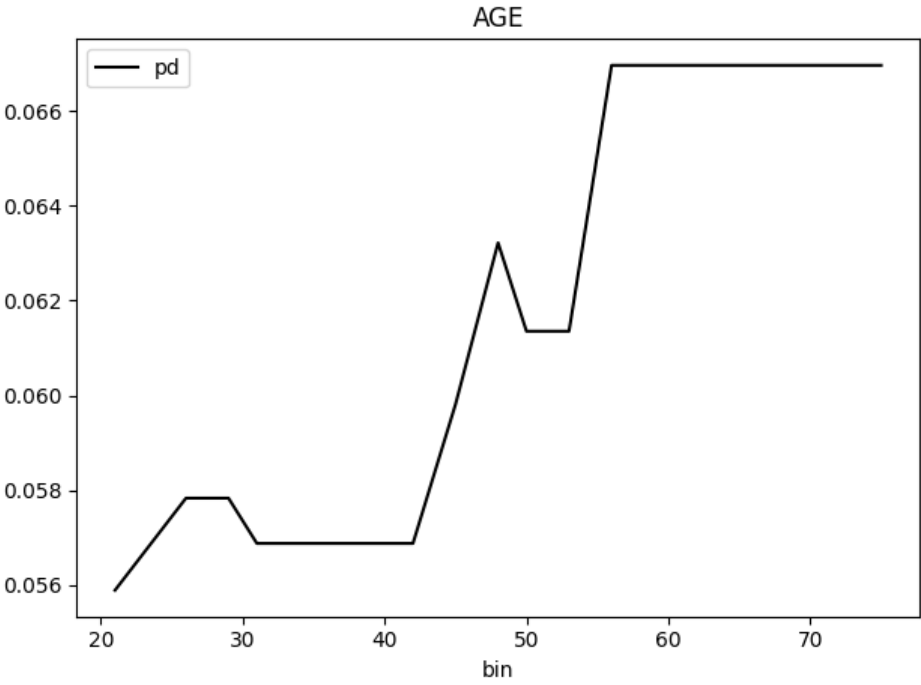
Partial Dependence Plot for the feature 'EDUCATION' and class '1':



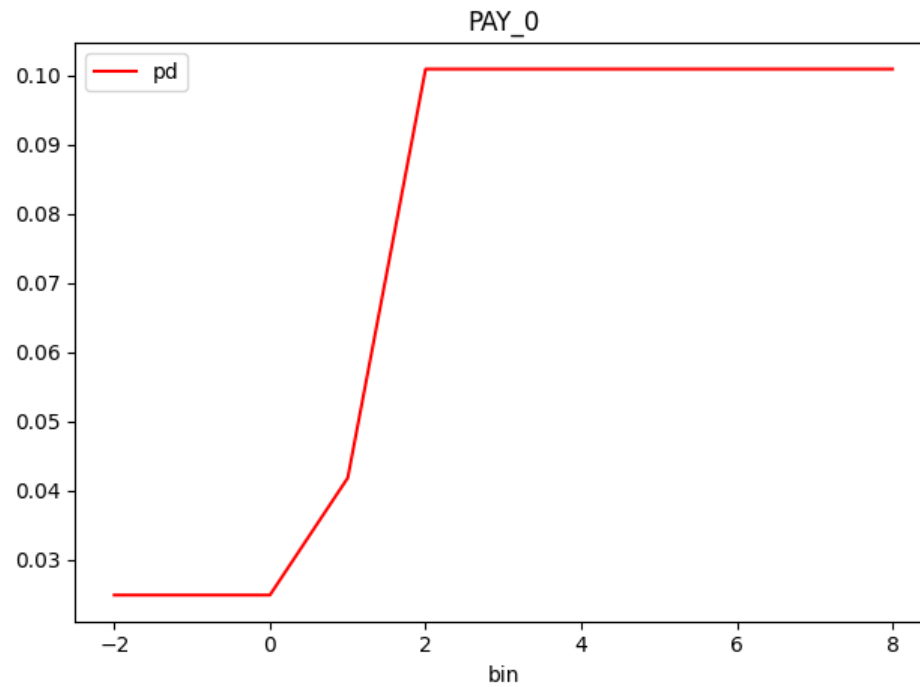
Partial Dependence Plot for the feature 'MARRIAGE' and class '1':



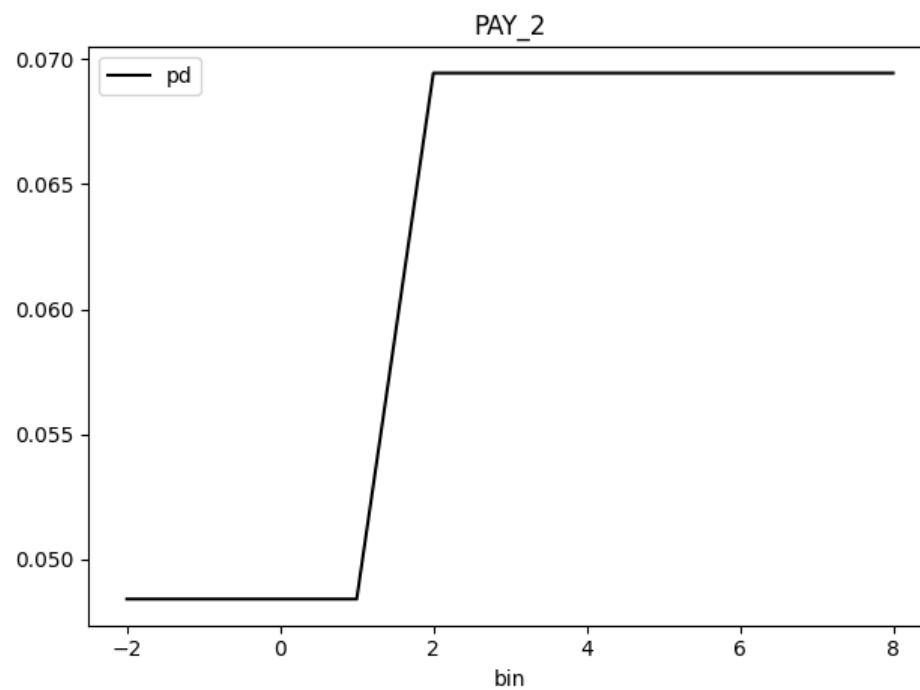
Partial Dependence Plot for the feature 'AGE' and class '1':



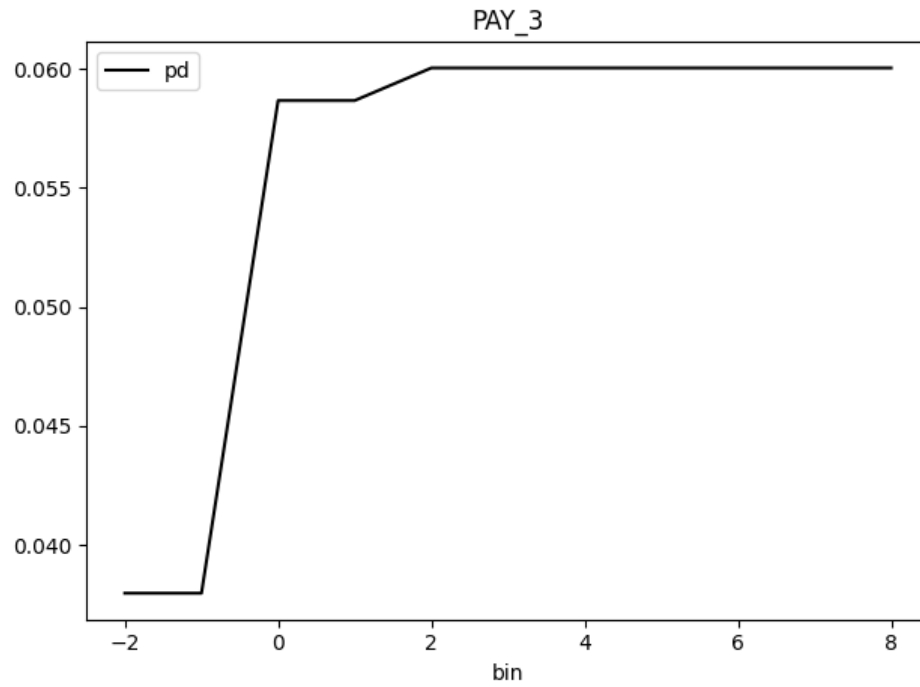
Partial Dependence Plot for the feature 'PAY_o' and class '1':



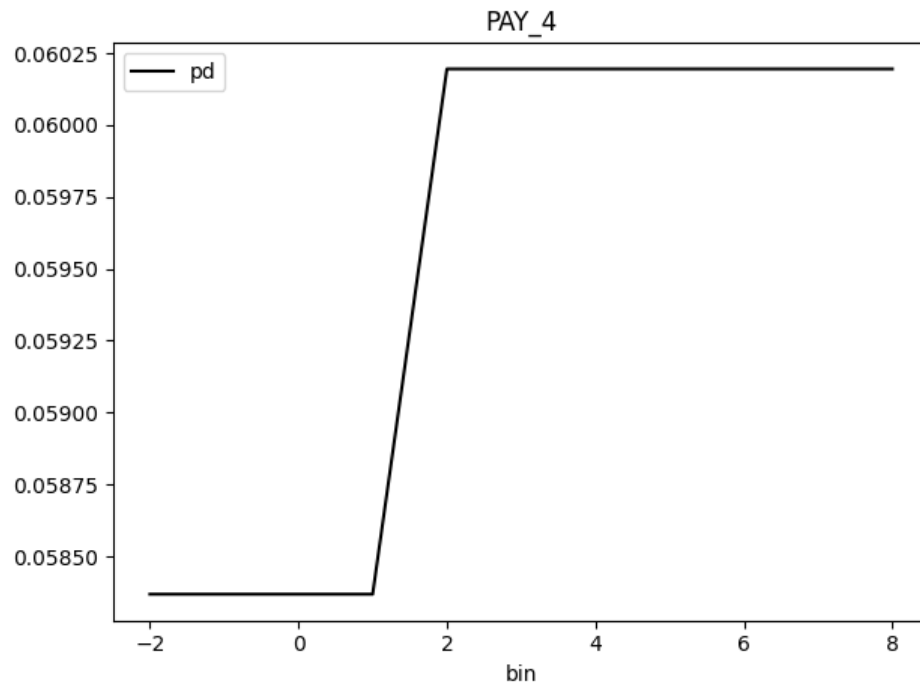
Partial Dependence Plot for the feature 'PAY_2' and class '1':



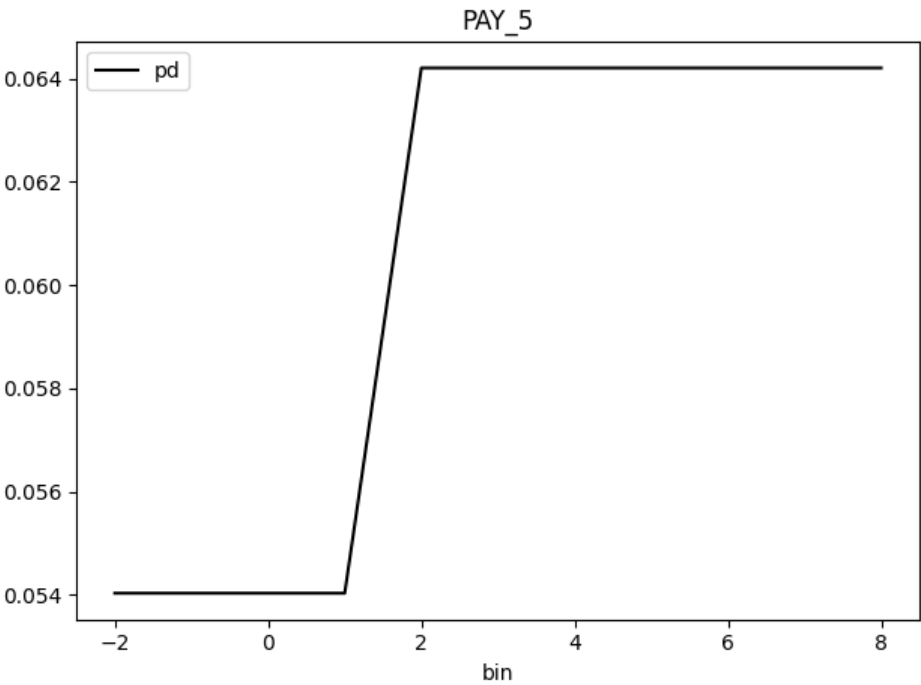
Partial Dependence Plot for the feature 'PAY_3' and class '1':



Partial Dependence Plot for the feature 'PAY_4' and class '1':



Partial Dependence Plot for the feature 'PAY_5' and class '1':



Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types :

- **Residual Partial Dependence Plot**
global-partial-dependence
 - [application/json](#)
- **Residual Individual Conditional Expectations (ICE)**
local-individual-conditional-explanation
 - [application/vnd.h2oai.json+datatable.jay](#)
- **Residual Partial Dependence Plot**
global-html-fragment
 - [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
sample_size	25000	Sample size for Partial Dependence Plot.	int	25000
max_features	10	Partial Dependence Plot number of features (to see all features used by model set to -1).	int	10
features	None	Partial Dependence Plot feature list.	list	None
oor_grid_resolution	0	Partial Dependence Plot number of out of range bins.	int	0
quantile-bin-grid-resolution	0	Partial Dependence Plot quantile binning (total quantile points used to create bins).	int	0
grid_resolution	20	Partial Dependence Plot observations per bin (number of equally spaced points used to create bins).	int	20

center	False	Center Partial Dependence Plot using ICE centered at 0.	bool	False
sort_bins	True	Ensure bin values sorting.	bool	True
histograms	True	Enable histograms.	bool	True
quantile-bins		Per-feature quantile binning (Example: if choosing features F1 and F2, this parameter is '{"F1": 2,"F2": 5}'. Note, you can set all features to use the same quantile binning with the 'Partial Dependence Plot quantile binning' parameter and then adjust the quantile binning for a subset of PDP features with this parameter).	str	
numcat_num_chart	True	Unique feature values count driven Partial Dependence Plot binning and chart selection.	bool	True
numcat_threshold	11	Threshold for Partial Dependence Plot binning and chart selection (<=threshold categorical, >threshold numeric).	int	11
debug_residuals	True			

Explainer metadata

The explainer can explain model types:

- regression
- binomial
- multinomial

Explainer keywords:

- can-add-feature
- explains-model-debugging
- h2o-sonar

Explainer run

Explainer run details:

- Status code: FINISHED
- Progress: 100%
- Started: 2023-11-03 11:39:54 T+0100
- Duration: 3.288s

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Shapley Values for Original Features (Kernel SHAP Method)

- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)

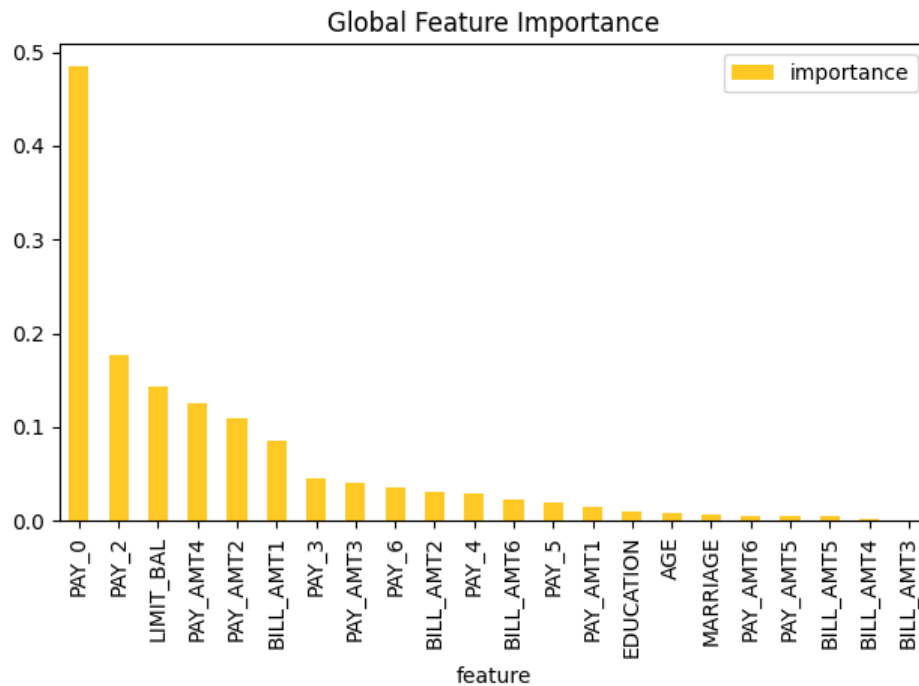
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

Shapley explanations are a technique with credible theoretical support that presents consistent global and local variable contributions. Local numeric Shapley values are calculated by tracing single rows of data through a trained tree ensemble and aggregating the contribution of each input variable as the row of data moves through the trained ensemble. For regression tasks, Shapley values sum to the prediction of the Driverless AI model. For classification problems, Shapley values sum to the prediction of the Driverless AI model before applying the link function. Global Shapley values are the average of the absolute Shapley values over every row of a dataset. Shapley values for original features are calculated with the Kernel Explainer method, which uses a special weighted linear regression to compute the importance of each feature. More information about Kernel SHAP is available at <http://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf>.

Explanations

Feature importance for the class '1':



The most important original feature of the class 1 is **PAY_0**.

Original feature importances for the class '1':

- 1. **PAY_0** feature with importance 0.48431425950676193
- 2. **PAY_2** feature with importance 0.17728837728463115
- 3. **LIMIT_BAL** feature with importance 0.14293027323329394
- 4. **PAY_AMT4** feature with importance 0.1259882615562616
- 5. **PAY_AMT2** feature with importance 0.10983877249736468
- 6. **BILL_AMT1** feature with importance 0.085668529046002
- 7. **PAY_3** feature with importance 0.04604692619013324
- 8. **PAY_AMT3** feature with importance 0.0405642732338496
- 9. **PAY_6** feature with importance 0.035128979444635125
- 10. **BILL_AMT2** feature with importance 0.03075326452456647
- 11. **PAY_4** feature with importance 0.029223238050425416
- 12. **BILL_AMT6** feature with importance 0.023093468860886066

- 13. PAY_5 feature with importance 0.02037994117215276
- 14. PAY_AMT1 feature with importance 0.014774042908039704
- 15. EDUCATION feature with importance 0.009833406838281324
- 16. AGE feature with importance 0.008313157282831355
- 17. MARRIAGE feature with importance 0.007252847269852645
- 18. PAY_AMT6 feature with importance 0.00588242508913081
- 19. PAY_AMT5 feature with importance 0.005415161620848719
- 20. BILL_AMT5 feature with importance 0.004646423485466219
- 21. BILL_AMT4 feature with importance 0.0014406579161764855
- 22. BILL_AMT3 feature with importance 0.0008182751896645635

Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types :

- **Shapley on Original Features (Kernel SHAP Method)**

global-feature-importance

- [application/vnd.h2oai.json+datatable.jay](#)
- [application/vnd.h2oai.json+csv](#)
- [application/json](#)

- **Shapley on Original Features (Kernel SHAP Method)**

local-feature-importance

- [application/vnd.h2oai.json+datatable.jay](#)

- **Shapley on Original Features (Kernel SHAP Method)**

global-html-fragment

- [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
sample_size	100000	Sample size.	int	100000
sample	True	Sample Kernel Shapley.	bool	True
nsample		Number of times to re-evaluate the model when explaining each prediction with Kernel Explainer. Default is determined internally.'auto' or int. Number of times to re-evaluate the model when explaining each prediction. More samples lead to lower variance estimates of the SHAP values. The 'auto' setting uses nsamples = 2 * X.shape[1] + 2048. This setting is disabled by default and runtime determines the right number internally.	int	
L1	auto	L1 regularization for Kernel Explainer. 'num_features(int)', 'auto' (default for now, but deprecated), 'aic', 'bic', or float. The L1 regularization to use for feature selection (the estimation procedure is based on a debiased lasso). The 'auto' option currently uses aic when less than 20% of the possible sample space is enumerated, otherwise it uses no regularization. The aic and bic options use the AIC and BIC rules for regularization. Using 'num_features(int)' selects a fix number of top features. Passing a float directly sets the alpha parameter of the sklearn.linear_model.Lasso model used for feature selection.	str	auto
max runtime	900	Max runtime for Kernel explainer in seconds.	int	900

fast_approx	True	Speed up predictions with fast predictions approximation.	bool	True
leakage_warning_threshold	0.95	The threshold above which to report a potentially detected feature importance leak problem.	float	0.95

Explainer metadata

The explainer can explain model types:

- regression
- binomial
- multinomial

Explainer keywords:

- explains-original-feature-importance
- is_slow
- h2o-sonar

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:39:57 T+0100**
- Duration: **1.411s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Partial Dependence Plot for Two Features

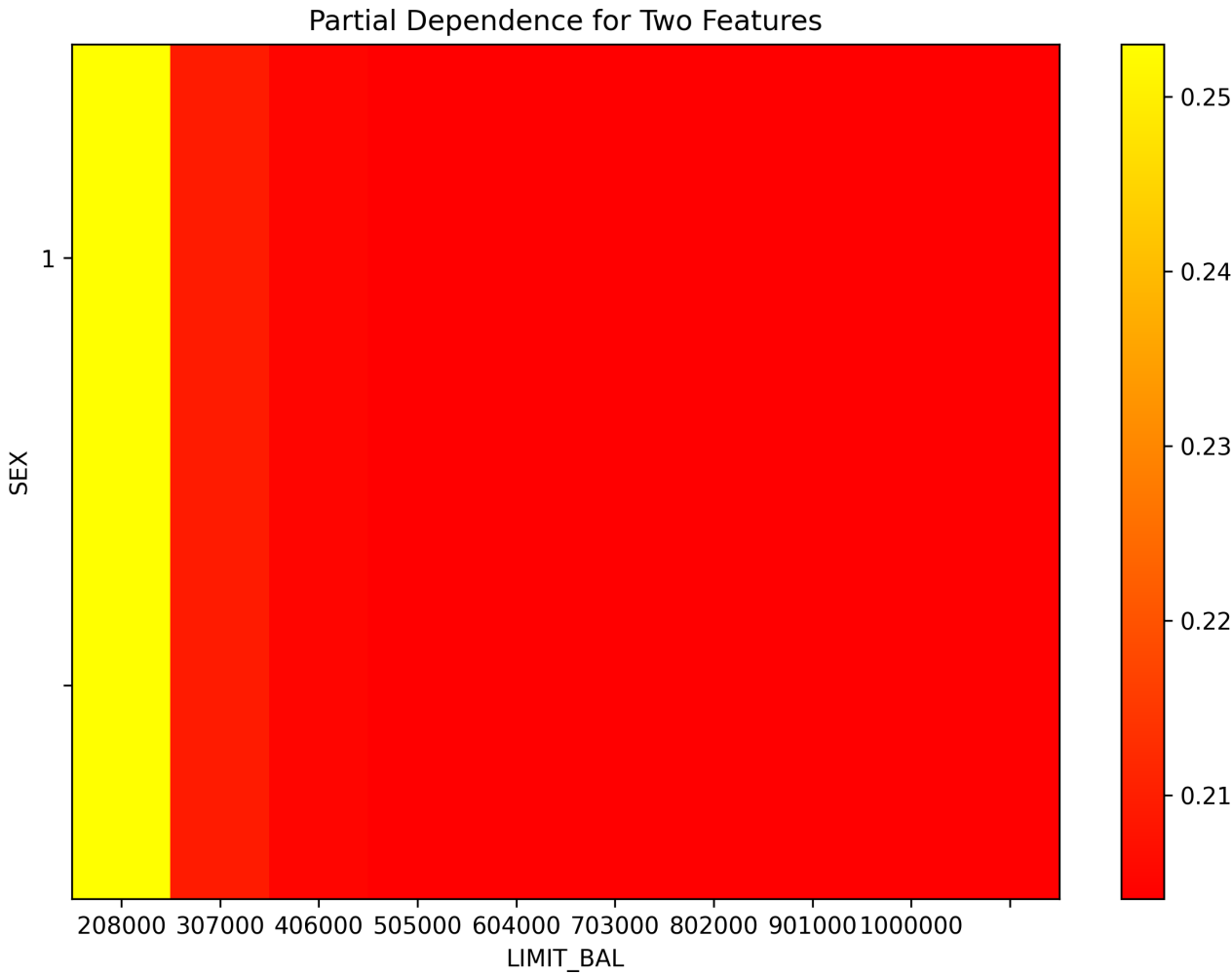
- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

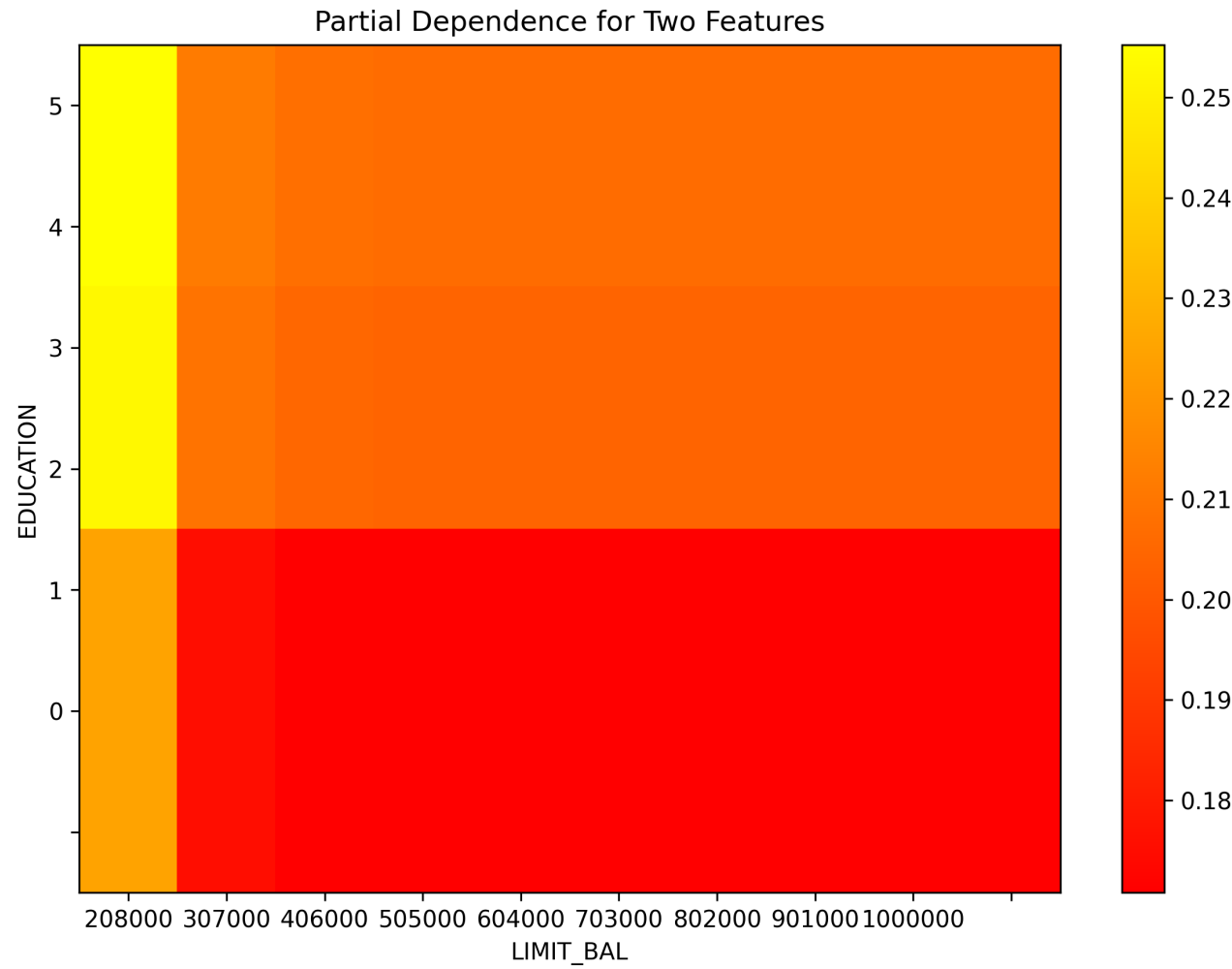
Partial dependence for 2 features portrays the average prediction behavior of a model across the domains of two input variables i.e. interaction of feature tuples with the prediction. While PD for one feature produces 2D plot, PD for two features produces 3D plots. This explainer plots PD for two features using heatmap, contour 3D or surface 3D.

Explanations

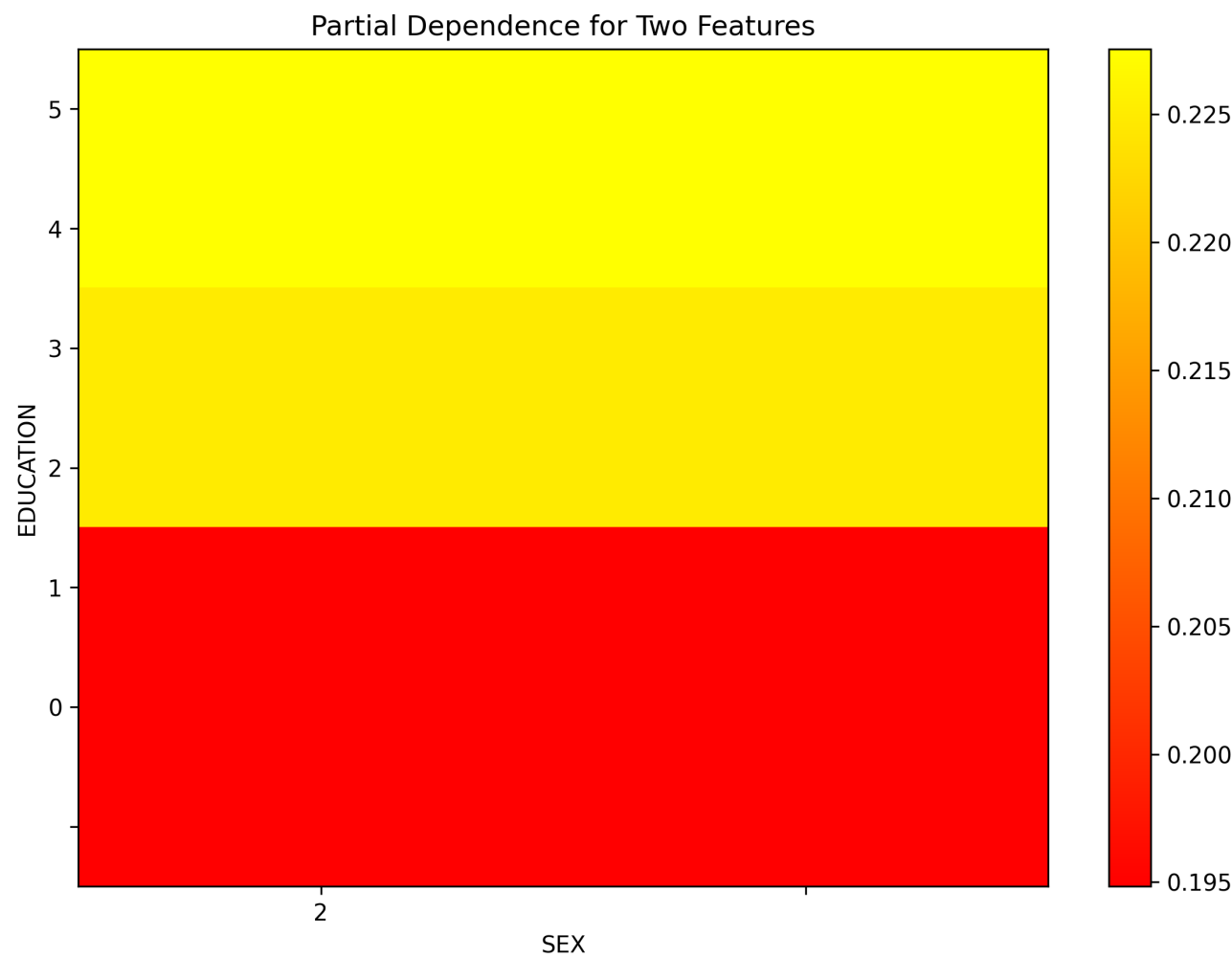
Partial dependence plot for features 'LIMIT_BAL' and 'SEX':



Partial dependence plot for features 'LIMIT_BAL' and 'EDUCATION':



Partial dependence plot for features 'SEX' and 'EDUCATION':



Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types :

- Partial Dependence Plot for Two Features
 - global-report*
 - [text/markdown](#)
- Partial Dependence Plot for Two Features
 - global-3d-data*
 - [application/json](#)
 - [application/vnd.h2oai.json+csv](#)
- Partial Dependence Plot for Two Features
 - global-html-fragment*
 - [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
sample_size	25000	Sample size for Partial Dependence Plot of 2 features.	int	25000

max_features	3	Partial Dependence Plot number of features.	int	3
features	None	List of features from which to choose pairs to compute PD for two features.	list	None
grid_resolution	10	Partial Dependence Plot observations per bin (number of equally spaced points used to create bins).	int	10
oor_grid_resolution	0	Partial Dependence Plot number of out of range bins.	int	0
quantile-bin-grid-resolution	0	Partial Dependence Plot quantile binning (total quantile points used to create bins).	int	0
plot_type	heatmap	Plot type.	str	heatmap

Explainer metadata

The explainer can explain model types:

- regression
- binomial

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:39:59 T+0100**
- Duration: **3.692s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Friedman's H-statistic

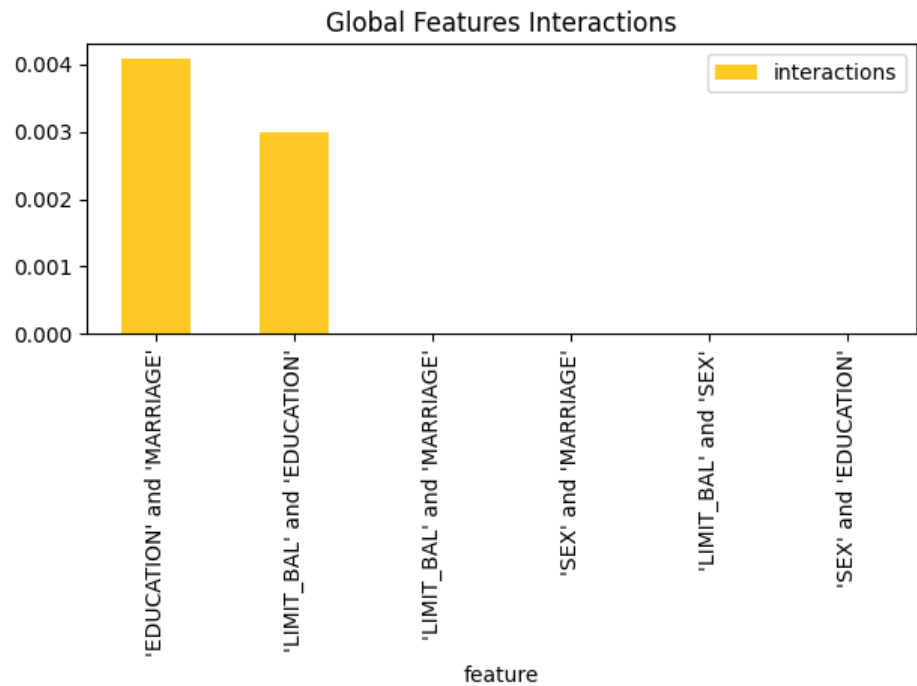
- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

Friedman's H-statistic describes the amount of variance explained by the feature pair. It's expressed with a graph where most important original features are nodes and the interaction scores are edges. When features interact with each other, then the influence of the features on the prediction does not have be additive, but more complex. For instance the contribution might be greater than the sum of contributions. Friedman's H-statistic calculation is computationally intensive and typically requires long time to finish - calculation duration grows with the number of features and bins.

Explanations

Feature importance for the class '1':



Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by media types :

- **Friedman's H-statistic**
global-feature-importance
 - [application/vnd.h2oai.json+datatable.jay](#)
 - [application/vnd.h2oai.json+csv](#)
 - [application/json](#)
- **Friedman's H-statistic report**
global-report
 - [text/markdown](#)
- **Friedman's H-statistic**
global-html-fragment
 - [text/html](#)

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
features_number	4	Number of features for which to calculate H-Statistic.	int	4
grid_resolution	3	Observations per bin (number of equally spaced points used to create bins).	int	3
features	None	Feature list - at least 2 features must be selected.	multilist	None
sample_size	25000	Sample size for Partial Dependence Plot	int	25000

Explainer metadata

The explainer can explain model types:

- regression
- binomial

Explainer keywords:

- explains-feature-behavior
- h2o-sonar

Explainer run

Explainer run details:

- Status code: **FINISHED**
- Progress: **100%**
- Started: **2023-11-03 11:40:02 T+0100**
- Duration: **1.521s**

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Dataset and model insights explainer

- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

The explainer checks the dataset and model for various issues. For example, it provides problems and actions for missing values in the target column and a low number of unique values across columns of a dataset.

Explanations

Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by [media types](#) :

- **Dataset and model insights explainer**
 - global-text-explanation*
 - [text/plain](#)

Explainer parameters

Explainer was run with the following parameters:

Explainer metadata

The explainer can explain model types:

- regression
- binomial
- multinomial

Explainer run

Explainer run details:

- Status code: `FINISHED`
- Progress: `100%`
- Started: `2023-11-03 11:40:04 T+0100`
- Duration: `0.284s`

Explainer log

Explainer log file:

- [explainer.log](#)

Explainer: Retrieval Augmented Generation Assessment (DeepEval)

- [Explainer description](#)
- [Explanations](#)
- [Explanations data](#)
- [Parameters](#)
- [Metadata](#)
- [Run](#)
- [Log](#)

Explainer description

Explanations

Explanations

Model explanations created by the explainer organized by explanation types with its formats (representations) identified by [media types](#) :

Explainer parameters

Explainer was run with the following parameters:

Parameter	Value	Description	Type	Default value
min_fcm_score	0.75			

Explainer metadata

The explainer can explain model types:

Explainer run

Explainer run details:

- Status code: **FAILED**
- Error: 'No module named 'deepeval''
- Progress: **30%**
- Started: **2023-11-03 11:40:04 T+0100**
- Duration: **0.009s**

Explainer log

Explainer log file:

- [explainer.log](#)

H2O Sonar

H2O Sonar is Python package that enables a holistic, low-risk, human-interpretable, fair, and trustable approach to machine learning by implementing various facets of Responsible AI.

Resources



H2O Sonar
GitHub repository



H2O.ai
Democratize AI with H2O.ai